Guide to Using AABS Software for Rotational Spectroscopy

(http://www.ifpan.edu.pl/~kisiel/aabs/aabs.htm#aabs**)**

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File Types (Kisiel recommended file terminations)

.ASF	Input file for ASFIT.exe with experimental measurements
.ASR	Single-molecule predicted stick spectrum
.BAK	Backup .PAR-type file for SPFIT (in case of undesired .PAR overwriting)
.CAT	Prediction stick spectrum, openable in ASCP_L.exe
.CON	Input file for CONVC.exe, which converts between variousreductions/representations, and for PLAN.exe, which determines correct error values for inertial defects from rotational constants (and other stuff)
CONVC.OUT	Output file from CONVC.exe
.DAT	Data files for various .GLE scripts – not directly edited by user
ECHO.ASC	Line frequency file for removing lines from SVIEW
.EGY	I don't know. We don't mess with that thing…
.FIT	SPFIT.exe least-squares fit output file, input for PIFORM.exe
.FRE	Peak-picked frequency file for Loomis-Wood plots obtained from SVIEW
.GLE	Script file for various spectral images and plots
.IN	Input files for ASROT to make predicted stick spectra
.INP	Batch input file for ASCP to display multiple stick spectra in single window (different from SVIEW_L.INP, below)
.INT	Required input file for SPCAT.exe, contains dipole moments
.LIN	Input file for SPFIT.exe containing experimental measurements; may contain multiple vibrational states

.OUT	Output file from THISTHING. See *PLAN.OUT* or *CONVC.OUT* thusly-named files.
.PAR	Input file for SPFIT.exe containing starting constants; may contain multiple vibrational states
PLAN.OUT	Output file from PLAN.exe
.RES	ASFIT or SPFIT.exe least-squares fit output file. See *.TXT*
.SPE	Experimentally obtained spectral file (may be binary or ASCII)
.STR	Another file I do not know what it does. Don't mess with it.
SVIEW_L.INP	File that links stick spectrum (ASCP), experimental spectrum (SVIEW), .FRE and .ASF file
.TXT	ASFIT or SPFIT.exe least-squares fit output file, also called *.RES*
.VAR	Output file from THISTHING.

.ASF Files

.ASF files are the input for least-squares fitting in asfit.exe (ASFIT). They include the following elements

Title – A descriptive title of the molecule/vibrational satellite

Number of transitions – This is irrelevant for ASFIT operation, but can be noted.

Number of constants – This value (MHz) indicates the constants to be fit to the program. The constant symbols listed after each value are not read by the program. Up to 35 constants (3 Rotational constants, as well as 5 quartic, 7 sextic, 9 octic, and 11 decadic distortion constants) may be fit. The A-reduction and S-reduction constants are provided below. The units specified in the table are the typical units for publication. All constants at the top of the .ASF file are given in MHz.

	A-reduction	**S-Reduction**	
1	A_0 (MHz)	A_0 (MHz)	*Rotational Constants*
2	B_0 (MHz)	B_0 (MHz)	
3	C_0 (MHz)	C_0 (MHz)	
4	Δ_J (kHz)	D_J (kHz)	*Quartic Distortion Constants*
5	Δ_{JK} (kHz)	D_{JK} (kHz)	
6	Δ_K (kHz)	D_K (kHz)	
7	δ_J (kHz)	d_1 (kHz)	
8	δ_K (kHz)	d_2 (kHz)	
9	Φ_J (Hz)	H_J (Hz)	*Sextic Distortion Constants*
10	Φ_{JK} (Hz)	H_{JK} (Hz)	
11	Φ_{KJ} (Hz)	H_{KJ} (Hz)	
12	Φ_K (Hz)	H_K (Hz)	
13	ϕ_J (Hz)	h_1 (Hz)	
14	ϕ_{JK} (Hz)	h_2 (Hz)	
15	ϕ_K (Hz)	h_3 (Hz)	
16	L_J (mHz)	L_J (mHz)	*Octic Distortion Constants*
17	L_{JJK} (mHz)	L_{JJK} (mHz)	
18	L_{JK} (mHz)	L_{JK} (mHz)	
19	L_{KKJ} (mHz)	L_{KKJ} (mHz)	
20	L_K (mHz)	L_K (mHz)	
21	l_J (mHz)	l_1 (mHz)	
22	l_{JK} (mHz)	l_2 (mHz)	
23	l_{KJ} (mHz)	l_3 (mHz)	
24	l_K (mHz)	l_4 (mHz)	

List of Constants – Ideally, the best current guess (from an *ab initio* calculation or previous work) or known constants (in MHz) are input as a starting point for the fit. They should be aligned at the decimal point and such that the last digit is in the last column allotted for constants (columns allotted for specific values are denoted by "-" and "+" symbols immediately above the list of constants). The constants are preceded by a value indicating whether the value should be optimized in the least-squares fit (1) or held constant (0). The list of constants must correspond to the "number of constants" described above in both number and order in which they are entered. The program does not read the symbol following the constant (these are included only for the user's convenience), but rather determines the constant by its number in the list. If more or fewer constants are listed than stated in "number of constants", ASFIT will err and not run.

1 = Fit constant
0 = Hold constant fixed

Constants, aligned at decimal and to last "+" column

Notes for user's convenience, not read by ASFIT

```
-----++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
    1      10030.6155891481703292811   A      Reduction-S, Representation-III.r
    1       9870.4829400000000000000   B
    1       4972.7564600000000000000   C
```

Reduction, Representation – This is a note included after A for the user's convenience and is not read by the program. The same file can be used for A-reduction or S-reduction fitting, though in some cases the input constants may need to change to avoid ASFIT finding a local rather than global minimum. The actual choice of reduction and representation is made from a keyboard input when ASFIT is run. The user should take care that fixed constants are in the correct reduction-representation before fitting.

List of measured lines – The list of experimentally measured lines. Each line includes the transition quantum numbers, frequency of the transition, max error value (0.05 standard) and value indicating how the lines is fit (0 = Excluded from fit, 1 = unique line, 2 = blended or degenerate line). Though, measured lines may be added manually, carefully following the proper format, it is recommended to have AABS automatically place line measurements in the file using the F1, F2, and F3 keys. This will be explained later using correctly linked ASCP_L.exe and SVIEW_L.exe programs. As with the constants, the "+" and "-" symbols above the list of lines denote the columns allotted for each of the labeled values. All values should end in the right-most column and frequencies should be decimal-aligned. The user is highly advised to keep the list of measured lines organized by frequency to avoid errors when running ASFIT, to more easily notice patterns and errors, and for ease of reading/searching. An efficient way to organize lines by frequency is explained in AAA, on page XX. The user may choose to add blank lines for ease of reading/visual organization (*e.g.* blank line every 10 GHz).

Transition [J'	$K'_{prolate}$	K'_{oblate}	←	J"	$K''_{prolate}$	Measured frequencies, aligned at decimal and to last "+" column	Max Error (Standard 0.05 MHz)	0 = Exclude 1 = Fit unique 2 = Fit degernate or blended	Notes for user's convenience; not read by ASFIT
J'	K-1'	K+1'		J	K-1	K+1	frequency	error	fit?	note/RI
10	8	3		10	8	2	23615.11	0.050000	1	Stiefvater
52	28	25		52	26	26	253048.741395	0.050000	2	
52	27	25		52	26	26	253048.741395	0.050000	2	

An excerpt of a sample file is provided below for 1H-1,2,3-triazole.

```
1H-123-triazole
----------------------+++++-------------------------------------------------------
number of transitions: 1000
number of constants  :   15
-----+++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
    1      10030.6155891481703292811   A      Reduction-S, Representation-III.r
    1       9870.4829464440808806103   B
    1       4972.7564624523311067605   C
    1          0.0035150000000000000   DJ
    1          0.0029659549759099707   DJK
    1         -0.0009162331568537175   DK
    1         -0.0005647367061063086   d1
    1         -0.0004493472371280167   d2
    0          0.0000000000000000000   HJ
    0          0.0000000000000000000   HJK
    1          0.0000000000000000000   HKJ
    1          0.0000000000000000000   HK
    0          0.0000000000000000000   h1
    0          0.0000000000000000000   h2
    0          0.0000000000000000000   h3
-------------------------------------------------------------------------------
   J' K-1' K+1'     J  K-1  K+1         frequency          error    fit?   note/RI
+++++-----+++++-----+++++-----+++++++++++++++++++++-----------++++++----------
   10     8     2   10     8     3         23615.11          0.050000     1
   52    28    25   52    26    26        253048.741395      0.050000     2
   52    27    25   52    26    26        253048.741395      0.050000     2
-------------------------------------------------------------------------------
-------------------------------------------------------------------------------
```

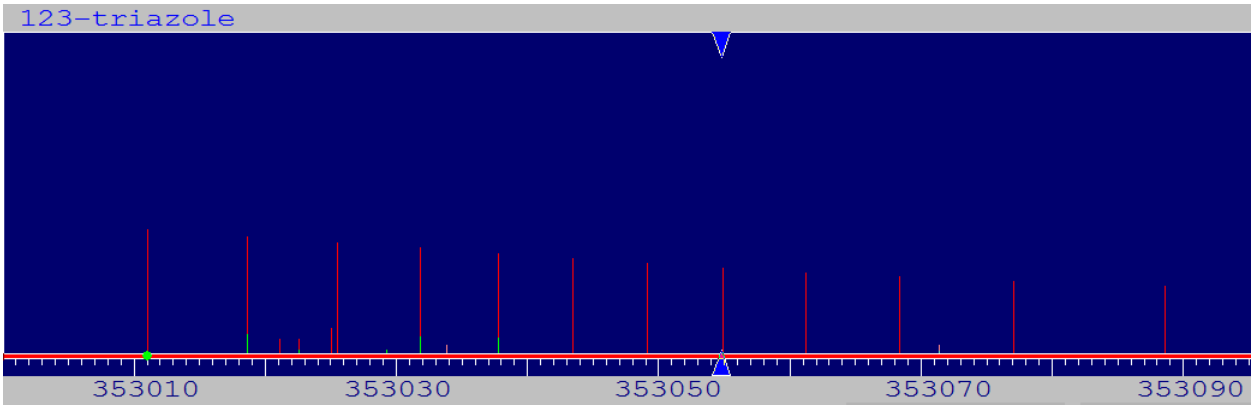
.ASR Files

.ASR files contain predicted stick spectra produced by asrot.exe (ASROT) using .IN files. They are typically viewed using ASCP_L.exe (ASCP), but may be opened by a plain text editor to view and search for specific transitions. When opening the .ASR file in ASCP, use code "1" for "Direct input from specified files". Multiple .ASR files can be opened either by keyboard input or as batch input (.INP files), which will be discussed later.

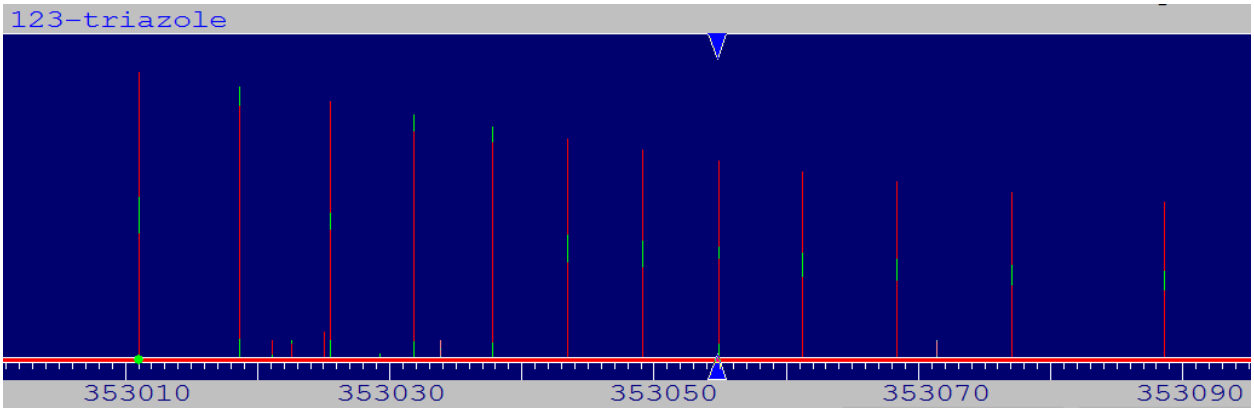
Different types of lines (R-, P-, and Q-branch transitions of each a- and b-type) appear in different colors for convenient identification. More color options are available when viewing multiple files. A description of keys for using ASCP is provided later in the section, *ASCP_L.exe*.

The window size can be adjusted in the aabs.cfg file (described in *ASCP_L.exe and SVIEW_L.exe*)
The *P key* toggles lines colors.
The *C key* toggles the background color.
The *Y key* can be used to toggle on/off the stacking of degenerate line intensities.
The *& key* (SHIFT+7) flips the stick spectrum vertically.

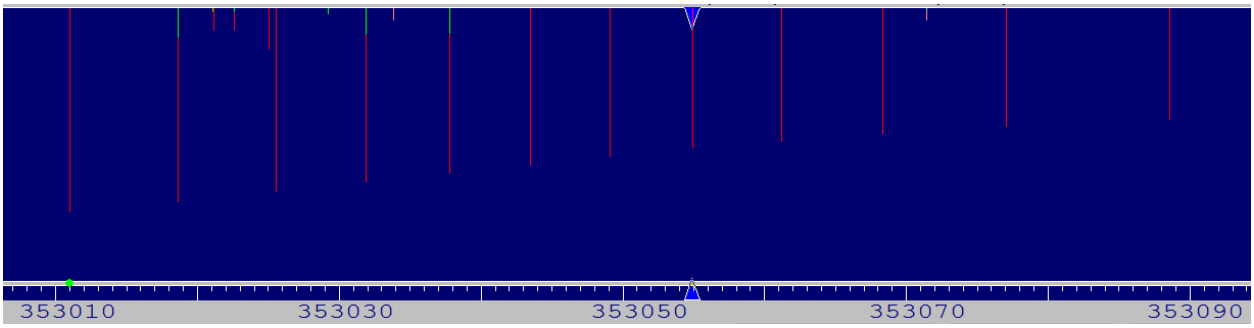
Default view, degenerate or blended transitions overlapped



Stacked view, degenerate or blended transitions stacked



Vertically flipped stick spectrum



.CAT Files

.CAT are predicted stick spectrum files output by SPCAT using .INT and .VAR files. .CAT files may contain one or multiple molecules or states, but otherwise behave like .ASR files. They are viewable using ASCP_L.exe, but may be opened and searched via plain text editor. However, if the .CAT file contains multiple states, the sticks of different states may not be differentiated by color using the P command in ASCP_L.exe and the text is somewhat less intuitive than that of .ASR files. Even if a .CAT file contains multiple states or molecules, it is opened with the 1 command ("Direct input from specified files") in ASCP_L.exe.

A few lines of a single-state .CAT file with no vibrational number assigned appears below; the frequency and transition are highlighted. In the example, a nondegenerate line lies between two degenerate pairs.

Frequency					Transition [J' $K'_{prolate}$ K'_{oblate}← J" $K''_{prolate}$ K''_{oblate}]
225127.7670	0.0049	-1.3551	3	73.9296 37	1 30318 414 17 413
225127.7670	0.0049	-1.3551	3	73.9296 37	1 30318 514 17 513
225139.7297	4.0012	-4.3382	3	1192.1222121	1 3036051 9 604912
225140.4928	0.0047	-1.4068	3	70.5904 35	1 30317 512 16 511
225140.4928	0.0047	-1.4068	3	70.5904 35	1 30317 612 16 611

A .CAT files with multiple states or assigned vibrational quantum numbers looks similar, but has the vibrational quantum number as the fourth value among the prime and double-prime quantum numbers.

Frequency					Transition [J' $K'_{prolate}$ K'_{oblate} ν'← J" $K''_{prolate}$ K''_{oblate} ν"]
224947.1260	0.0385	-4.5524	3	213.3442 67	11404332311 1 331914 1
224968.8382	0.0241	-7.4268	3	67.0826 39	11404191 5 5 0 19 910 0
224987.6626	2.4995	-3.9881	3	1520.7128181	11404905337 1 905338 1
224987.6626	2.4995	-3.9881	3	1520.7128181	11404905437 1 905238 1
225038.7918	27.5748	-4.7376	3	953.4516137	11404686 5 4 0 6863 5 0

.CON Files

.CON (constant) files are used as input files for both CONVC.exe and PLAN.exe. CONCV.exe allows the user to convert between constants in different reductions and representations while PLAN.exe is mainly used to calculate the correct error values for inertial defects from the rotational constants. The file format follows.

NCON – This is the number of constants included

List of constants and errors – This list is formatted in the same way as at the top of the .TXT ASFIT output file, so they may easily be copy-pasted and the "+-" replaced with two spaces. (If in a column-highlighting text editor, it is typically most efficient to copy two columns of spaces adjacent to the "+-" symbols and then paste over the two columns of "+-" symbols.

Reduction and Representation – This is a note for the user's convenience and is not read by PLAN.exe or CONVC.exe

```
NCON =   10
 A    =     10030.7989132911989145214         0.0002570922104606597
 B    =      9870.6933116967466048663         0.0002607921576414931
 C    =      4972.9417134999202971812         0.0001647771968440177
  DJ  =         0.0020667188362214792         0.0000004128773257324
  DJK =        -0.0008241358220703684         0.0000014205958453726
  DK  =         0.0023328037257342288         0.0000007611002590506
  dJ  =         0.0008321047525198883         0.0000002007540751156
  dK  =         0.0013323075777698669         0.0000002291479375699
 HKJ  =        -0.0000000088634847209         0.0000000002544170517
 HK   =         0.0000000144890858113         0.0000000002336395411
  |
   ---------------------- Reduction-A,  Representation-I.r
```

CONVC.OUT

CONVC.OUT is the output file from CONVC.exe. It contains all of the conversions completed since the file was initially created upon the first run of CONVC.exe.

Each conversion includes the original constants and reduction/representation on the left and the constants along with new reduction/representation on the right. Each conversion is separated from other by a line of dashes. Multiple conversions are appended to the end of the file. A sample is presented below.

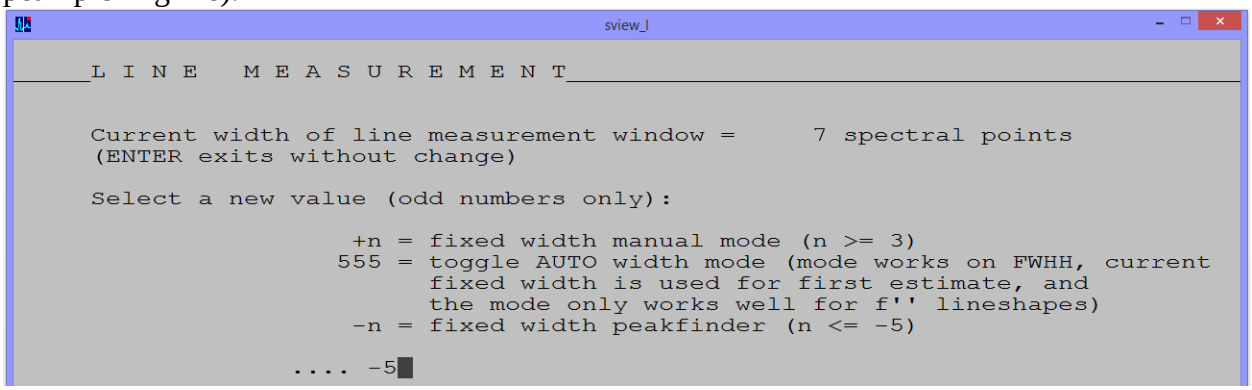
```
          Watson A,I.r          --->   Watson S,I.r


A    /MHz       10030.798913(257)       A    /MHz       10030.802039(257)
B    /MHz        9870.693311(260)       B    /MHz        9870.689396(260)
C    /MHz        4972.941713(164)       C    /MHz        4972.943127(164)
 DJ /kHz           2.066718(412)        DJ /kHz           1.441446(426)
 DJK/kHz          -0.82413(142)         DJK/kHz           2.92749(156)
 DK /kHz           2.332803(761)        DK /kHz          -0.793556(931)
 dJ /kHz           0.832104(200)        d1 /kHz          -0.832104(200)
 dK /kHz           1.332307(229)        d2 /kHz          -0.3126360(537)
HJ  / Hz          -0.008863(254)       HJ  / Hz          -0.008943(254)
HJK / Hz           0.014489(233)       HJK / Hz           0.014763(233)

 ------------------------------------------------------------------------------


          Watson A,I.r          --->   Watson S,I.r


A    /MHz       10252.028397(291)       A    /MHz       10252.031313(291)
B    /MHz        9776.991464(203)       B    /MHz        9776.987500(203)
C    /MHz        5002.457741(112)       C    /MHz        5002.459372(112)
 DJ /kHz           2.126158(331)        DJ /kHz           1.542930(366)
 DJK/kHz          -1.04465(159)         DJK/kHz           2.45471(184)
 DK /kHz           2.31739(129)         DK /kHz          -0.59874(150)
 dJ /kHz           0.861604(160)        d1 /kHz          -0.861604(160)
 dK /kHz           1.398570(372)        d2 /kHz          -0.2916140(776)
HJ  / Hz           0.001693(513)       HJ  / Hz           0.001489(513)
HJK / Hz          -0.011480(953)       HJK / Hz          -0.010465(953)
HKJ / Hz           0.016814(561)       HKJ / Hz           0.016490(561)

 ------------------------------------------------------------------------------
```

ECHO.ASC

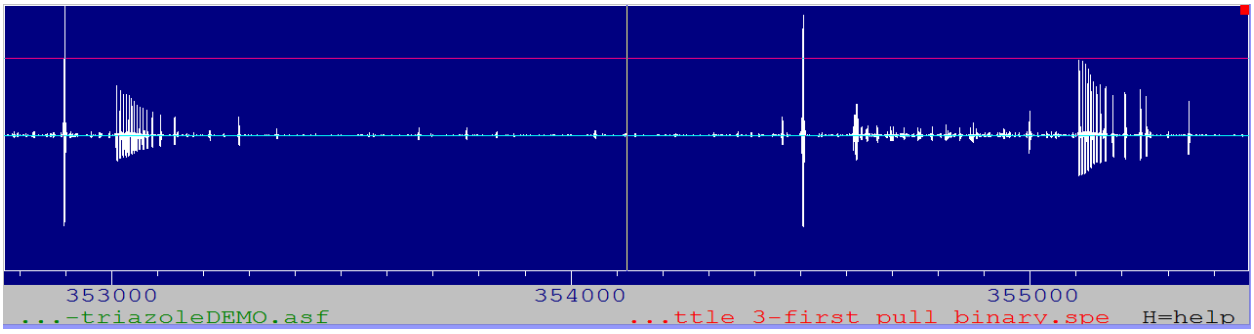
The ECHO.ASC file contains line frequencies to be removed from the experimental spectrum. This feature may be used to remove lines that are above a certain intensity set by the user or to remove lines belonging to molecules or states that have predictive .ASR files.

Creating an ECHO.ASC file to remove lines over a certain intensity

Open the desired .SPE in SVIEW. Scroll to a region including the lines of maximum intensity to be left in. Press "-" (dash/underscore key), "0" (zero), "-5", and ENTER (similar to a peak-picking file).



Set the baseline and the upper limit just above the most intense lines to be kept using the Z (down) and W (up). Press ENTER to set. In this example, the spectrum contains lines not from the target molecule that are much more intense than the molecule's most intense lines, around 355100 MHz.



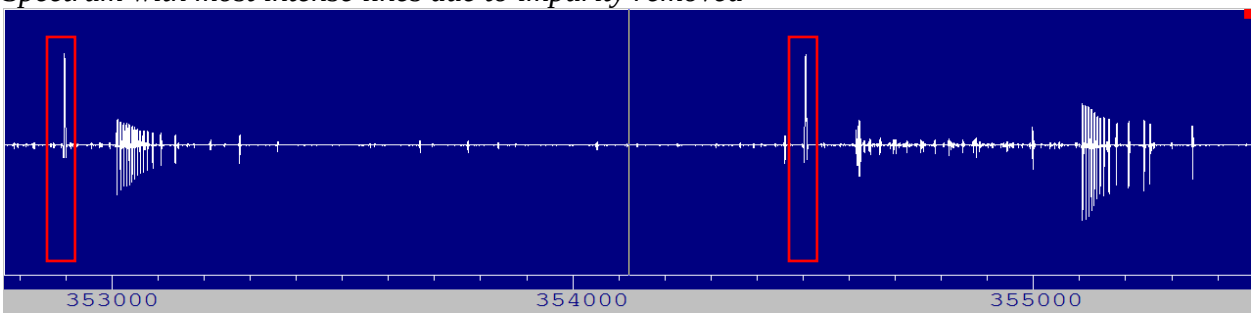
Select option "1" and ENTER for full spectrum peak-picking. Name the output file *ECHO.ASC.* The program will prompt the user whether or not to overwrite if there is already an existing ECHO.ASC file.

Creating an ECHO.ASC file to remove lines of given molecules and/or states

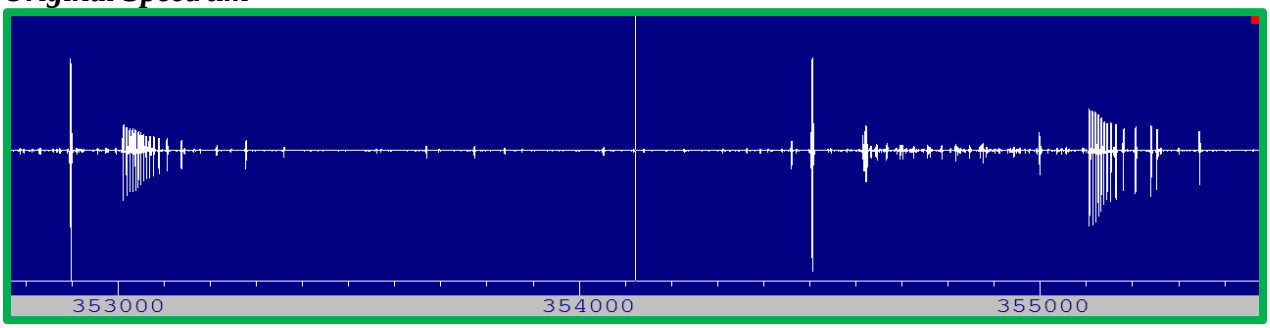
The .ASR file must be predictive for the molecule or state to be removed.
Open the .ASR file for the molecule or state to be removed in a text editor.
Copy-paste the list of frequencies (the column headers and information after the lines frequency may optionally be included; lines removal will work either way). Save the file as *ECHO.ASC*

Frequencies belonging to multiple molecules and/or states may be combined. The user may consider matching the intensity cutoff of the .ASR file (via the .IN file) to match that of the experimental spectrum.
Samples of the two types of removal are provided below with a comparison to the original spectrum.

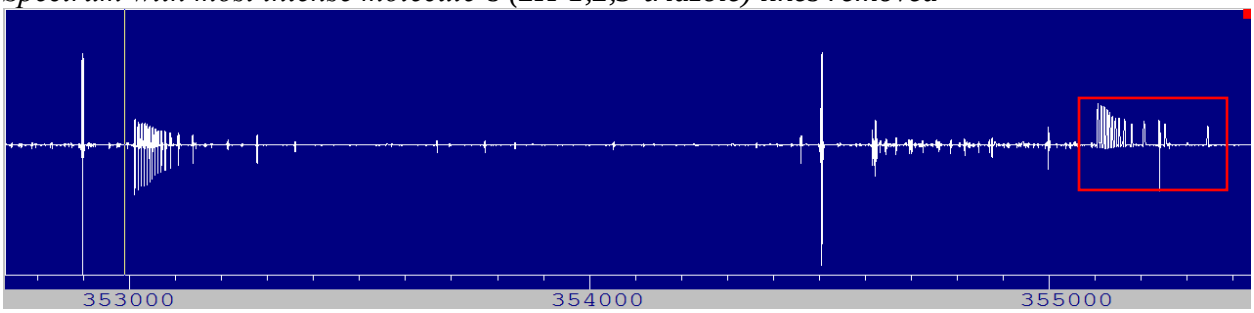
Spectrum with most intense lines due to impurity removed



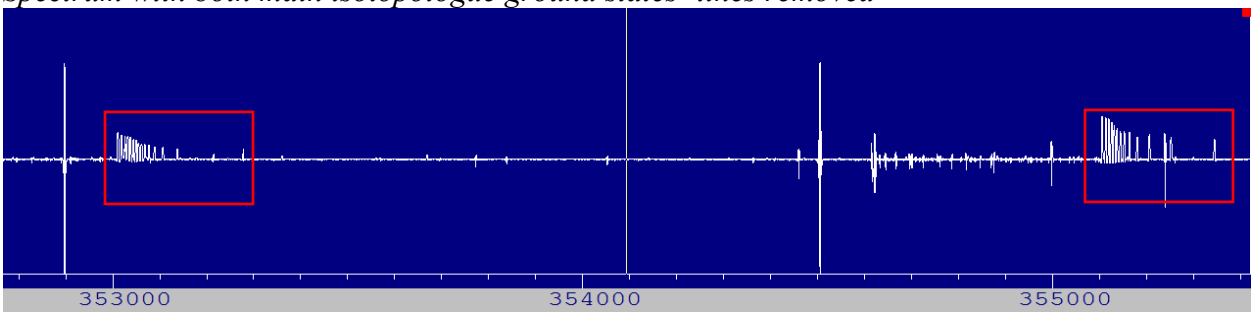
Original Spectrum



Spectrum with most intense molecule's (2H-1,2,3-triazole) lines removed



Spectrum with both main isotopologue ground states' lines removed



.FIT Files

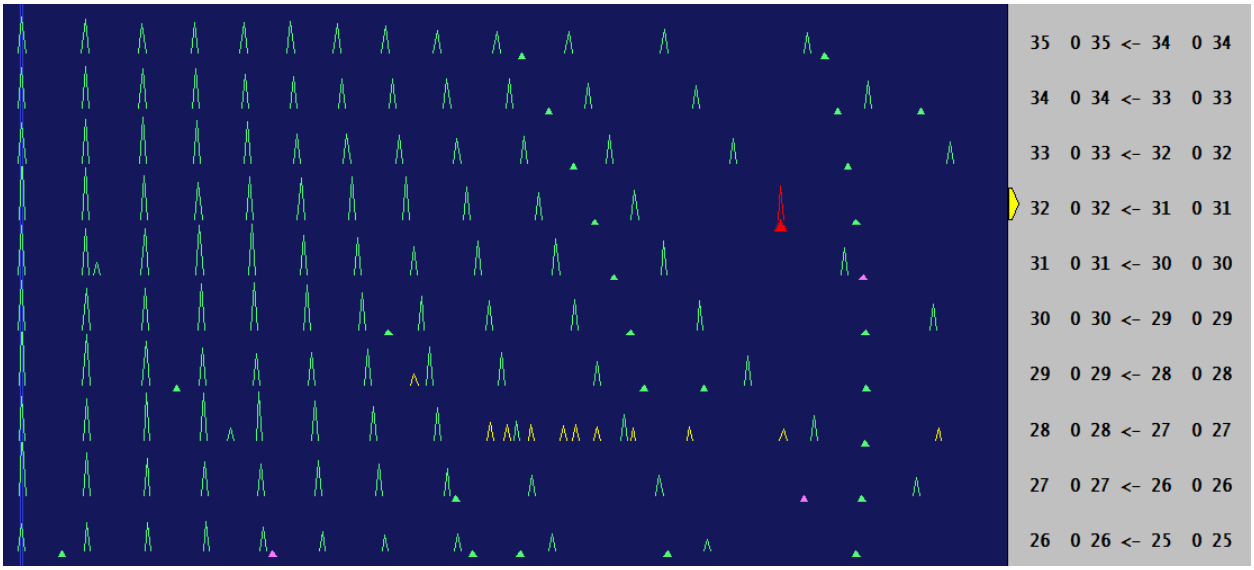
.FIT files are the least-squares fitting output file of SPFIT. It is highly recommended that the user immediately run PIFORM and transform the output file into a more reader-friendly output form, labeled with a .TXT or .RES extension.

.FRE Files

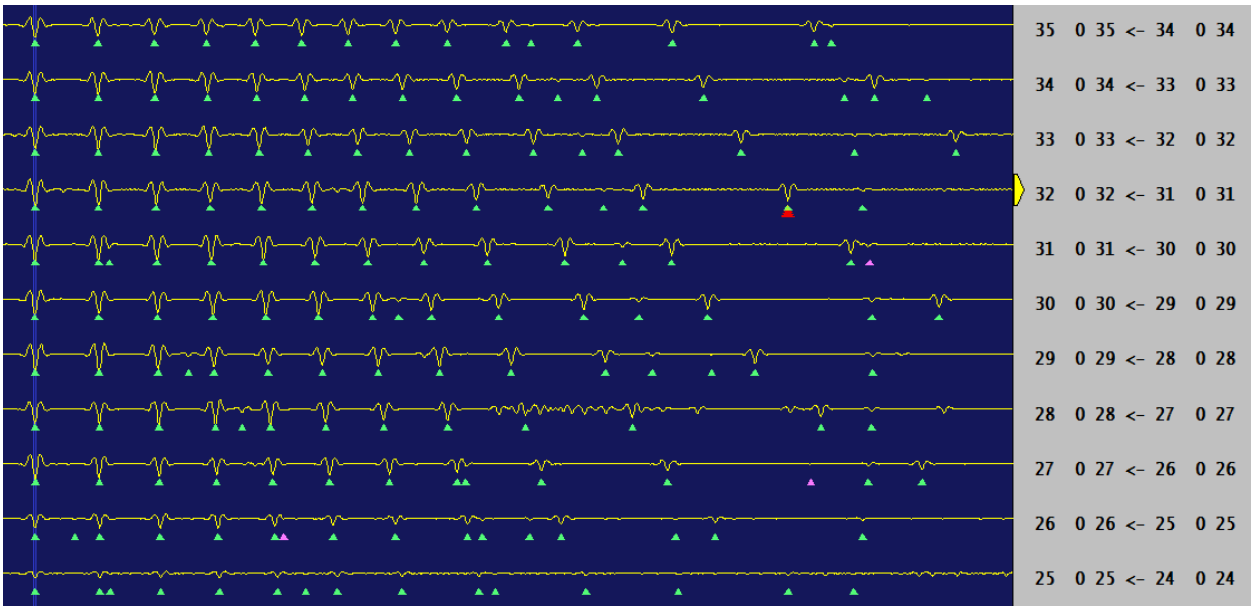
The .FRE file allows the user to properly view Loomis-Wood plots using markers (triangles, cones, or sticks), move the cursor properly among peaks, and take experimental measurements directly from the Loomis-Wood plot when properly linked using SVIEW_L.INP. A sample view of this type of file is presented below, although the user does not need to manipulate the text.

```
-------------------------------------------------------------------------------
 SVIEW PeakFinder RESULTS FOR SPECTRUM:
                   C:\Rot\123-triazoles_8mTorr_250-360_GHz_binary.spe

   cutoff:     4936243                 fitting width:    5
 baseline:     4621232
-------------------------------------------------------------------------------

    Frequency     Error   Intensity     FWHH

251279.4064659 +- 0.00609  365656    0.36806  |-----
251281.0766349 +- 0.00524  377850    0.33546  |-----
251284.0960407 +- 0.00591  374140    0.36656  |-----
251321.9128233 +- 0.00714  338408    0.33761  |----
251485.1510451 +- 0.00689  323943    0.35149  |----
253597.5244971 +- 0.00322  484962    0.36939  |------
253605.3912547 +- 0.00425  457974    0.35920  |------
253612.7916648 +- 0.00587  413157    0.38475  |-----
253619.9388339 +- 0.00367  444711    0.38850  |------
253627.0140137 +- 0.00380  380909    0.38003  |-----
253642.1993136 +- 0.00777  325647    0.38124  |----
254019.3192740 +- 0.00432  340387    0.35339  |----
255102.5510225 +- 0.00485  895014    0.32141  |-----------
255109.9676186 +- 0.00392  784836    0.34035  |----------
255117.0560949 +- 0.00393  824538    0.32193  |----------
255124.2261641 +- 0.00594  749121    0.33399  |---------
255132.0766268 +- 0.00338  670620    0.33683  |--------
```

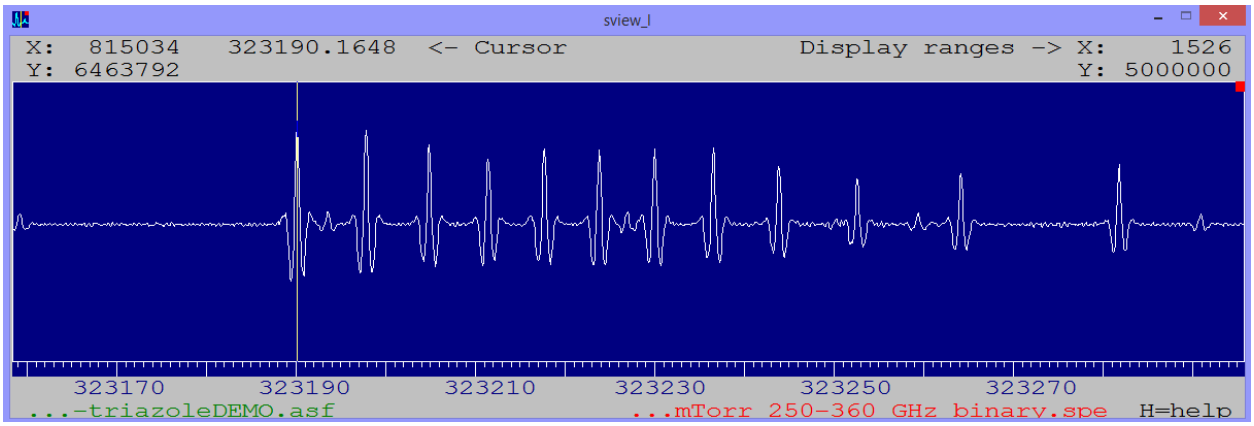
Loomis-Wood plot in default mode



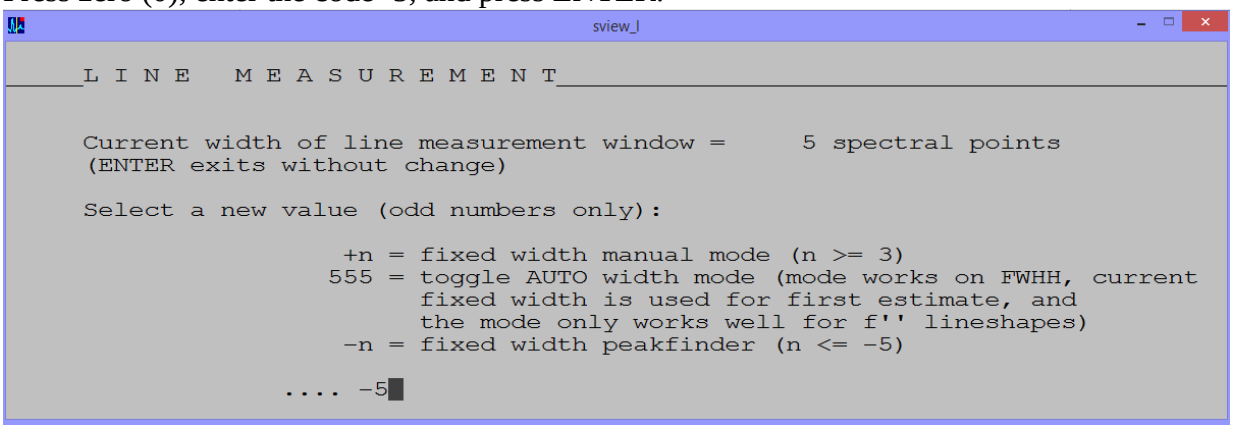
Loomis-Wood plot in spectral view mode



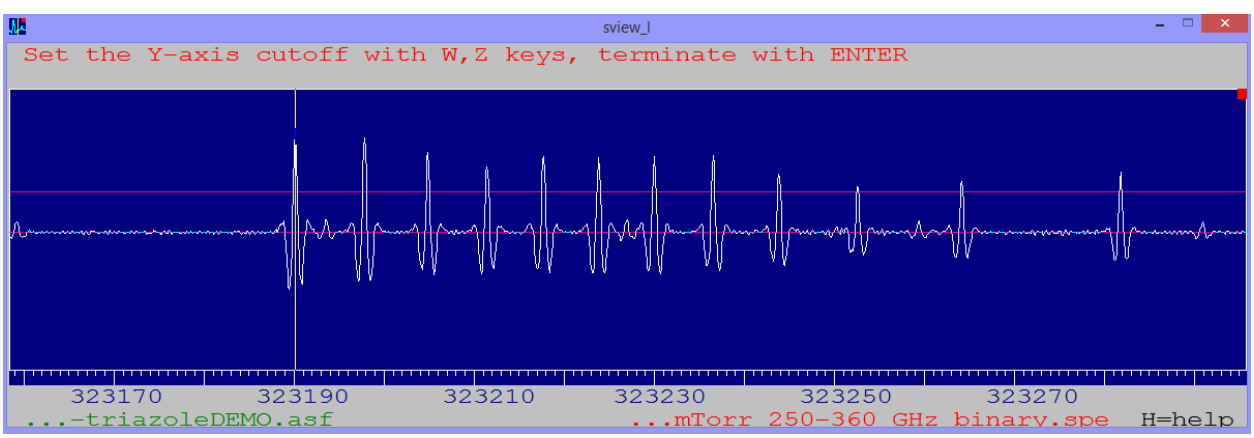
To generate a .FRE file, open the .SPE in SVIEW. Scroll to the desired region (typically and band). Press the dash/underscore (-) key to flip the spectrum, such that all peaks are positive in amplitude.



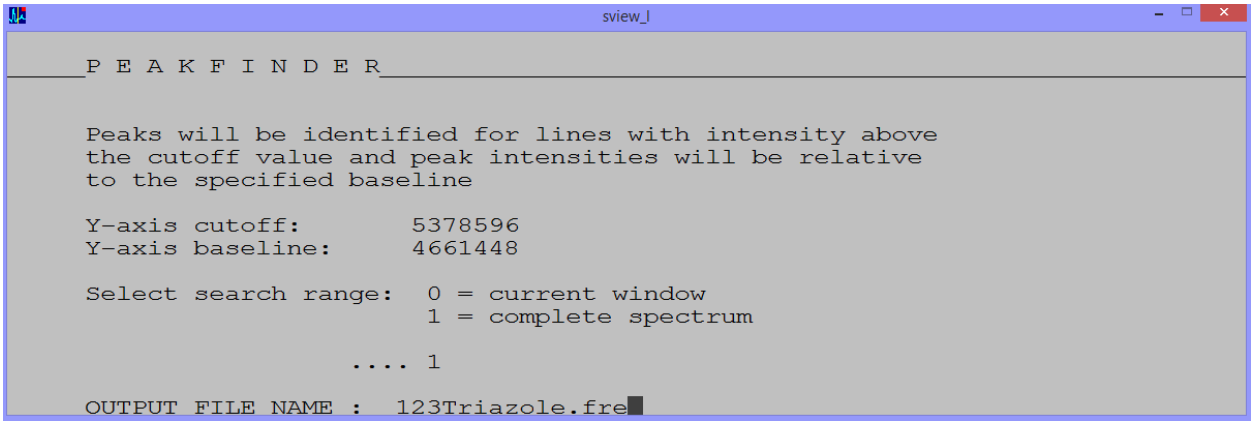
Press zero (0), enter the code -5, and press ENTER.



Use the Z (down) and W (up) keys to select the baseline level. Press ENTER.
Do the same for the Y-axis cutoff (peaks that reach above this line will be marked in the Loomis-Wood plot. Peaks that are below the line will not be recorded in the .FRE file and, accordingly, will not appear in symbol view of Loomis-Wood plots (as opposed to spectral view).



To generate the .FRE for the entire spectrum, select “1” and ENTER.
Type a file name ending in “.fre” and press ENTER. *The file name is limited to 21 characters, including “.fre”. (17 characters allotted to the name, excluding extension.)*



If a file with this name already exists in the directory, SVIEW will inquire whether or not to replace it.
Make sure to update the .FRE file in SVIEW_L.INP and refresh the ASCP and SVIEW windows using the ESC key.

Using spectra from which certain molecules and/or states have been removed using the ECHO.ASC file and line removal, the user may create .FRE files that exclude the ECHO-excluded molecules/states from the symbolic Loomis-Wood plots, as well.

.GLE Files

This type of file is automatically generated by ASCP and SVIEW. It contains a script to draw plots or spectral images, viewable in QGLE.exe and directly editable using a text editor. An edited .GLE file is available in the AABS Suite folder for sample changes to improve the appearance of the plot.

Common Functions

! – Comments out line

amove – sets the position of an object (*e.g.* text)

Graph function – makes a plot, drawing data from .DAT files

```
begin graph
  size 10 10    Plot size
  !"syn formyl azide"
    data "formyl azide.dat" d1=c1,c2
  !"CO"
    data "formyl azide.dat" d2=c3,c4
  !"HN3"
    data "formyl azide.dat" d3=c5,c6          Pull data from .DAT files
  !"HNCO"
    data "formyl azide.dat" d4=c7,c8
  !"HCN"
    data "formyl azide.dat" d5=c9,c10
  !"N2O"
    data "formyl azide.dat" d6=c11,c12

    d1 lstyle 1 marker cross msize 0.15 color black
    d2 lstyle 2 marker fcircle msize 0.15 color blue        Define appearance
    d3 lstyle 3 marker otimes msize 0.15 color green        of lines/marker for
    d4 lstyle 4 marker fsquare msize 0.15 color red         each data series
    d5 lstyle 5 marker star msize 0.15 color purple
    d6 lstyle 6 marker ftriangle msize 0.15 color brown

    title "Intensity of Transitions versus Time (sec)"      Label axes and title
    xtitle "Time  (sec)"
    ytitle "Relative Intensity"

    xaxis  min 0    max 9000    dticks 1000   Define appearance of axes
    yaxis  min 0 max 7 dticks 1

    key pos bl compact offset 0.2 -1.2 nobox
    d1 lstyle 1 marker cross msize 0.15 color black key "syn formyl azide"
    d2 lstyle 2 marker fcircle msize 0.15 color blue key "CO"
    d3 lstyle 3 marker otimes msize 0.15 color green key "HN_3"      Draw
                                                                     built-in
end graph                                                            legend
```

Pull data from .DAT files

data "[.DAT file]" [data series]#=[column 3 containing x-values],[column containing y-axis values]

Define appearance of lines/marker for each data series

this is not required if all of the data series appear in the built-in legend, but a separate key may be used for aesthetic reasons

```
d1 lstyle 1 marker cross msize 0.15 color black
```

[data series]# lstyle [lstyle#] marker [plot marker name] [plot marker size] color [color]

Draw build-in legend

```
key pos bl compact offset 0.2 -1.2 nobox
```

Positions – bl = bottom left, br = bottom right, tl = top left, tr = top right, bc = bottom center, tc = top center

compact – makes the legend more compact than default (command may be omitted)

offset – moves the legend relative the position defined (command may be omitted)

nobox – removes box from around legend (command may be omitted)

Key function – creates new legend, separate from that built into graph function

```
begin key
  nobox compact
  position bl offset 4.2 -1.2
  text "HNCO" lstyle 4 marker fsquare msize 0.15 color red
  text "HCN" lstyle 6 marker ftriangle msize 0.15 color brown
  text "N_2O" lstyle 6 marker ftriangle msize 0.15 color brown
end key
```

text "[Data series label]" lstyle [#] marker [marker name] msize [marker size] color [color]

Colors

Black	●	Lightgreen	●	Blue	●
Brown	●	Green	●	Navy	●
Red	●	Lime	●	Purple	●
Orange	●	Cyan	●	Violet	●
Gold	●	Turquoise	●	Pink	●
Yellow	●	Lightblue	●	Magenta	●

Line Types

lstyle	Appearance
1	
2	
3	
4	
5	
6	
7	
8	
9	
11 - 19	to (Decreasingly dense dot-dashes)
21 - 29	to (Decreasingly dense dashes)
Following this pattern, dashes get increasingly longer until lstyle 99	
101 - 109	to
111 - 119	to
121 - 129	to

Plot Markers

Marker Type	Appearance
circle, fcircle	○ , ●
square, fsquare	□ , ■
triangle, ftriangle	△ , ▲
diamond, fdiamond	◇ , ◆
heart	♡
cross	×
otimes	⊗
star	★
plus	+
club	♣
flower	❀
spade	♠

.IN Files

.IN Files are the input for ASROT to create a predicted stick spectrum (.ASR) file. The main aspects that users typically address are highlighted in non-black colors in the sample file below. Take care to change the name to the appropriate file. Also note that ASROT only takes I^r representations of the molecule in either the S- or A-reduced Hamiltonian.

```
1H-123-triazole.asr
-1,   type of calc. -1 = S
0,JMIN
100,JMAX
100,QMAX
100,K-1MAX
1,   sort?
1,   discard unsorted?
from previous
0,   are constants on file?
6,   order of Hamiltonian
10030.8021070390041131759,A
9870.6893623339419718832,B
4972.9431722476883805939,C
0.0014417038633415338,DJ
0.0029256518823671746,DJK
-0.0007916911865305491,DK
-0.0008321381426553891,d1
-0.0003125555219046655,d2
0.0,HJ
0.0,HJK
-0.0000000095043234784,HKJ
0.0000000154199860489,HK
0.0,h1
0.0,h2
0.0,h3
4.105,mu.A
1.591,mu.B
0.,mu.C
0.2,   cutoff
223.,T
100000.,FMIN
550000.,FMAX
```

Name of ASR output file

Type of calculation ≡ Reduction

Min and max values for which to predict transitions (300 max)

Up to which constants – sextic, octic, etc.

Constants used for prediction in the order listed above

Dipole moments μ_A, μ_B, and μ_C in Debye

Intensity cutoff & Temperature of gas sample

Minimum and maximum frequencies between which to predict lines

The user is advised to update the .IN and .ASR files frequently, especially when first beginning to fit a new molecule or vibrational satellite. Doing so expedites the data collection by minimizing scrolling to lines and helps avoid mis-assignment errors from poorly-predicted lines coinciding with peaks belonging to other transitions.

.INP Files – Batch input for multi-molecule/state stick spectra

This type of .INP file allows the user to view multiple fits in a single ASCP stick spectrum window when opened using ASCP. *SVIEW_L.INP will be discussed in a separate section below.*

Intensity Cutoff allows the user to set the minimum intensity of lines to appear in the stick spectrum. Adjusting this value can help the user weed transitions that are too weak to be seen experimentally out of the stick spectrum to avoid visual confusion/noise.

Parent .ASR file (typically the ground vibrational state and/or main isotopologue) is given first and other file intensities may be adjusted relative to that molecule.

Intensity rel. to parent sets the intensity of lines from the corresponding .ASR file relative to the parent, given in percent. It is recommended that the relative intensities are set to a value based upon the dipole moment and Boltzmann distribution (such that intensities reflect experimentally observed intensities).

$$Intensity \propto e^{-hc\nu/kT}$$

$$Intensity \propto \mu^2$$

$h = 6.62608 \times 10^{-34}$ J•s
$c = 2.99792458 \times 10^{10}$ cm/s
ν = vibrational frequency in cm^{-1}
$k = 1.38065 \times 10^{-23}$ J/K
T = temperature of the sample in K
μ = dipole moment

Line code allows the user to view each of the species in a different color in the stick spectrum viewer. A list of numbers and corresponding colors is provided on the following page.

More data should be 1 if other files follow and 0 if that is the last .ASR file to be included. If files follow a line with "more data" marked 0, they will not be read. Up to 50 files may be listed. When opening an INP file in ASCP, select "2" for "Batch input".

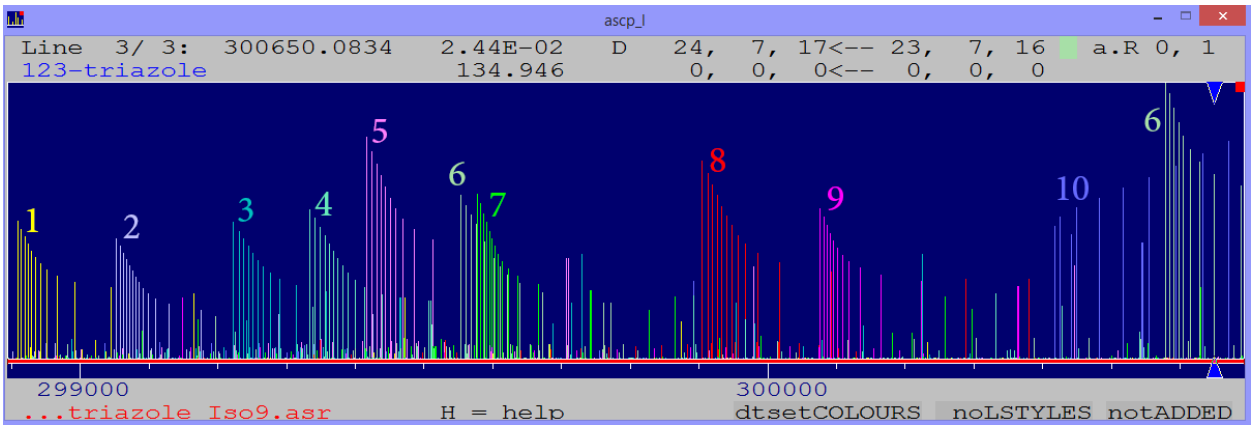
```
          filtering (1/0):   1
         frequency limits:   200000.0,360000.0
         intensity cutoff:   1.E-9
          echo file (1/0):   0
-------------------------------------------------
       parent data file :   1H-123-triazole.asr
more data (1), end (0)  :   1
-------------------------------------------------
name of data file       :   2H-123-triazole.asr
intensity rel. to parent:   85.0
line code (1-10)        :   2
more data (1), end (0)  :   1
-------------------------------------------------
name of data file       :   1D-123-triazole.asr
intensity rel. to parent:   1.3
line code (1-10)        :   3
more data (1), end (0)  :   0
-------------------------------------------------
```

The window size can be adjusted in the aabs.cfg file (described in *ASCP_L.exe and SVIEW_L.exe*)

The *P key* toggles lines colors.
The *C key* toggles the background color.
The *Y key* can be used to toggle on/off the stacking of degenerate line intensities.
The *R key* targets all lines belonging to the molecule presently selected by the cursor

Line codes and corresponding colors.

INP file viewed in ASCP with different numbers in line code for different species, marked with the corresponding line code numbers.

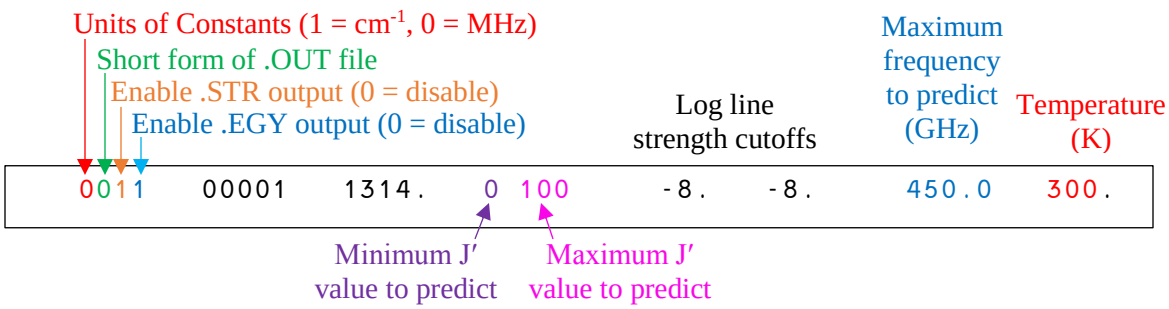


.INT Files

.INT files are input for SPCAT containing the dipole moments along the principle rotational axes. They may contain one or multiple sets of dipole moments, one for each state or molecule to be included in the .CAT file. Even if the dipole moments are the same (*e.g.* as for vibrational states of the same isotopologue), *a set of dipole moments must be included for each state or molecule*.

Title – The first line of the file is for the user's reference, not read by the program

Command Line This value tells the program the units of the constants (1 = cm^{-1}, 0 = MHz), a command to the short form of the .OUT file (0), a command to enable .STR output (1), and a value to enable .EGY output (1).



Dipole Moments – The first triplet of values in the line indicates the molecule or state (using the vibrational quantum number assignment: 00 for single-state, 11, 22, etc.) and the symmetry (corresponding to the rotational axis, 1 = A, 2 = B, 3 = C). This value is followed by the dipole moment in Debye. Information after the slash is for the user's use, not read by the program.

A single-state .INT

```
2H-123-triazole
   0011    00001    1314.    0 100      -8.    -8.      450.0    300.
 001     0.52       / a dipole moment v13
 002     0.00       / b dipole moment v13
 003     0.00       / c dipole moment v13
```

A multi-state .INT

```
Pyridazine v13 and v16 combined
   0011    00001    1314.    0 100      -8.    -8.      450.0    300.
 001     4.22       / a dipole moment v13
 002     0.00       / b dipole moment v13
 003     0.00       / c dipole moment v13
 111     4.22       / a dipole moment v16
 112     0.00       / b dipole moment v16
 113     0.00       / c dipole moment v16
```

.LIN Files

.LIN files contain experimental line measurements. They work similarly to .ASF files in that they may be linked to SVIEW and ASCP via the SVIEW_L.INP file to measure lines directly from the spectrum using the function keys. However, the different keys result in different "allowable" error values and relative weights for overlapping or degenerate lines (rather than fit blend, fit unique, or do not fit line codes). They are used as an input file for least-squares fitting in SPFIT. The name of the .LIN file must be the same (except extension) as that of the corresponding .PAR file.

Quantum Numbers – There are 12 quantum numbers, six rotational and six vibrational:

$$J' \;\; K'_{prolate} \;\; K'_{oblate} \;\; \nu' \;\; J'' \;\; K''_{prolate} \;\; K''_{oblate} \;\; \nu'' \;\; 0 \;\; 0 \;\; 0 \;\; 0$$

Three columns are devoted to each number. They should be aligned to the rightmost column allotted. The vibrational quantum numbers can be used for multi-state least-squares fitting (such as for coupled states). The states are assigned ν values consecutively upward beginning with the lowest state, ν'' = ν' = 0. See the two-state example below.

Frequency – Experimental frequency (MHz if positive value, wavenumbers if negative value). The list should be in order of frequency. See "Efficient Re-Ordering of Line List by Frequency" on page XX for efficient sorting by frequency.

Error – Allowable experimental error (MHz if positive value, wavenumbers in negative value). The standard is 0.05 MHz. A value of zero (0) will exclude the line from the least-squares fit.

Relative Weight – This line gives the relative weight of lines at the same frequency (overlapping or degenerate). The programs change normalize this value.

WARNING: There should be no space/blank lines between measured lines, otherwise lines beyond the break will not be included in least-squares fitting.

Transition [J'	K'prolate	K'oblate	ν'	J"	K"prolate	K"oblate	ν"	0	0	0	Measured frequencies, aligned at decimal MHz > 0 or cm^{-1} < 0	Max Error (Standard 0.05 MHz)	Relative Weight
51	10	42	0	51	8	43	0	0	0	0	258368.23670	0.05000	1.00E+00
51	9	42	0	51	9	43	0	0	0	0	258368.23670	0.05000	1.00E+00
49	7	42	0	49	7	43	0	0	0	0	258432.61400	0.05000	1.00E+00
49	8	42	0	49	6	43	0	0	0	0	258432.61400	0.05000	1.00E+00
49	1	49	1	48	1	48	1	0	0	0	302095.52720	0.05000	1.00E+00
49	0	49	1	48	0	48	1	0	0	0	302095.52720	0.05000	1.00E+00
30	20	11	1	29	20	10	1	0	0	0	302162.44000	0.05000	1.00E+00
30	19	11	1	29	19	10	1	0	0	0	302163.24000	0.05000	1.00E+00

Degenerate or overlapping lines are indicated by having the lines listed adjacent to one another with the same frequency and error values. Lines that are formatted incorrectly are automatically excluded from least-squares fitting.

Single-molecule/state .LIN files are created by linking and measuring lines into a .LIN file from ASCP and SVIEW or from .ASF files using ASFIT (see section *ASFIT.exe* on page XX). Multi-state .LIN files are created manually by copying and pasting the lines of upper state .LIN files after lower state lines into a single .LIN file. The different states must have different vibrational numbers, as described above and the file should be named appropriately (the name must match that of all other corresponding files).

.PAR Files

.PAR files are one of the input files for SPFIT and contain the starting constants to be used in least-squares fitting or (after running SPFIT) the output constants, as SPFIT overwrites previous files. The name of the .PAR file must be the same (except extension) as that of the corresponding .LIN file. A single-state .PAR file may automatically be generated using ASFIT (see page XX). A multi-state .PAR file can be generated by copy-pasting constants from individual state .PAR files with different assigned quantum numbers, then adding the relative energies of the vibrational states and the interaction terms manually.

First Row

Title – The first line of the file is the title.

Second Row

Number of Constants – The first value in the second row is the number of constants in the .PAR file. It is critical. If the number is larger than the actual number of constants in the file, SPFIT will automatically correct it. If, however, the number is less than the actual number of constants, those constants beyond that number will be excluded.

Number of lines – The (maximum) number of lines in the file is the second value in the second line of the file. This value, like the number of constants, can be greater than the actual number of lines.

Maximum number of Iterations – The maximum number of least-squares fitting iterations to run; third value in the second line. Using a negative value here results in line assignment diagnostics in the output.

Number of Constants to Exclude – This number of parameters is excluded from the end of the list. It is typically zero (0); used for fitting special lines. See Kisiel's Crib-Sheet (http://www.ifpan.edu.pl/~kisiel/asym/pickett/crib.htm) or Kisiel's annotated SPFIT documentation (http://info.ifpan.edu.pl/~kisiel/asym/pickett/spinv15_annotated.pdf)

Maximum observed-to-allowed error – the sixth value in the first row is the maximum (observed frequency – predicted frequency) / allowed error value listed, typically set to 100.

Third Row, the "Option Line" (potentially multiple option lines/rows)

If spaces are left blank, they are delineated by commas. An example of a file with multiple "third" rows is provided below.

Reduction – the first character in this row is the reduction, specificied using "a" or "s" for A reduction or S reduction, respectively.

Spin degeneracy – the second value in the third or analogous rows; a negative value indicates symmetric rotor, positive indicates asymmetric rotor. The sign is read by the program only in the third row and not in following analogs, if included. For additional information, see the annotated SPFIT documentation (http://info.ifpan.edu.pl/~kisiel/asym/pickett/spinv15_annotated.pdf).

Number of Vibrational States – third value in the third row: a positive value indicates a prolate top (z = a, y = b, x = c), while a negative value indicates an oblate top (z = c, y = b, x = a). The value indicates the number of vibrational states; this value should equal the *largest vibrational quantum number + 1*, otherwise lines with the vibrational quantum number larger than this value will be deemed "bad line" and excluded from fitting. If multiple analogs of the third row are used, this value indicates the vibrational state to which the following parameters of the third row analog pertain.

K minimum – fourth value in the third row or its analogs. Minimum value of *K*, often set to zero.

K maximum – fifth value in the third row or its analogs. Maximum value of *K*, often left blank.

Statistical Weight for Even State – the eighth value in this row gives the vibronic statistical weight for the even state belonging to the vibrational state associated with the third value in the corresponding row

Statistical Weight for Odd State – the ninth value in this row gives the vibronic statistical weight for the odd state belonging to the vibrational state associated with the third value in the corresponding row

Vibronic Symmetry – the tenth value in the option line gives information regarding the vibronic symmetry (see annotated documentation:). This line defines whether there is one or whether there are more option lines – a value of 0 or 1 indicates a single option line; a value of -1 indicates multiple option lines.

Example top of .PAR file with multiple Option Lines

```
Benzonitrile 2nu22 (11), nu22+nu33 (22) and 2nu33 (33)
 114 90968    1    0    0.0000E+000    3.0000E+001    1.0000E+000 1.0000000000
a   1  6 0,     ,   , 1 ,  5.    ,  3.    , -1  ,      , 0  ,,,,
a   1  1 0,     ,   , 1 ,  5.    ,  3.    , -1  ,      , 0  ,,,,
a   1  3 0,     ,   , 1 ,  5.    ,  3.    , -1  ,      , 0  ,,,,
a   1  2 0,     ,   , 1 ,  5.    ,  3.    ,     ,      , 0  ,,,,
```

Fourth Row and Beyond (or rows after option lines)

List of constants – The list of constants is column-aligned.

- The first value in a row is the code associated with a given constant. The last two digits in the code are the assigned quantum number(s) of the state(s) to which the constant pertains (see below). Note that many of the codes are the same for both S- and A-reductions, however, they differ for d_2 (S reduction) vs. δ_k (A reduction). A table of commonly-used constants and their codes in the A and S reductions is provided below.
- The code is followed by the predicted constant value in MHz.
- The value listed after the predicted constant value is essentially an allowed error – the amount the least-squares fit may vary the constant. To fix a constant, this value should be set to 1×10^{-37} (see example below).
- The final item in a row is the constant's symbol for the user's convenience.

Abridged list of constants for multi-state fit

	Predicted Constant Value (MHz)	Allowed Variation (MHz)	Constant Symbol
10000	6.240767057135967E+003	1.00000000E+002	/A
10011	6.236173120667400E+003	1.00000000E+002	/A
0	0.000000000000000E+000	1.00000000E-037	/relEnu13
11	2.078716411781962E+005	1.00000000E+004	/relEnu16
600001	7.004901590276861E+000	1.00000000E+001	/Gc
600101	0.000000000000000E+000	1.00000000E-037	/GcJ
601001	-7.544851654650010E-004	1.00000000E+000	/GcK

These values are fixed

Constant code ending in assigned vibrational quantum number(s) to which constant pertains (*i.e.* two of the same digit for rotation constants, distortion constants, and relative energies; different digits for perturbation/interaction terms)

For a multi-state fit, the user must enter the relative energies of the vibrational states, as well as the desired perturbation terms (included above).

Warnings:

No row in the .PAR files may exceed 82 columns/characters (even if the characters are spaces), otherwise SPFIT will crash.

Everything in the .PAR file past a completely blank line will be deleted when SPFIT is run and the .PAR file is rewritten.

The .PAR files is automatically overwritten by SPFIT when the program is run; to avoid overwriting the .PAR file, the user may have the .PAR file open in a text-editing program, select "No" if asked to update the file, and immediately save the .PAR file (overwriting the newly SPFIT-generated .PAR file).

A fit is converged when three criteria are met:
(1) SPFIT runs only one iteration (when more are allowed),
(2) the Marquardt parameter is zero, and
(3) the Trust Expansion is one.
Single quantum vibrational states' distortion constants are not expected to be more than an order of magnitude different from that of the ground state and the distortion constants of multiple quanta vibrational states should not be more than an order of magnitude off from those predicted. Constants should be examined after each fitting to ensure that they appear reasonable.

The .PAR files may be used as a sort of lab notebook to track changes in the fitting process over time. To do so, below the line of dashes, the user may use an exclamation point (!) to comment out lines, followed by a copy of a previous .PAR file. Information typically included in the comments are the maximum error allowed in the fitting, the number of lines excluded, and the error of the fit, along with any meaningful comments of changes made to the data. A simple way to do this is to state the max error and copy-paste the fitting "overview" from the .RES file, as in the example below.

An example of the .PAR file used to document previous .PAR files.

```
[Current .PAR file above]
!
!-----------------------------------------------------------------
! Removed GaK, added both As -- convergence!
! Questionable HJK nu22 and A's
! Sigma = 10 45 lines rejected
!
! MICROWAVE       lines fitted       lines    lines      RMS      RMS ERROR
!             total   dv=0 dv.ne.0   UNFITTD  e>900
! v"= 1       3877   3853     24       23    174       0.107869     1.69146
! v"= 2       3968   3944     24       22    361       0.112199     1.49245
! ---------------------------------------------------------------------------
! total:      7845   7797     48       45    535       0.110081     1.59391
!
Benzoisonitrile nu22 (11) and nu33 (22) A-red          Wed Nov 28 12:09:37 2018
  75 93246   10    0    0.0000E+000    1.0000E+001    1.0000E+000 1.0000000000
a   1  3 0,    ,  , 1 ,  5.   ,  3.   , -1  ,     , 0  ,,,,
a   1  1 0,    ,  , 1 ,  3.   ,  5.   , -1  ,     , 0  ,,,,
a   1  2 0,    ,  , 1 ,  3.   ,  5.   ,     ,     , 0  ,,,,
        10000  5.659375490515413E+003 1.00000000E-037 /A
        20000  1.639772007713838E+003 1.00000000E-037 /B
        30000  1.271153735987853E+003 1.00000000E-037 /C [Other constants/rest
of .PAR file, followed by exclamation point, line of dashes, and additional
previous .PAR files]
```

Sample multi-state .PAR file

```
Pyridazine     nu13 and nu 16
  35 3161    10     0     0.0000E+000     1.0000E+002     1.0000E+000 1.0000000000
s 1   2 0,   ,    ,   ,      ,      ,   ,   ,,,,,
         10000  6.240767057135967E+003 1.00000000E+002 /A
         20000  5.956370153582256E+003 1.00000000E+001 /B
         30000  3.052136054197582E+003 1.00000000E+000 /C
           200 -4.946916138600777E-004 1.00000000E+000 /-DJ
          1100 -1.482108809033521E-003 1.00000000E+000 /-DJK
          2000  5.443641484260985E-004 1.00000000E+000 /-DK
         40100 -3.222076089625772E-004 1.00000000E+000 /d1
         50000 -1.396358213628550E-004 1.00000000E+000 /d2
           300  7.324036677385456E-008 1.00000000E+000 /HJ
          1200 -4.468309591574684E-007 1.00000000E+000 /HJK
          2100  5.224486208855604E-007 1.00000000E+000 /HKJ
          3000 -2.297213339970856E-007 1.00000000E+000 /HK
         40200  2.532316073200000E-008 1.00000000E-037 /h1
         50100 -9.113021921124342E-009 1.00000000E+000 /h2
         60000  2.011623864944738E-009 1.00000000E+000 /h3
         10011  6.236173120667400E+003 1.00000000E+002 /A
         20011  5.961555725524318E+003 1.00000000E+001 /B
         30011  3.052186169232155E+003 1.00000000E+000 /C
           211 -3.902891303248850E-004 1.00000000E+000 /-DJ
          1111 -2.136758698016926E-003 1.00000000E+000 /-DJK
          2011  1.096128896708658E-003 1.00000000E+000 /-DK
         40111 -2.918806899156741E-004 1.00000000E+000 /d1
         50011 -1.616145058526003E-004 1.00000000E+000 /d2
           311 -7.332200540298196E-008 1.00000000E+000 /HJ
          1211  3.565933490806007E-007 1.00000000E+000 /HJK
          2111  4.388748027131581E-008 1.00000000E+000 /HKJ
          3011 -3.624394736141817E-007 1.00000000E+000 /HK
         40211 -2.360728509450052E-008 1.00000000E-037 /h1
         50111  1.651795610906949E-008 1.00000000E+000 /h2
         60011  3.284750351401198E-009 1.00000000E+000 /h3
             0  0.000000000000000E+000 1.00000000E-037 /relEnu13
            11  2.078716411781962E+005 1.00000000E+004 /relEnu16
        600001  7.004901590276861E+000 1.00000000E+001 /Gc
        600101  0.000000000000000E+000 1.00000000E-037 /GcJ
        601001 -7.544851654650010E-004 1.00000000E+000 /GcK
'
```

Coriolis Coupling Term	Code	
G_a	*2000XY*	
G_b	*4000XY*	*First Order*
G_c	*6000XY*	
F_{bc}	*2100XY*	
F_{ca}	*4100XY*	*Second Order*
F_{ab}	*6100XY*	
Fermi Coupling Term		
F	*XY*	

X and Y are the vibrational quantum number assigned to the lower and upper (respectively) interacting states.

A-reduction Constant	A-Reduction Code	S-Reduction Constant	S-Reduction Code
A_0	*100***	A_0	*100***
B_0	200**	B_0	200**
C_0	300**	C_0	300**
Δ_J	*2***	D_J	*2***
Δ_{JK}	*11***	D_{JK}	*11***
Δ_K	*20***	D_K	*20***
δ_J	*401***	d_1	*401***
δ_K	*410***	d_2	*500***
Φ_J	*3***	H_J	*3***
Φ_{JK}	*12***	H_{JK}	*12***
Φ_{KJ}	*21***	H_{KJ}	*21***
Φ_K	*30***	H_K	*30***
ϕ_J	*402***	h_1	*402***
ϕ_{JK}	*411***	h_2	*501***
ϕ_K	*420***	h_3	*600***
L_J	*4***	L_J	*4***
L_{JJK}	*13***	L_{JJK}	*13***
L_{JK}	*22***	L_{JK}	*22***
L_{KKJ}	*31***	L_{KKJ}	*31***
L_K	*40***	L	*40***
l_J	*403***	l_1	*403***
l_{JK}	*412***	l_2	*502***
l_{KJ}	*421***	l_3	*601***
l_K	*430***	l_4	*700***

** The asterisks should be replaced with the vibrational quantum number assigned to the state (*e.g.* 00 for a state with 0 as the assigned vibrational quantum number, 11 for a state with 1 as the assigned vibrational quantum number)

Other constants and codes available in the Crib-Sheet at *http://info.ifpan.edu.pl/~kisiel/asym/pickett/crib.htm#asymc*

A note from Prof. Kisliel: "Fitting DE, F, G_c and F_{ab} simultaneously is a daunting proposition. DE and F are normally very strongly correlated, and so is G_c with F_{ab}. In many cases the

information accounted for by G_c and its centrifugal distortion expansion is completely exchangeable with that carried by F_{ab} and its cd expansion. So my advice would be to use just one of these (I suggest G_c) and its J and $K_{centrifugal}$ dependence. In any case try shifting from using the highest order centrifugal distortion constants to centrifugal distortion type terms on the interaction constants.

As for F and DE, again it depends on the available transition set and the shape of the normal modes. In the acrylonitrile saga with really abundant data it turned out that the most prominent A' - A' interaction between g.s. and ν_{11} did not allow fitting F (there called W_F) and G_c, but only F_{ab} and terms in the W+- sequence. On the other hand with glycolic acid (attached) we could fit WF and Gc reasonably well. There is no sure-fire general advice here, in the end the fit is found to be experimentally driven."

PLAN.OUT

PLAN.OUT is the output file from PLAN.exe. It contains the output of all calculations run using PLAN since the creation of the file, each calculation delineated from others by a line of dashes.

Inertial defect errors are the primary piece of information typically sought in these files.

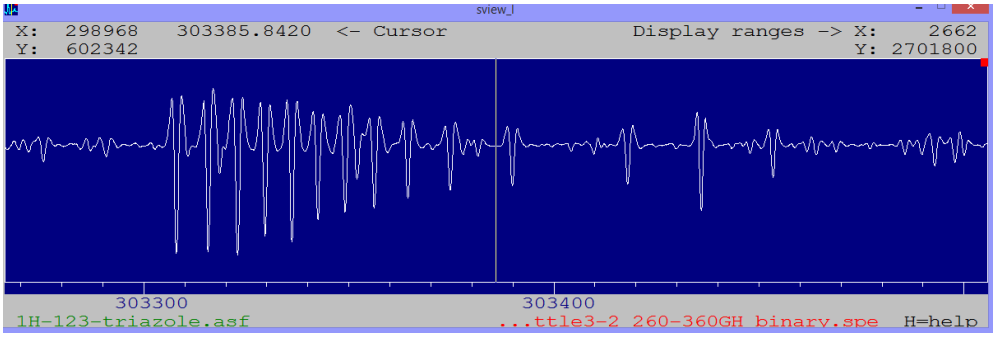
```
INSERT GOOD EXAMPLE OF FILE HERE!!!
```

.RES Files

.RES files are results files from a variety of programs. Commonly, results .FIT files from SPFIT that are converted through PIFORM are renamed to be .RES files to differentiate the file types. For the description of such output files, see *.TXT or .RES (PIFORM output) Files*, on page XX. Output files from ASFIT are also sometimes deemed .RES files. For the description of these files, see *.TXT or .RES (ASFIT output) Files*, on page XX. Other programs will automatically output .RES files, such as PISLIN.exe, when outputting a lines file with re-ordered split blends, will write the new lines file to a file called pislin.res.

.SPE Files

.SPE files are experimental spectra viewable by SVIEW_L.exe. They are generated directly by the LabView spectrometer program in ASCII, in binary by saving from an ASCII file in SVIEW_L.exe (SVIEW), or may be created manually using a right column of equally-spaced frequencies and left column of intensities, delimited by a space (in ASCII).



While any of the file types are viewable and will work for fitting with ASCP a(n) ASCII file is required for use with Autofit

Binary file (1) will open noticeably faster than an ASCII file for broadband scans

(2) is required to view and use Loomis-Wood plots in spectral view mode

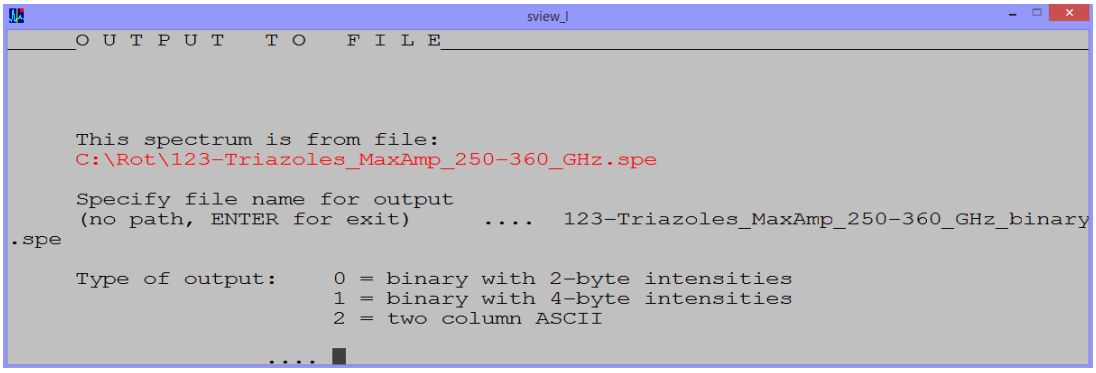
(3) is required for the user to append multiple spectra using the *& key*, described in

To generate a binary .SPE file, open the spectrum in SVIEW and press "M". Enter a file name; it is highly recommended that the file name the same as the original spectrum with a binary designation:

[original_spectrum_name]_binary.spe or *[original_spectrum_name]_bin.spe*.

The file name is limited to a total of 50 characters, including extension. (46 characters to the name itself.)

Press ENTER and the program will prompt the user for a file type. Typically, a binary file with 4-byte intensities (best for spectra with high signal : noise ratio) is used – select "1" and press ENTER.



Unless you wish to select a new frequency range, press "0" and ENTER.

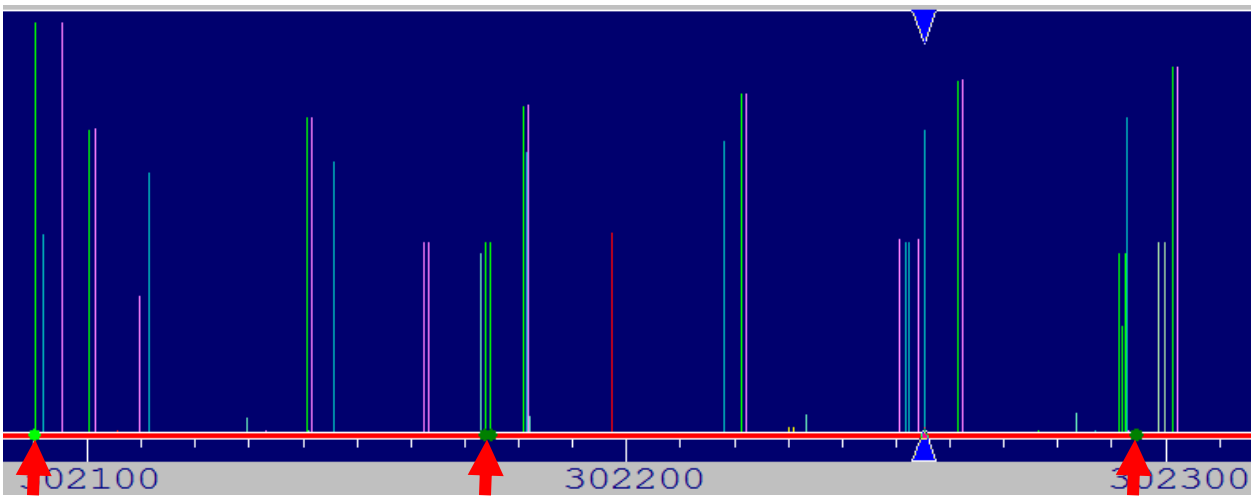
SVIEW_L.INP File – Linking spectra and ASF

This SVIEW_L.INP file links the spectra viewed in ASCP and SVIEW to a .ASF file and .FRE file, marking the measured lines in the stick spectrum and allowing the user to add new measurements to the ASF file using function keys. It should be updated prior to each change in the fit being examined/fit/refined.

```
!-------------------------------------------------------------------------!
!                                                                         !
!   This is a control file for linked operation of ASCP and SVIEW         !
!                                                                         !
!-------------------------------------------------------------------------
!
!   The fitting data file can be:
!          1/ either a .LIN file for PICKETT's SPFIT program
!          2/ or an .ASF file for ASFIT
!
!   Frequency of BEEP signal is in Hz, duration is in milliseconds.
!
!   Peakfinder file is to be generated by the SVIEW_L peakfinder
!
!   Please modify only the entries starting from the column indicated
!   below
!_________________________________________
!                                         |
            Name of fitting data file:  pyridazine_2nu13.asf
Nominal error of measured frequencies:  0.05
     Frequency,duration of F1,F2 beep:  3000,80
     Peakfiner output for Loomis-Wood:  pyridazine.fre
Gle version:  <=4.0.7 (=0), >4.1 (=1):  1
!-------------------------------------------------------------------------
```

*Sounds for the program can be disabled by changing the "`Frequency,duration of F1,F2 beep:`" to 0,0

Proper linking of files via SVIEW_L.INP results in green dots appearing in the stick spectrum at the frequencies that have been measured in the linked .ASF file. **Light green** dots indicate measurements are included in the fit (*i.e.* Have a 1 or 2 “fit?” code), while **dark green** dots indicate measurements that are excluded from the fit (*i.e.* Have a 0 “fit?” code).

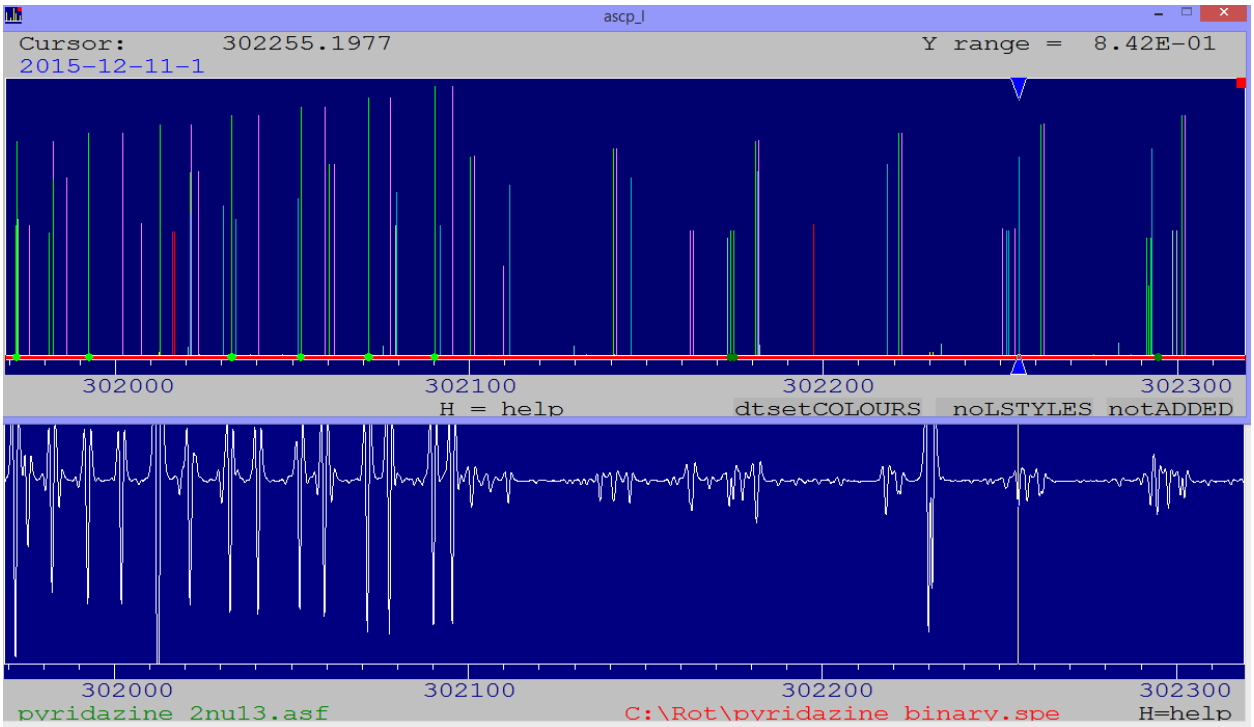


Moreover, proper linking allows the user to automatically add measurements with their correct quantum numbers to the .ASF file by using the function keys. By setting the cursor in the stick spectrum on the desired line and adjusting the cursor in SVIEW to the approximate minimum of the desired peak,

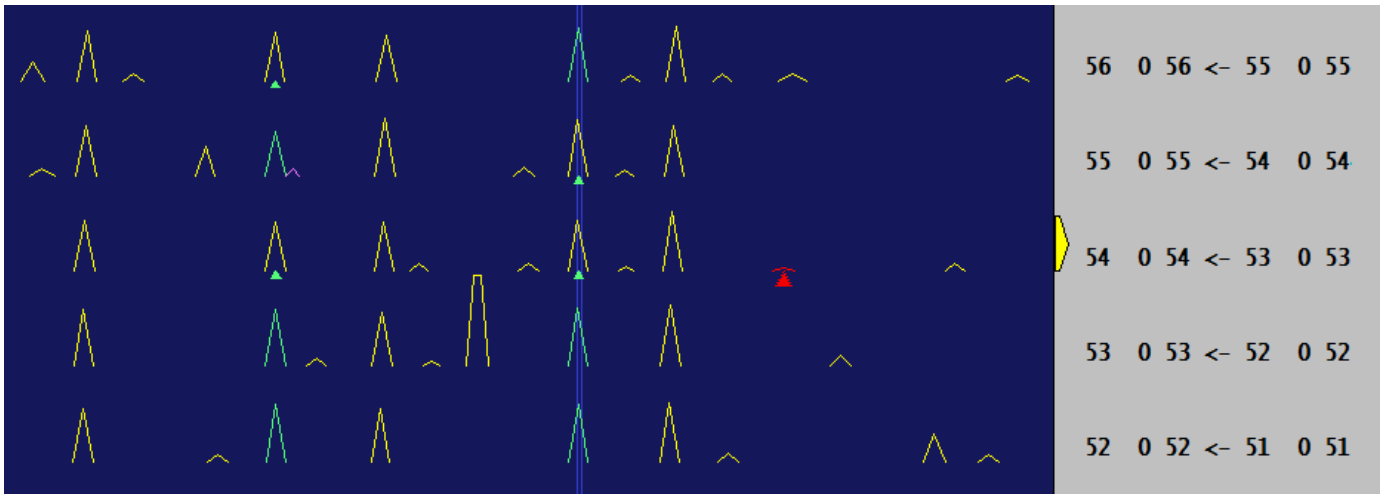
F1 enters measurement to be fit (1)

F2 enters measurement to be fit as a blended or degenerate line (2)

F3 enters the measurement, but excludes it from the fit (0)



Properly linked .ASF and .FRE files result in cleanly depicted triangles in the Loomis-Wood plot, as well as the ability to move properly from peak to peak using the arrows and measuring lines directly from the Loomis-Wood plot view. Green symbols indicate lines measured and included in the least-squares fit (*i.e.* marked with 1 or 2 in the .ASF file) and purple symbols indicate lines measured, but excluded from the least-squares fit (*i.e.* marked with a 0).



.TXT or .RES (ASFIT output) Files

The output file from ASFIT is a plain text file. It includes the title of the ASF file, followed by the fit constants, error, weighted error, a list of the measured lines with their individual errors, the number of distinct lines in the fit, the number of degrees of freedom, a correlation matrix of the constants, a list of transitions that were most influential in the fit, and a list of worst-fitting lines. A sample output is provided on the following page, although the list of lines has been abridged to 5 transitions to fit the rest of the file on a single page (for viewing purposes). A macro (ASFIT_Line_Analysis) has been written to count the types of lines, notify the user of improperly-entered lines, and extract constants.

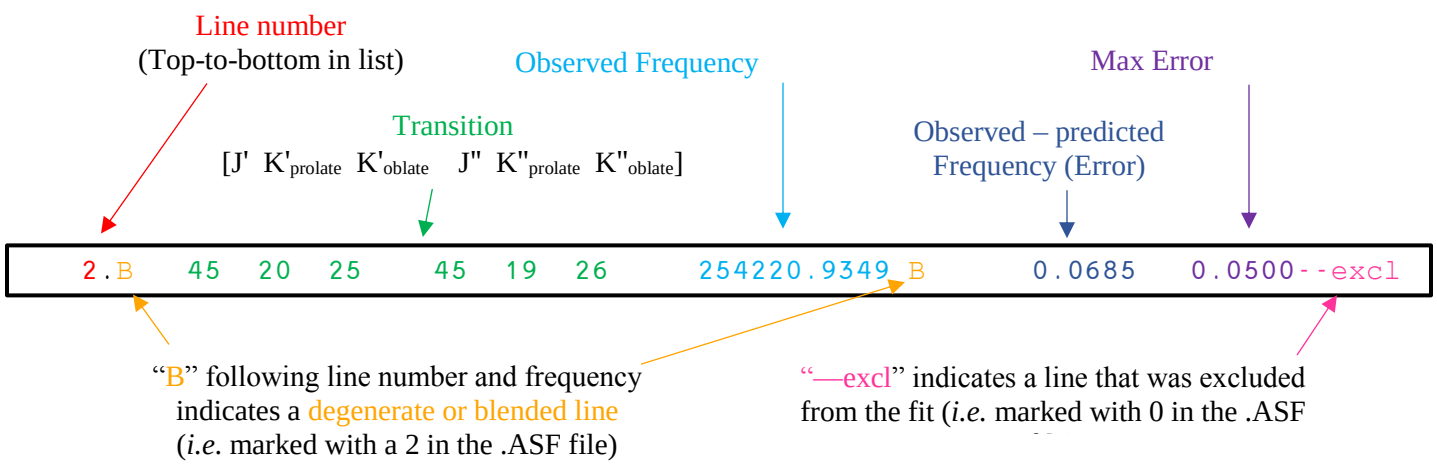
Title – the title from the ASFIT file

Reduction/Representation – the output states the reduction (A or S) and representation (I.r or III.r) that were used for the fit.

Constants – The fit constants and their errors are provided at the top of the file in MHz, as well as in varying units near the bottom of the file. If a value was input, but held constant, it will have no corresponding error value.

Deviation of Fit and Weighted Deviation of Fit – These are the error values, σ and weighted σ, respectively, of the fit.

Transitions –This is a list of all of the lines in the .ASF file.



Correlation Matrix – This tells the user the extent to which the least-squares fit constants are dependent on one another. Values will be between zero and one.

0 = No correlation and a perfectly determined value.

1 = A completely undetermined variable dependent on the other constant values.

Distribution of Correlation Coefficients – Depending upon the output selected when creating the .TXT/.RES file, it may or may not contain a distribution of correlation coefficients. This visual allows the user to quickly gauge how well constants are determined overall, rather than looking at each individual value in the correlation matrix. The x-axis, listed at the top, is the number of correlation coefficients that falls in a certain range. The y-axis is the correlation coefficient range in increments of 0.05, increasing downward. The leftmost value is the number of correlation coefficients in the corresponding range (*i.e.* the x-axis value and the number of corresponding asterisks). The more asterisks at the top of the distribution (at lower correlation coefficient values), the more constants are better determined.

```
Distribution of values of |C.ij|, i.ne.j:

                 10        20        30        40        50        60
          ....,....|....,....|....,....|....,....|....,....|....,....|....,..
12 <0.05 ************
 3 <0.10 ***
 1 <0.15 *
 1 <0.20 *
 4 <0.25 ****
 4 <0.30 ****
 3 <0.35 ***
 0 <0.40
 1 <0.45 *
 2 <0.50 **
 3 <0.55 ***
 1 <0.60 *
 0 <0.65
 1 <0.70 *
 1 <0.75 *
 3 <0.80 ***
 1 <0.85 *
 0 <0.90
 2 <0.95 **
 2 <1.00 **
```

Most Influential Transitions – This list tells the user the lines that had the greatest impact on the least-squares fit. Lines of which there are very few types are often listed here, as are lines that are mis-measured, lines that are mis-assigned, lines that are affected by perturbation, lines with blending, etc.

Worst Fitting Lines – This list gives ~7 lines included in the fit with the highest error values in descending error magnitude, referred to by line number. The value given is a ratio of observed-to-max.desirable error, calculated by the equation below. The user should be aware that this value depends on the "error" value listed in the .ASF file (standard 0.05 MHz). A complete fit should have all low error ratio values in the worst fitting lines list (< 2.0 to 1.6 depending on the number of lines and molecule).

$$Value\ displayed = \frac{Observed\ frequency - Predicted\ frequency}{\text{Error value listed in .ASF file}}$$

```
                                                    ASFIT version 14.IV.2008
_____________________________________________________________________________

 2H-123-triazole Found in the Benzonitrile spectrum
_____________________________________________________________________________

 Fit of Watson's Hamiltonian --> reduction-S, representation-I.r  :


 A    =     10252.0313514390327327419   +-    0.0002925294284363126
 B    =      9776.9874612635521771153   +-    0.0002043523831158803
 C    =      5002.4593770795427190023   +-    0.0001124397612906131
  DJ  =         0.0015429277798590590   +-    0.0000004229942472579
  DJK =         0.0024542303816698016   +-    0.0000023190419342024
  DK  =        -0.0005978684141513159   +-    0.0000020139065656241
  d1  =        -0.0008615666515251647   +-    0.0000001606747288254
  d2  =        -0.0002915805226630309   +-    0.0000000779885670494
 HJK  =         0.0000000010069026067   +-    0.0000000005151890701
 HKJ  =        -0.0000000109997553355   +-    0.0000000009549263606
 HK   =         0.0000000178981747742   +-    0.0000000005619707997

          Deviation of fit =       0.021462
 Weighted deviation of fit =       0.429249

-----------------------------------------------------------------------------------------------------------
                 TRANSITION              Observed      Obs-Calc    Error  Note        Blend components
-----------------------------------------------------------------------------------------------------------

   1.    14  11   4    13  10   3     251488.5275        0.0168   0.0500
   2.B   45  20  25    45  19  26     254220.9349 B      0.0685   0.0500           2.4E-05   254220.8664
   3.B   45  21  25    45  20  26     254220.9349 B      0.0685   0.0500           2.4E-05   254220.8664
   4.B   43  18  25    43  17  26     254394.8790 B     -0.0179   0.0500           3.2E-05   254394.8969
   5.B   43  19  25    43  18  26     254394.8790 B     -0.0179   0.0500           3.2E-05   254394.8969
-----------------------------------------------------------------------------------------------------------

 Sigma   =       0.021462                              483 Distinct lines in fit
 Sigma.w =       0.429249                              472 Degrees of freedom

 A    /MHz       10252.03135(29)         LJK /mHz           0.0
 B    /MHz        9776.98746(20)         LKKJ/mHz           0.0
 C    /MHz        5002.45937(11)         LK  /mHz           0.0
  DJ  /kHz           1.54292(42)        l1   /mHz           0.0
  DJK /kHz           2.4542(23)         l2   /mHz           0.0
  DK  /kHz          -0.5978(20)         l3   /mHz           0.0
  d1  /kHz          -0.86156(16)        l4   /mHz           0.0
  d2  /kHz          -0.291580(77)       PJ   /microHz       0.0
 HJ   / Hz           0.                 PJJK /microHz       0.0
 HJK  / Hz           0.00100(51)        PJK  /microHz       0.0
 HKJ  / Hz          -0.01099(95)        PKJ  /microHz       0.0
 HK   / Hz           0.01789(56)        PKKJ /microHz       0.0
 h1   / Hz           0.0                PK   /microHz       0.0
 h2   / Hz           0.0                p1   /microHz       0.0
 h3   / Hz           0.0                p2   /microHz       0.0
  LJ  /mHz           0.0                p3   /microHz       0.0
  LJJK/mHz           0.0                p4   /microHz       0.0
                                        p5   /microHz       0.0

 CORRELATION COEFFICIENTS, C.ij:

          A        B        C         DJ       DJK      DK       d1       d2

 A        1.0000
 B       -0.4840   1.0000
 C        0.0290   0.2645   1.0000
  DJ     -0.7047   0.8695   0.2334   1.0000
  DJK     0.6763  -0.6595  -0.1292  -0.9050   1.0000
  DK     -0.3060   0.3938   0.1115   0.6441  -0.8782   1.0000
  d1      0.8006  -0.8461   0.0004  -0.9142   0.7446  -0.3631   1.0000
  d2     -0.2543   0.4860   0.1420   0.7011  -0.8521   0.9493  -0.4098   1.0000
 HJK      0.2529   0.2761   0.1607   0.1712   0.0694  -0.0747  -0.0886   0.2207
 HKJ     -0.0424  -0.3442  -0.2194  -0.2933   0.0375   0.0613   0.2258  -0.2292
 HK      -0.1595   0.4337   0.2203   0.4129  -0.1442  -0.0455  -0.3856   0.2127

          HJK      HKJ      HK

 HJK      1.0000
 HKJ     -0.9575   1.0000
 HK       0.8530  -0.9624   1.0000

 Mean value of |C.ij|, i.ne.j =   0.4056
 Mean value of  C.ij,  i.ne.j =  -0.0182

_____________________________________________________________________________

 There are no influential transitions with hjj > 0.500

                                                mean of all values = 0.0210

 WORST FITTING LINES (line number=(o-c)/error):
   460=  1.9  295= -1.8  610=  1.7  612= -1.6  695= -1.4    2=  1.4  451=  1.4
_____________________________________________________________________________
```

.TXT or .RES (PIFORM output) Files

The PIFORM output file is written so as to be fairly similar and easy-to-read as the ASFIT ouput file. Beware that the number of iterations in the run of SPFIT that created the .FIT file input into PIFORM is the number of fits/iterations that will be printed in the PIFORM output file. The first line is the title, but here the line list precedes the constants.

Multi-state or Vibrational Quantum Number Assigned Line List (file without assigned vibrational quantum umber simply lacks the vibrational quantum number)

Line Number - Slash (/) indicates that (observed – predicted) > 3*Error

Transition [J' K'$_{prolate}$ K'$_{oblate}$ v'← J" K"$_{prolate}$ K"$_{oblate}$ v"]

Measured Frequency

Observed - Predicted

Error Set

Weighted Average (Obs – Pred)

Weight of blended line

```
2085/ 29 18 11  1   28 18 10  1          289908.9500   0.0340  0.050   0.2312 0.50
2086: 29 19 11  1   28 19 10  1          289908.9500   0.4284  0.050   0.2312 0.50
2087: 27 25  3  1   26 25  2  1          290042.0235   3.3919900.050
2088: 26 20  6  1   25 20  5  1          290059.7219 271.836 UNFITTD  289787.8850
```

"UNFITTD" → line rejected from least-squares fitting due to high error

Predicted Frequency if rejected line

Below the line list, the number of lines rejected from the least-squares fit is given, followed by a table of "PARAMETERS IN FIT (values truncated)". The user is advised, however, to examine the error values and parameters below that, beginning with the line "for standard parameter errors previous errors are multiplied by:", as these have been corrected.

Least-squares fitting summary – For each vibrational state, as well as the total, the summary gives the total number of lines included in the least-squares fit (fitted), the number of lines rejected from the fit (UNFITTD), and the number of lines excluded from the fit (e>900). This table also lists the root mean squared (RMS) error for each of the states and for the overall least-squares fit. The observed *J*, $K_{prolate}$, and frequency ranges are provided for each state.

The example provided is for a multi-state least-squares fit, but that for a single state looks similar.

```
MICROWAVE      lines fitted      lines     lines     RMS      RMS ERROR   J range   Ka range    freq. range
         total    dv=0 dv.ne.0   UNFITTD   e>900
v"= 0      820     820      0       87      213    1.290144    0.79879     0  78     0  31      0   359643
v"= 1      822     822      0       58      195   14.675244    0.67667    16  79     0  31      0   359078
-----------------------------------------------------------------------------------------------------------
total:    1642    1642      0      145      408   10.423233    0.74018
```

List of Constants with corresponding error values – the list of constants with their corresponding errors is provided beneath the summary table. The list gives the code associated with the constant, the symbol in Roman letters, the units in which it is listed, and the fitted constant with error in parentheses. It should be noted that the only apparent way to tell in which reduction the least-squares fit was run is to examine the lowercase symbols. If the letter is followed by a letter, the least-squares fit is in the A reduction; if it is followed by a number, the least-squares fit is in the S

reduction. (*e.g.* d1 → distortion constant in the S reduction; dj → distortion constant in the A reduction). In a multi-state fit, the last two digits on the constant's code are the vibrational constant assigned to the vibrational state, while constants that apply to multiple states will include two states' assigned vibrational quantum numbers as the last two digits in the code. Constants in brackets are fixed.

```
10000          A  /MHz      6240.678150(1278)
20000          B  /MHz      5956.438990(1164)
30000          C  /MHz      3052.133440(2588)
  200         DJ  /kHz         0.521310(1663)
 1100        DJK  /kHz         1.542200(5509)
 2000         DK  /kHz        -0.814180(1694)
40100         d1  /kHz        -0.3469900(8004)
50000         d2  /kHz        -0.1498590(1164)
  300         HJ  /Hz          0.034830(1101)
 1200        HJK  /Hz         -0.079400(3846)
 2100        HKJ  /Hz          0.012800(5405)
 3000         HK  /Hz         -0.112200(4262)
40200         h1  /Hz          0.0209700(5509)
50100         h2  /Hz        [ 0.004494556264]
60000         h3  /Hz          0.00094700(3742)
```

Constants for vibrational state with assigned quantum number 0.

```
10011          A  /MHz      6236.204900(1736)
20011          B  /MHz      5961.518960(1445)
30011          C  /MHz      3052.178070(2578)
  211         DJ  /kHz         0.392240(1590)
 1111        DJK  /kHz         2.00010(1580)
 2011         DK  /kHz        -0.87850(1985)
40111         d1  /kHz        -0.2862000(4262)
50011         d2  /kHz        -0.1537100(7485)
  311         HJ  /Hz         -0.066140(1112)
 1211        HJK  /Hz          0.310100(9980)
 2111        HKJ  /Hz         -0.031000(8628)
 3011         HK  /Hz         -0.19430(1278)
40211         h1  /Hz        [-0.023607285]
50111         h2  /Hz          0.0112800(6445)
60011         h3  /Hz          0.0018190(1091)
```

Constants for vibrational state with assigned quantum number 1.

```
    0   relEnu13  /MHz       [ 0.]

   11   relEnu16  /MHz    157901.90(1465)

600001         Gc  /MHz       -21.400(9668)
600101        GcJ  /MHz       [ 0.]
601001        GcK  /MHz         0.008670(3056)
```

Perturbation constants between vibrational states with assigned quantum numbers 0 and 1.

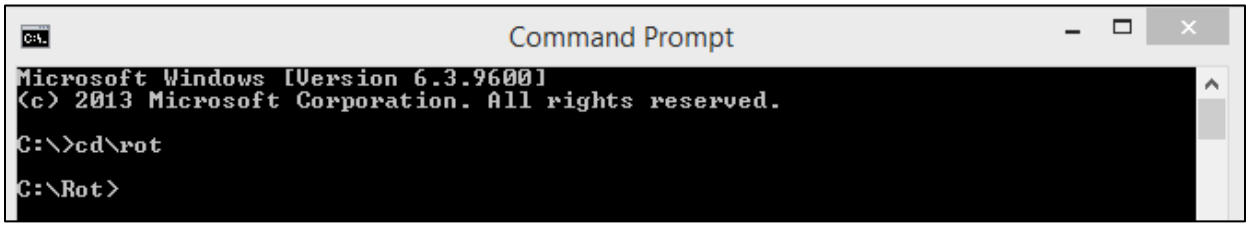
Constant Correlation Tables – Just as in the ASFIT output, the correlation values between constants are provided. The closer the correlation value is to zero (0), the less the fitted values of the constants depend on one another, indicating that they are better determined. The closer the correlation value is to one (1), the higher the constants' interdependence and poorer their determination.

List of worst-fitting lines – The final portion of the PIFORM output gives a list of worst-fitting lines along with their observed-to-accepted error values: (observed frequency – predicted frequency) / allowed error listed in .LIN file. In addition to a list of such lines, the last few lines of the file provide the line number, transition, measured frequency, observed error, and allowed error of the ten worst-fitting lines.

Command Prompt Accessible Programs

Open the command prompt and enter your ROT directory by using the command
"*cd C:\rot*"

Note throughout use that if the user has already entered a line of text in this Command Prompt instance, the user may press the up arrow to automatically display previously entered text and thus reduce typing.



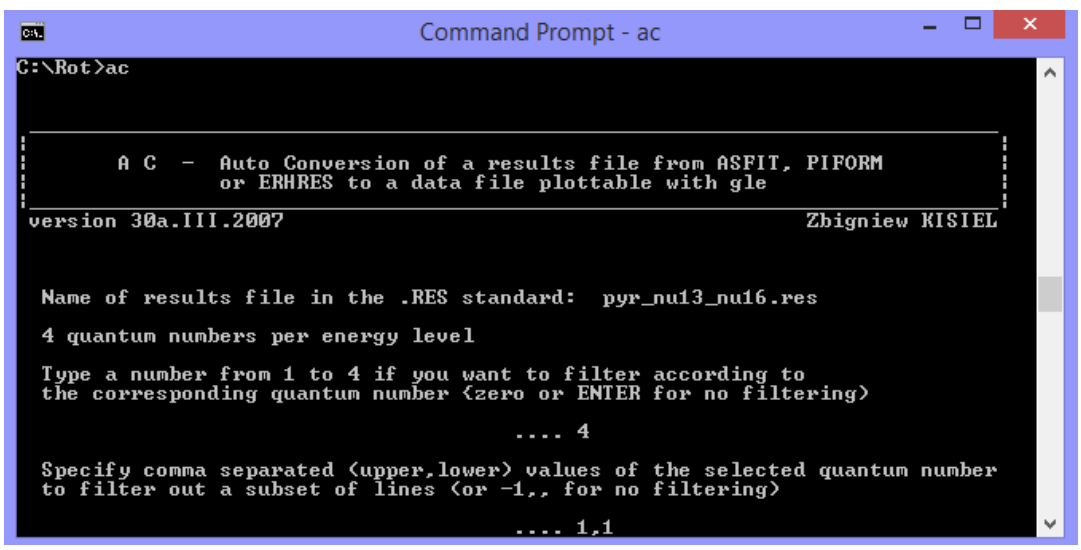
Note: All command prompt accessible programs may be aborted by pressing CTRL+C.

AC.exe
(http://www.ifpan.edu.pl/~kisiel/aabs/aabs.htm#aabs)

AC generates a plot of J″ vs. $K_{prolate}$ (K_{-1}). Type "AC" and press ENTER. The program will prompt the user for a file name. *The program only takes ASFIT output files or PIFORM-edited output files from SPFIT.* Enter the file name, including file extension and press ENTER.

If the file includes a vibrational quantum number, the program will inquire whether to sort by the lines by quantum number. To decline, press ENTER. To sort, select the desired quantum number (1 = J, 2 = $K_{prolate}$, 3 =K_{oblate} 4 = vibrational quantum number) and press ENTER. If you have chosen to sort by quantum number, enter the highest and lowest values to include, separated be a comma (*i.e.* [high],[low]) and press ENTER.

In the example below, a multi-state PIFORM output file is used with the quantum numbers 0 and 1 corresponding to the lower and upper states, respectively. The lines were sorted by 4 (vibrational quantum number) ranging from 1 to 1, thus selecting only the lines of the upper vibrational state.

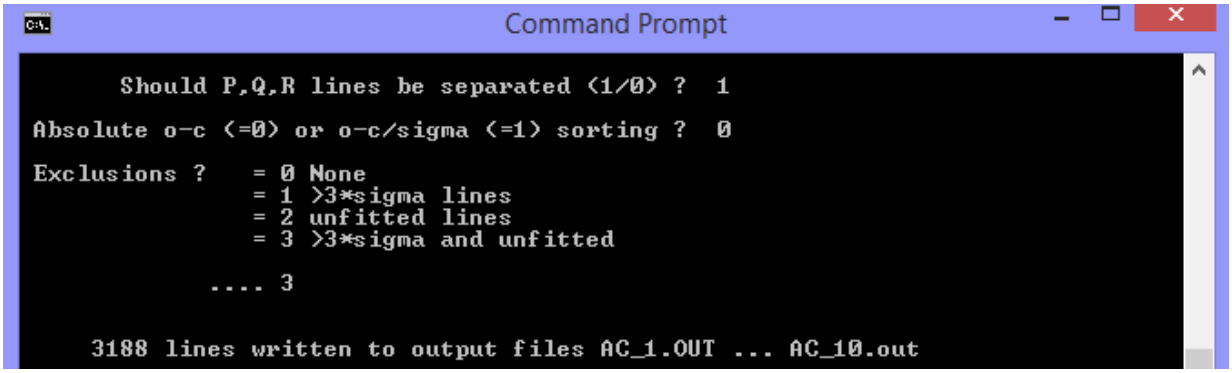


The program may prompt the user to enter the error for unfitted lines from a PIFORM output file. Enter a large value (*e.g.* 3) and press ENTER to assign an error value to lines that were excluded from the fit.

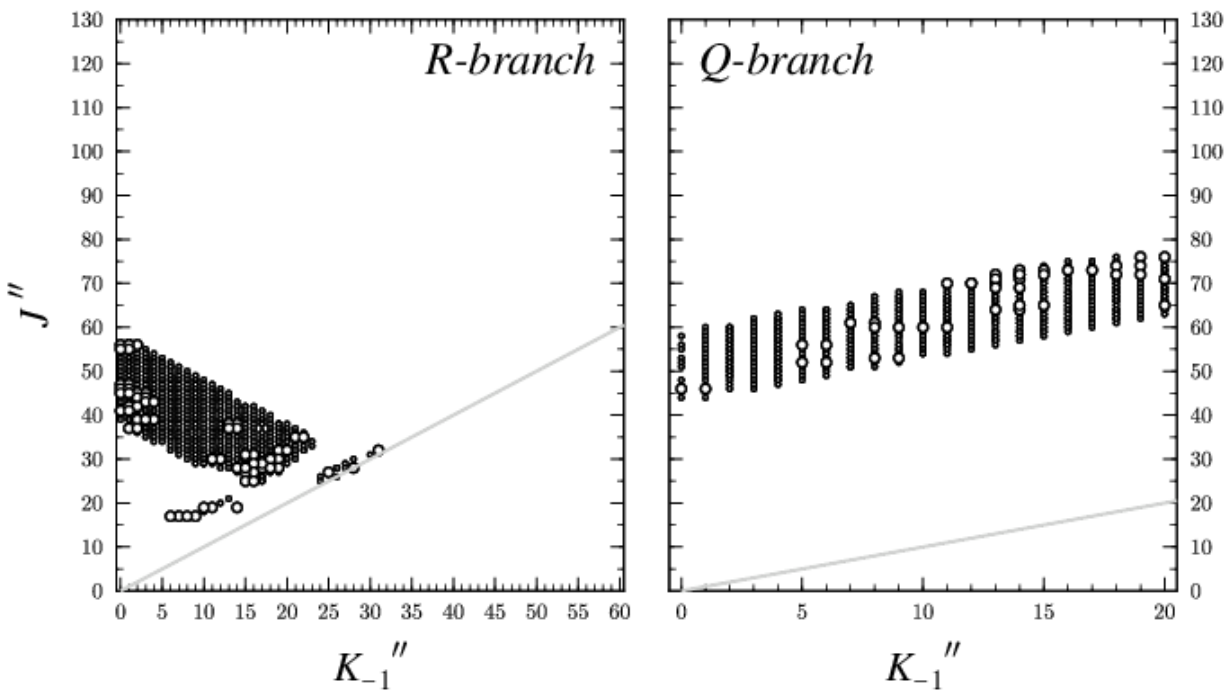
The program then inquires…

- Whether to separate P-type, R-type, and Q-type lines (1 = Yes, 0 = No)
- Whether to sort by absolute error (0) or the absolute-to-overall error ratio (1)
- Which lines to exlude (0 = None, 1 = Lines with error more than three times the overall error, 2 = unfitted lines, or 3 = both unfitted and high-error lines)

After each prompt, make a selection and press ENTER.



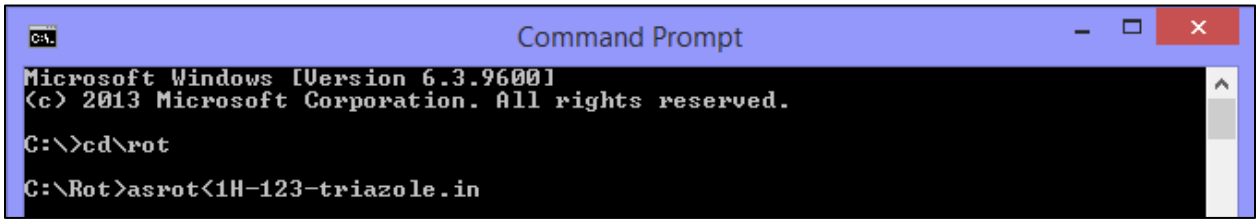
The program's output is viewed by opening PQR.gle.



ASROT.exe

(http://www.ifpan.edu.pl/~kisiel/asym/asym.htm#asrot)

To generate a stick spectrum, type "ASROT<" followed by the IN file name (including extension). It is critical to utilize the *less than* symbol, pointing the .IN file as input for the program and not the other way around. Hit ENTER. A new .ASR file will be generated in the same directory, named as specified in the first line of the .IN file.



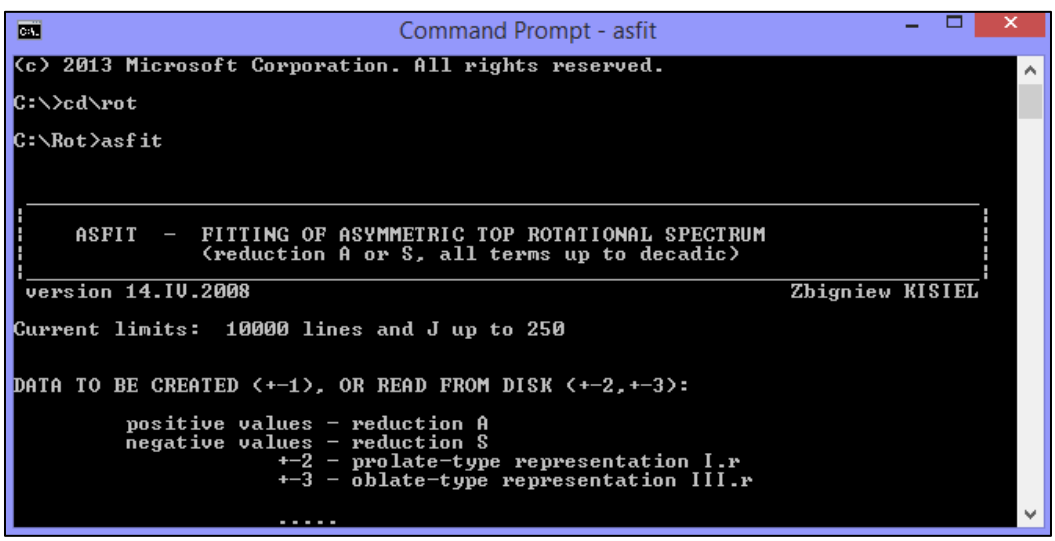
ASFIT.exe
(http://www.ifpan.edu.pl/~kisiel/asym/asym.htm#asfit)

To run a least-squares fit, type "ASFIT" into the command prompt. The program will prompt the user for the reduction and representation. The S-reduction is generally preferred for nearly symmetric asymmetric tops, but all four reduction-representations should work reasonably well.

Number Code	Reduction & Representation
2	A-Reduction, I^r representation
-2	S-Reduction, I^r representation
3	A-Reduction, III^r representation
-3	S-Reduction, III^r representation
I ≡ Prolate definition of A, B, and C	
III ≡ Oblate definition of A, B, and C	

In case constants are fixed, those in the .ASF file should correspond to the constants for the reduction that is to be fit. It is possible that the user will need to change constants or set all distortion terms to zero to switch S- to A- for difficult cases where a local rather than global minimum is found. This is more likely to occur for higher order Hamiltonians, octic or decadic.

Note: The program may be aborted at any point by pressing CTRL+C.

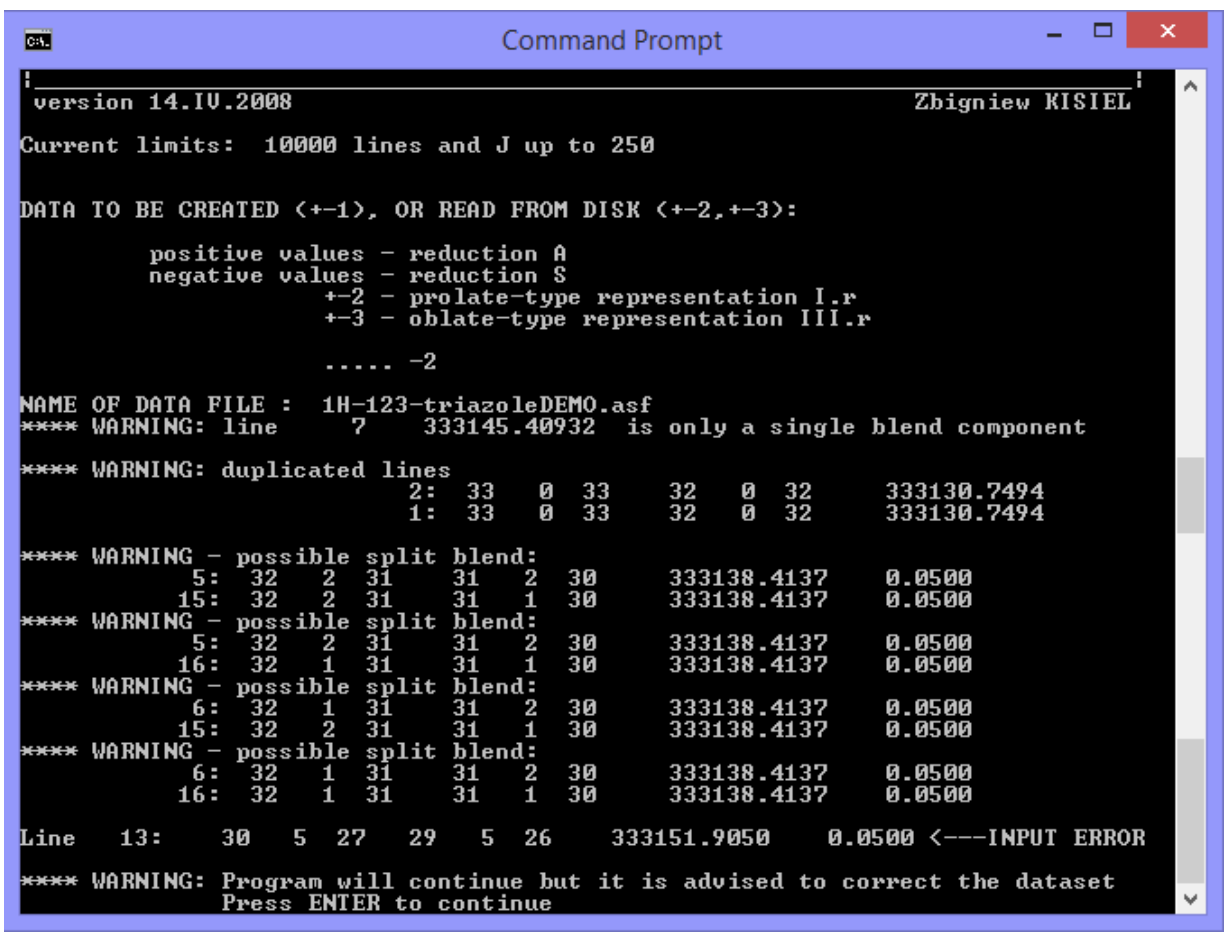


Next, the program will prompt the user for the ASF file. Type the name of the file, including the extension (.ASF) and press ENTER.

The program may display warnings if lines have been duplicated, if lines marked degenerate (2) do not have another transition assigned to the same peak, or other errors are found. Conveniently, the program states the error and displays the line(s) to which it applies. Should this occur, the user is advised to abort the fit by pressing CTRL+C and correct the .ASF file according to the errors noted. Then attempt to fit again.

A description of each of the four errors and how to correct them is provided in the next section (page numbers?). *Note that the image below was edited to simultaneously show four types of errors. In actuality, ASFIT presents each type of error one at a time. That is, it will list lines to which a single type of error applies. Once those are all corrected and the user attempts to run ASFIT again, it will list lines with the next type of error, should another type be present. (e.g. If a user has duplicated lines and split blend components, the list of duplicated lines will appear the first time ASFIT is run. Once those are corrected and another instance of ASFIT is run, the list of split blend components will be displayed.)*

The user should also note that while ASFIT will catch certain types of quantum number errors (e.g. K_{oblate} *and* $K_{prolate}$ *sum to a value other than J or J+1), it cannot necessarily discern if the upper and lower states have been swapped in the order they are written. In this case, the ASFIT_Line_Analysis Macro may be of use to the user (page XX).*



If all goes well, the program will display the starting constants and options to proceed. Proceed to fit by typing “0” and pressing ENTER as directed.

```
Command Prompt - asfit
                    ..... -2

NAME OF DATA FILE :  1H-123-triazole.asf

1H-123-triazole

CURRENT CONSTANTS :

A     =    10030.615589148170329281    1
B     =     9870.482946444080880610    1
C     =     4972.756462452331106761    1
 DJ   =        0.003515000000000000    1
 DJK  =        0.002965954975909971    1
 DK   =       -0.000916233156853718    1
 d1   =       -0.000564736706106309    1
 d2   =       -0.000449347237128017    1
HKJ   =        0.000000000000000000    1
HK    =        0.000000000000000000    1

TRANSITIONS IN DATA SET:  1839

SELECT AN OPTION:        = -1 list lines page at a time
                         =  0 proceed to fit
                         =  1 list lines
                         =  2 modifications      ....
```

When asked, “Do you want to interrupt the fit (1/0)?”, press “0” and ENTER to decline and proceed with fitting. Continue to respond in this manner until one of two results.
If the program asks “MORE CYCLES?”, type “1” and ENTER to proceed fitting.

```
Command Prompt - asfit
  1   1   2   3   4   5   6   7   8   9  10  11  12  13  14  15  16  17  18  19
 20  21  22  23  24  25  26  27  28  29  30  31  32  33  34  35  36  37  38  39
 40  41  42  43  44  45  46  47  48  49  50  51  52  53  54  55  56  57  58  59
 60  61  62  63  64  65  66  67


ITERATION STEP NO. 3          Deviation of fit   =      0.032441
                       Weighted deviation of fit =      0.648828

RESULTS - constant, change, standard deviation :

A     =     10030.802055946848        -0.000000021572         0.000257164941
B     =      9870.689384851248         0.000000024525         0.000261164517
C     =      4972.943132184941        -0.000000001188         0.000164908502
 DJ   =         0.001441469355         0.000000000101         0.000000451465
 DJK  =         0.002927227871        -0.000000000520         0.000001842255
 DK   =        -0.000793186164         0.000000000354         0.000001277453
 d1   =        -0.000832094732        -0.000000000036         0.000000200948
 d2   =        -0.000312616967         0.000000000015         0.000000053855
HKJ   =        -0.0000000094760217    -0.0000000000000017     0.0000000002550394
HK    =         0.0000000153975937     0.0000000000000042     0.0000000002343404

Do you want to interrupt the fit (1/0) ?  0

MORE CYCLES ?
```

After enough cycles, the program will achieve a fit and display the constants with further options. The most common action is to continue by pressing “0” and ENTER.

```
Command Prompt - asfit

1H-123-triazole

Fit of Watson's Hamiltonian --> reduction-S, representation-I.r  :

 A     =     10030.8020559379947371781   +-    0.0002571650601296228
 B     =      9870.6893848597555916058   +-    0.0002611643350722092
 C     =      4972.9431321855327041703   +-    0.0001649085045728148
  DJ   =         0.0014414693770573841   +-    0.0000004514653194291
  DJK  =         0.0029272277788363007   +-    0.0000018422553206026
  DK   =        -0.0007931861273990097   +-    0.0000012774527976271
  d1   =        -0.0008320947416627051   +-    0.0000002009477499472
  d2   =        -0.0003126169649977892   +-    0.0000000538548469806
 HKJ   =        -0.0000000094760259248   +-    0.0000000002550394039
 HK    =         0.0000000153975983739   +-    0.0000000002343402430

          Deviation of fit =       0.032441
 Weighted deviation of fit =       0.648828

F I T   C O M P L E T E D:   0 = continue
                             1 = scroll through fitted lines
                            -1 = as above, screen at a time    ....
```

At this point, the user may choose among a variety of options, the most common of which are to

- Create a .TXT or .RES output file by pressing "1" and ENTER (contains correlation coefficient matrix)
- Create a .TXT or .RES output file with a distribution plot of correlation coefficients in addition to the matrix
- Generate .LIN and .PAR (input) files for SPFIT (a similar fitting program capable of multi-state fits) by pressing "6" and ENTER

```
Command Prompt - asfit

 C     =      4972.9431321855327041703   +-    0.0001649085045728148
  DJ   =         0.0014414693770573841   +-    0.0000004514653194291
  DJK  =         0.0029272277788363007   +-    0.0000018422553206026
  DK   =        -0.0007931861273990097   +-    0.0000012774527976271
  d1   =        -0.0008320947416627051   +-    0.0000002009477499472
  d2   =        -0.0003126169649977892   +-    0.0000000538548469806
 HKJ   =        -0.0000000094760259248   +-    0.0000000002550394039
 HK    =         0.0000000153975983739   +-    0.0000000002343402430

          Deviation of fit =       0.032441
 Weighted deviation of fit =       0.648828

F I T   C O M P L E T E D:   0 = continue
                             1 = scroll through fitted lines
                            -1 = as above, screen at a time    .... 0

OUTPUT OF RESULTS - CHOOSE ONE OF:

        0 = no output
        1 = standard output
        2 = above + distribution plot of corr. coeff.
        3 = above + freq. contributions or hjj/DFBETAS
        4 = constants file for ASROT
        5 = frequency listing in deposition format
        6 = generate .LIN and .PAR files for SPFIT          ....
```

Following either of the above-named options, the program prompts the user for an output file name. At this time, it may be convenient to press the up arrow until the ASF file name appears on screen and then change to the appropriate extension (.txt for an ASFIT output file, no extension necessary for .lin and .par creation). *The file name is limited to 25 characters.*

For generating .LIN and .PAR files (option 6), the program then gives the option to add a vibrational quantum number. This value is required for multi-state least-squares fitting in SPFIT. Type "1" to do so or "0" to forgo, followed by ENTER.

The vibrational quantum member may be a one-digit, non-negative integer (0-9); press ENTER to select. If forgone, the vibrational quantum number will be zero.

Respond "0" to "DATA FILE TO BE WRITTEN TO DISK (1/0)?" for both .TXT/.RES output and .LIN and .PAR output.
"555" to exit ASFIT after writing a .TXT/.RES file.

ASFIT Errors and How to Correct Them

Single Blend Component

```
Command Prompt
version 14.IV.2008                                          Zbigniew KISIEL

Current limits:  10000 lines and J up to 250


DATA TO BE CREATED (+-1), OR READ FROM DISK (+-2,+-3):

         positive values - reduction A
         negative values - reduction S
                     +-2 - prolate-type representation I.r
                     +-3 - oblate-type representation III.r

                     ..... -2

NAME OF DATA FILE :  1H-123-triazoleDEMO.asf
**** WARNING: line      7    333145.40932  is only a single blend component

**** WARNING: Program will continue but it is advised to correct the dataset
             Press ENTER to continue forrtl: error (200): program aborting due
to control-C event
```

This type of error occurs when a line whose observed frequency is unique in the .ASF file is listed as blend component (*i.e.* has a 2 under "fit?").

J'	K-1'	K+1'	J	K-1	K+1	frequency	error	fit?	note/RI
33	0	33	32	0	32	333130.749429	0.050000	2	3.16E-02
33	0	33	32	0	32	333130.749429	0.050000	2	3.16E-02
33	1	33	32	0	32	333130.749429	0.050000	2	4.75E-03
33	0	33	32	1	32	333130.749429	0.050000	2	4.75E-03
32	2	31	31	2	30	333138.413664	0.050000	2	2.98E-02
32	1	31	31	2	30	333138.413664	0.050000	2	4.47E-03
31	2	29	30	2	28	333145.409316	0.050000	2	2.82E-02
30	5	27	29	5	26	333151.904951	0.050000	2	5.32E-02
30	3	27	29	4	26	333151.904951	0.050000	2	3.98E-03
30	4	27	29	3	26	333151.904951	0.050000	2	3.98E-03
30	3	27	29	3	26	333151.904951	0.050000	2	5.32E-02
30	3	27	29	4	26	333151.904951	0.050000	2	3.98E-03
30	4	27	29	3	26	333151.904951	0.050000	2	3.98E-03
30	3	27	29	3	26	333151.904951	0.050000	2	5.32E-02
32	2	31	31	1	30	333138.413664	0.050000	2	4.47E-03
32	1	31	31	1	30	333138.413664	0.050000	2	2.98E-02

To correct the error, find the erring line in the .ASF file. If this is a non-degenerate, non-overlapping line, change the “fit?” value to a one (1). If this line is degenerate or overlaps with other line(s), enter these lines into the .ASF file adjacent to the erring line.

In the example provided, the erring line was one of a set of four degenerate lines.

J'	K-1'	K+1'	J	K-1	K+1	frequency	error	fit?	note/RI
33	0	33	32	0	32	333130.749429	0.050000	2	3.16E-02
33	0	33	32	0	32	333130.749429	0.050000	2	3.16E-02
33	1	33	32	0	32	333130.749429	0.050000	2	4.75E-03
33	0	33	32	1	32	333130.749429	0.050000	2	4.75E-03
32	2	31	31	2	30	333138.413664	0.050000	2	2.98E-02
32	1	31	31	2	30	333138.413664	0.050000	2	4.47E-03
31	2	29	30	2	28	333145.409316	0.050000	2	2.82E-02
31	3	29	30	3	28	333145.409316	0.050000	2	2.82E-02
31	2	29	30	3	28	333145.409316	0.050000	2	4.22E-03
31	3	29	30	2	28	333145.409316	0.050000	2	4.22E-03
30	5	27	29	5	26	333151.904951	0.050000	2	5.32E-02
30	3	27	29	4	26	333151.904951	0.050000	2	3.98E-03
30	4	27	29	3	26	333151.904951	0.050000	2	3.98E-03
30	3	27	29	3	26	333151.904951	0.050000	2	5.32E-02
32	2	31	31	1	30	333138.413664	0.050000	2	4.47E-03
32	1	31	31	1	30	333138.413664	0.050000	2	2.98E-02

Duplicated Lines

```
Command Prompt
version 14.IV.2008                                          Zbigniew KISIEL

Current limits:  10000 lines and J up to 250


DATA TO BE CREATED (+-1), OR READ FROM DISK (+-2,+-3):

         positive values - reduction A
         negative values - reduction S
                     +-2 - prolate-type representation I.r
                     +-3 - oblate-type representation III.r

                     ..... -2

NAME OF DATA FILE :  1H-123-triazoleDEMO.asf

**** WARNING: duplicated lines
                          17:  33   0  33    32   0  32     333158.0192
                           1:  33   0  33    32   0  32     333130.7494

**** WARNING: duplicated lines
                           2:  33   0  33    32   0  32     333130.7494
                           1:  33   0  33    32   0  32     333130.7494

**** WARNING. Program will continue but it is advised to correct the dataset
              Press ENTER to continue
```

The error occurs when the same transition quantum numbers are listed for any two lines. They may have different frequencies, as in the first example above, or they may have the same frequency, as in the latter example.

J'	K-1'	K+1'	J	K-1	K+1	frequency	error	fit?	note/RI
33	0	33	32	0	32	333130.749429	0.050000	2	3.16E-02
33	0	33	32	0	32	333130.749429	0.050000	2	3.16E-02
33	1	33	32	0	32	333130.749429	0.050000	2	4.75E-03
33	0	33	32	1	32	333130.749429	0.050000	2	4.75E-03
32	2	31	31	2	30	333138.413664	0.050000	2	2.98E-02
32	1	31	31	2	30	333138.413664	0.050000	2	4.47E-03
32	2	31	31	1	30	333138.413664	0.050000	2	4.47E-03
32	1	31	31	1	30	333138.413664	0.050000	2	2.98E-02
31	2	29	30	2	28	333145.409316	0.050000	2	2.82E-02
31	3	29	30	3	28	333145.409316	0.050000	2	2.82E-02
31	2	29	30	3	28	333145.409316	0.050000	2	4.22E-03
31	3	29	30	2	28	333145.409316	0.050000	2	4.22E-03
30	5	27	29	5	26	333151.904951	0.050000	2	5.32E-02
30	3	27	29	4	26	333151.904951	0.050000	2	3.98E-03
30	4	27	29	3	26	333151.904951	0.050000	2	3.98E-03
30	3	27	29	3	26	333151.904951	0.050000	2	5.32E-02
33	0	33	32	0	32	333158.019204	0.050000	2	5.02E-02
29	4	25	28	5	24	333158.019204	0.050000	2	3.75E-03
29	5	25	28	4	24	333158.019204	0.050000	2	3.75E-03
29	5	25	28	5	24	333158.019204	0.050000	2	5.02E-02

If the same transition is listed at two different frequencies, the user needs to determine which (if any) of the lines is listed correctly. The other line(s) should either be deleted or its correct quantum numbers determined and corrected in the .ASF file.

In the example, the incorrectly entered line was corrected.

J'	K-1'	K+1'	J	K-1	K+1	frequency	error	fit?	note/RI
33	0	33	32	0	32	333130.749429	0.050000	2	3.16E-02
33	0	33	32	0	32	333130.749429	0.050000	2	3.16E-02
33	1	33	32	0	32	333130.749429	0.050000	2	4.75E-03
33	0	33	32	1	32	333130.749429	0.050000	2	4.75E-03
32	2	31	31	2	30	333138.413664	0.050000	2	2.98E-02
32	1	31	31	2	30	333138.413664	0.050000	2	4.47E-03
32	2	31	31	1	30	333138.413664	0.050000	2	4.47E-03
32	1	31	31	1	30	333138.413664	0.050000	2	2.98E-02
31	2	29	30	2	28	333145.409316	0.050000	2	2.82E-02
31	3	29	30	3	28	333145.409316	0.050000	2	2.82E-02
31	2	29	30	3	28	333145.409316	0.050000	2	4.22E-03
31	3	29	30	2	28	333145.409316	0.050000	2	4.22E-03
30	5	27	29	5	26	333151.904951	0.050000	2	5.32E-02
30	3	27	29	4	26	333151.904951	0.050000	2	3.98E-03
30	4	27	29	3	26	333151.904951	0.050000	2	3.98E-03
30	3	27	29	3	26	333151.904951	0.050000	2	5.32E-02
29	4	25	28	4	24	333158.019204	0.050000	2	5.02E-02
29	4	25	28	5	24	333158.019204	0.050000	2	3.75E-03
29	5	25	28	4	24	333158.019204	0.050000	2	3.75E-03
29	5	25	28	5	24	333158.019204	0.050000	2	5.02E-02

The error where the exact same line is listed twice (with the same frequency) typically occurs as a result of measuring degenerate lines directly from ASCP and SVIEW using the function keys. It is easily corrected if the degenerate or overlapping lines are listed together (one reason why it is useful to list lines in order of frequency). Determine the missing quantum numbers and correct one of the multiply listed lines.

J'	K-1'	K+1'	J	K-1	K+1	frequency	error	fit?	note/RI
33	0	33	32	0	32	333130.749429	0.050000	2	3.16E-02
33	1	33	32	1	32	333130.749429	0.050000	2	3.16E-02
33	1	33	32	0	32	333130.749429	0.050000	2	4.75E-03
33	0	33	32	1	32	333130.749429	0.050000	2	4.75E-03

Split Blend Components

```
Command Prompt
version 14.IV.2008                                                    Zbigniew KISIEL

Current limits:  10000 lines and J up to 250


DATA TO BE CREATED (+-1), OR READ FROM DISK (+-2,+-3):

          positive values - reduction A
          negative values - reduction S
                      +-2 - prolate-type representation I.r
                      +-3 - oblate-type representation III.r

                      ..... -2

NAME OF DATA FILE :  1H-123-triazoleDEMO.asf

**** WARNING - possible split blend:
            5:  32   2  31    31   2  30     333138.4137    0.0500
           15:  32   2  31    31   1  30     333138.4137    0.0500
**** WARNING - possible split blend:
            5:  32   2  31    31   2  30     333138.4137    0.0500
           16:  32   1  31    31   1  30     333138.4137    0.0500
**** WARNING - possible split blend:
            6:  32   1  31    31   2  30     333138.4137    0.0500
           15:  32   2  31    31   1  30     333138.4137    0.0500
**** WARNING - possible split blend:
            6:  32   1  31    31   2  30     333138.4137    0.0500
           16:  32   1  31    31   1  30     333138.4137    0.0500

**** WARNING: Program will continue but it is advised to correct the dataset
              Press ENTER to continue
```

ASFIT requires that degenerate or overlapping lines be adjacent to one another in the line list (major reason to keep the list ordered by frequency). If they are not, the split blend error occurs.

J'	K-1'	K+1'	J	K-1	K+1	frequency	error	fit?	note/RI
33	0	33	32	0	32	333130.749429	0.050000	2	3.16E-02
33	1	33	32	1	32	333130.749429	0.050000	2	3.16E-02
33	1	33	32	0	32	333130.749429	0.050000	2	4.75E-03
33	0	33	32	1	32	333130.749429	0.050000	2	4.75E-03
32	2	31	31	2	30	333138.413664	0.050000	2	2.98E-02
32	1	31	31	2	30	333138.413664	0.050000	2	4.47E-03
31	2	29	30	2	28	333145.409316	0.050000	2	2.82E-02
31	3	29	30	3	28	333145.409316	0.050000	2	2.82E-02
31	2	29	30	3	28	333145.409316	0.050000	2	4.22E-03
31	3	29	30	2	28	333145.409316	0.050000	2	4.22E-03
30	5	27	29	5	26	333151.904951	0.050000	2	5.32E-02
30	3	27	29	4	26	333151.904951	0.050000	2	3.98E-03
30	4	27	29	3	26	333151.904951	0.050000	2	3.98E-03
30	3	27	29	3	26	333151.904951	0.050000	2	5.32E-02
32	2	31	31	1	30	333138.413664	0.050000	2	4.47E-03
32	1	31	31	1	30	333138.413664	0.050000	2	2.98E-02

If there are only a couple of lines separated, it is fastest to cut and paste them to the proper location. If many lines are out of frequency order, a column-highlighting text editor (*e.g.* PSPad, Notepad++) and spreadsheet editing program (*e.g.* Microsoft Excel) may be used to efficiently bring order to the .ASF file. (See below.)

J'	K-1'	K+1'	J	K-1	K+1	frequency	error	fit?	note/RI
33	0	33	32	0	32	333130.749429	0.050000	2	3.16E-02
33	1	33	32	1	32	333130.749429	0.050000	2	3.16E-02
33	1	33	32	0	32	333130.749429	0.050000	2	4.75E-03
33	0	33	32	1	32	333130.749429	0.050000	2	4.75E-03
32	2	31	31	2	30	333138.413664	0.050000	2	2.98E-02
32	1	31	31	2	30	333138.413664	0.050000	2	4.47E-03
32	2	31	31	1	30	333138.413664	0.050000	2	4.47E-03
32	1	31	31	1	30	333138.413664	0.050000	2	2.98E-02
31	2	29	30	2	28	333145.409316	0.050000	2	2.82E-02
31	3	29	30	3	28	333145.409316	0.050000	2	2.82E-02
31	2	29	30	3	28	333145.409316	0.050000	2	4.22E-03
31	3	29	30	2	28	333145.409316	0.050000	2	4.22E-03
30	5	27	29	5	26	333151.904951	0.050000	2	5.32E-02
30	3	27	29	4	26	333151.904951	0.050000	2	3.98E-03
30	4	27	29	3	26	333151.904951	0.050000	2	3.98E-03
30	3	27	29	3	26	333151.904951	0.050000	2	5.32E-02

Efficient Re-ordering of Line List by Frequency

To use this technique, the user requires a text editor that allows column highlighting (*e.g.* PSPad, NotePad++) and a spreadsheet editor (*e.g.* Microsoft Excel) that has a sort function.

Open the .ASF file using the text editor and highlight all of the frequencies only. If the list is long, the user may avoid long scrolling by beginning to highlight at the leftmost column, using the scroll bar to jump to the bottom of the list, and then clicking after the rightmost frequency column of the last frequency in the list while holding down SHIFT.

J'	K-1'	K+1'	J	K-1	K+1	frequency	error	fit?	note/RI
33	0	33	32	0	32	333130.749429	0.050000	2	3.16E-02
30	5	27	29	5	26	333151.904951	0.050000	2	5.32E-02
31	3	29	30	2	28	333145.409316	0.050000	2	4.22E-03
33	1	33	32	1	32	333130.749429	0.050000	2	3.16E-02
31	2	29	30	2	28	333145.409316	0.050000	2	2.82E-02
32	2	31	31	2	30	333138.413664	0.050000	2	2.98E-02
31	2	29	30	3	28	333145.409316	0.050000	2	4.22E-03
32	1	31	31	2	30	333138.413664	0.050000	2	4.47E-03
31	3	29	30	3	28	333145.409316	0.050000	2	2.82E-02
30	3	27	29	3	26	333151.904951	0.050000	2	5.32E-02
33	0	33	32	1	32	333130.749429	0.050000	2	4.75E-03
30	3	27	29	4	26	333151.904951	0.050000	2	3.98E-03
32	1	31	31	1	30	333138.413664	0.050000	2	2.98E-02
32	2	31	31	1	30	333138.413664	0.050000	2	4.47E-03
33	1	33	32	0	32	333130.749429	0.050000	2	4.75E-03
30	4	27	29	3	26	333151.904951	0.050000	2	3.98E-03

Copy and paste the frequencies into a column of the spreadsheet editor. Each frequency should have its own row. Next, highlight all of lines in the line list to be ordered. The SHIFT highlight trick described above may again be used. *It is critical that the user make no changes in the order or number of rows between the frequency and line highlighting*. Leave the lines highlighted in the text editor after copying.

J'	K-1'	K+1'	J	K-1	K+1	frequency	error	fit?	note/RI
33	0	33	32	0	32	333130.749429	0.050000	2	3.16E-02
30	5	27	29	5	26	333151.904951	0.050000	2	5.32E-02
31	3	29	30	2	28	333145.409316	0.050000	2	4.22E-03
33	1	33	32	1	32	333130.749429	0.050000	2	3.16E-02
31	2	29	30	2	28	333145.409316	0.050000	2	2.82E-02
32	2	31	31	2	30	333138.413664	0.050000	2	2.98E-02
31	2	29	30	3	28	333145.409316	0.050000	2	4.22E-03
32	1	31	31	2	30	333138.413664	0.050000	2	4.47E-03
31	3	29	30	3	28	333145.409316	0.050000	2	2.82E-02
30	3	27	29	3	26	333151.904951	0.050000	2	5.32E-02
33	0	33	32	1	32	333130.749429	0.050000	2	4.75E-03
30	3	27	29	4	26	333151.904951	0.050000	2	3.98E-03
32	1	31	31	1	30	333138.413664	0.050000	2	2.98E-02
32	2	31	31	1	30	333138.413664	0.050000	2	4.47E-03
33	1	33	32	0	32	333130.749429	0.050000	2	4.75E-03
30	4	27	29	3	26	333151.904951	0.050000	2	3.98E-03

Copy and paste the lines into the spreadsheet editor in a new column with each line in its own row corresponding to the previously copied frequencies.

	A	B	C
1	Freq	Line	
2	333130.749429	33 0 33 32 0 32 333130.749429 0.050000 2 3.16E-02	
3	333151.904951	30 5 27 29 5 26 333151.904951 0.050000 2 5.32E-02	
4	333145.409316	31 3 29 30 2 28 333145.409316 0.050000 2 4.22E-03	
5	333130.749429	33 1 33 32 1 32 333130.749429 0.050000 2 3.16E-02	
6	333145.409316	31 2 29 30 2 28 333145.409316 0.050000 2 2.82E-02	
7	333138.413664	32 2 31 31 2 30 333138.413664 0.050000 2 2.98E-02	
8	333145.409316	31 2 29 30 3 28 333145.409316 0.050000 2 4.22E-03	
9	333138.413664	32 1 31 31 2 30 333138.413664 0.050000 2 4.47E-03	
10	333145.409316	31 3 29 30 3 28 333145.409316 0.050000 2 2.82E-02	
11	333151.904951	30 3 27 29 3 26 333151.904951 0.050000 2 5.32E-02	
12	333130.749429	33 0 33 32 1 32 333130.749429 0.050000 2 4.75E-03	
13	333151.904951	30 3 27 29 4 26 333151.904951 0.050000 2 3.98E-03	
14	333138.413664	32 1 31 31 1 30 333138.413664 0.050000 2 2.98E-02	
15	333138.413664	32 2 31 31 1 30 333138.413664 0.050000 2 4.47E-03	
16	333130.749429	33 1 33 32 0 32 333130.749429 0.050000 2 4.75E-03	
17	333151.904951	30 4 27 29 3 26 333151.904951 0.050000 2 3.98E-03	

Highlight the two columns and sort by frequency, lowest to highest value. (Use "Custom Sort" in Microsoft Excel). Check briefly that the frequencies are ordered as desired. Select and copy ONLY the cells containing lines. Use SHIFT to highlight as necessary.

Assuming the user left the lines copied from the text editor highlighted, simply make the text editor active and paste the lines copied from the spreadsheet. Save the .ASF file.

J'	K-1'	K+1'	J	K-1	K+1	frequency	error	fit?	note/RI
33	0	33	32	0	32	333130.749429	0.050000	2	3.16E-02
33	1	33	32	1	32	333130.749429	0.050000	2	3.16E-02
33	0	33	32	1	32	333130.749429	0.050000	2	4.75E-03
33	1	33	32	0	32	333130.749429	0.050000	2	4.75E-03
32	2	31	31	2	30	333138.413664	0.050000	2	2.98E-02
32	1	31	31	2	30	333138.413664	0.050000	2	4.47E-03
32	1	31	31	1	30	333138.413664	0.050000	2	2.98E-02
32	2	31	31	1	30	333138.413664	0.050000	2	4.47E-03
31	3	29	30	2	28	333145.409316	0.050000	2	4.22E-03
31	2	29	30	2	28	333145.409316	0.050000	2	2.82E-02
31	2	29	30	3	28	333145.409316	0.050000	2	4.22E-03
31	3	29	30	3	28	333145.409316	0.050000	2	2.82E-02
30	5	27	29	5	26	333151.904951	0.050000	2	5.32E-02
30	3	27	29	3	26	333151.904951	0.050000	2	5.32E-02
30	3	27	29	4	26	333151.904951	0.050000	2	3.98E-03
30	4	27	29	3	26	333151.904951	0.050000	2	3.98E-03

Bad Quantum Numbers

```
Command Prompt - asfit
 version 14.IV.2008                                          Zbigniew KISIEL

Current limits:  10000 lines and J up to 250


DATA TO BE CREATED (+-1), OR READ FROM DISK (+-2,+-3):

         positive values - reduction A
         negative values - reduction S
                     +-2 - prolate-type representation I.r
                     +-3 - oblate-type representation III.r

                     ..... -2

NAME OF DATA FILE :  1H-123-triazoleDEMO.asf

1H-123-triazole

CURRENT CONSTANTS :

A     =    10030.615589148170329281    1
B     =     9870.482946444080880610    1
C     =     4972.756462452331106761    1
 DJ   =        0.003515000000000000    1
 DJK  =        0.002965954975909971    1
 DK   =       -0.000916233156853718    1
 d1   =       -0.000564736706106309    1
 d2   =       -0.000449347237128017    1
HKJ   =        0.000000000000000000    1
HK    =        0.000000000000000000    1

TRANSITIONS IN DATA SET:    36

The following lines have errors in quantum numbers and have been
automatically excluded from fit:

Line   13:    30   5  27   29   5  26    333151.9050    0.0500 <---INPUT ERROR

**** WARNING: Program will continue but it is advised to correct the dataset
              Press ENTER to continue
```

The error occurs when ASFIT detects incorrectly input quantum numbers (*e.g.* when the sum of $K_{prolate}$ and K_{oblate} does not sum to J or J+1).

30	5	27	29	5	26	333151.904951	0.050000	2 5.32E-02
30	3	27	29	3	26	333151.904951	0.050000	2 5.32E-02
30	3	27	29	4	26	333151.904951	0.050000	2 3.98E-03
30	4	27	29	3	26	333151.904951	0.050000	2 3.98E-03

The quantum numbers should be corrected or the line completely deleted from the .ASF file.

30	4	27	29	4	26	333151.904951	0.050000	2 5.32E-02
30	3	27	29	3	26	333151.904951	0.050000	2 5.32E-02
30	3	27	29	4	26	333151.904951	0.050000	2 3.98E-03
30	4	27	29	3	26	333151.904951	0.050000	2 3.98E-03

Note that ASFIT cannot necessarily discern if the upper and lower states have been swapped in the order they are written. In this case, the ASFIT_Line_Analysis Macro may be helpful (page XX).

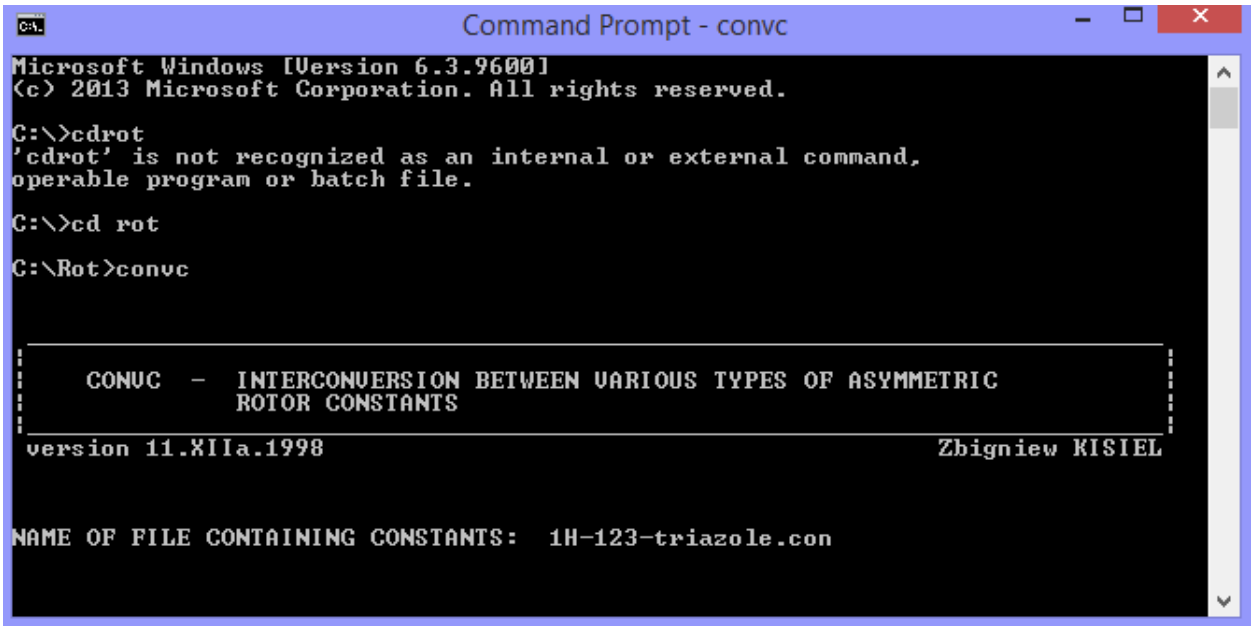
CONVC.exe

(http://www.ifpan.edu.pl/~kisiel/asym/asym.htm#convc)

CONVC.exe is a program that converts constants from one reduction and representation to another.

Type "CONVC" and press ENTER to launch the program.

Type in the name of the .CON file containing the constants. Press ENTER.



Select a conversion option.

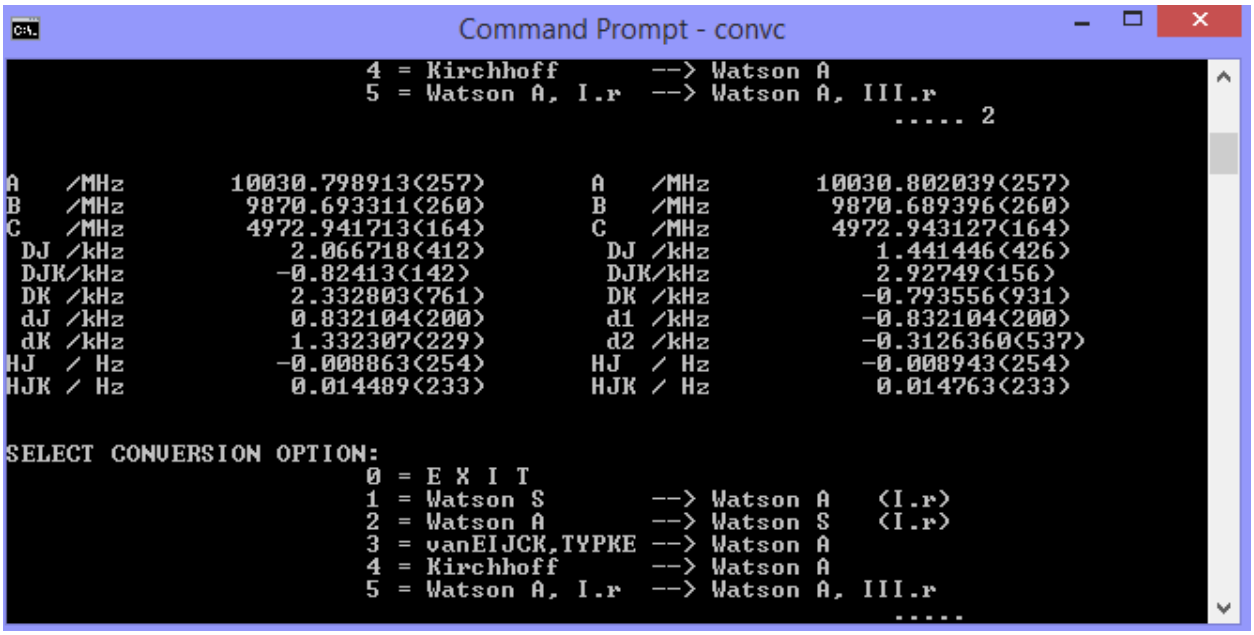
Option Value	Converting from…		Convert to…
1	Watson S-Reduction, I^r	→	Watson A-Reduction, I^r
2	Watson A-Reduction, I^r	→	Watson S-Reduction, I^r
3	vanEIJCK,TYPKE	→	Watson A-Reduction
4	Kirchhoff	→	Watson A-Reduction
5	Watson A-Reduction, I^r	→	Watson A-Reduction, III^r

I ≡ Prolate definition of A, B, and C

III ≡ Oblate definition of A, B, and C

The program must be exited and run again in order to convert the constants of other molecules or states (*i.e.* using a different file of constants).

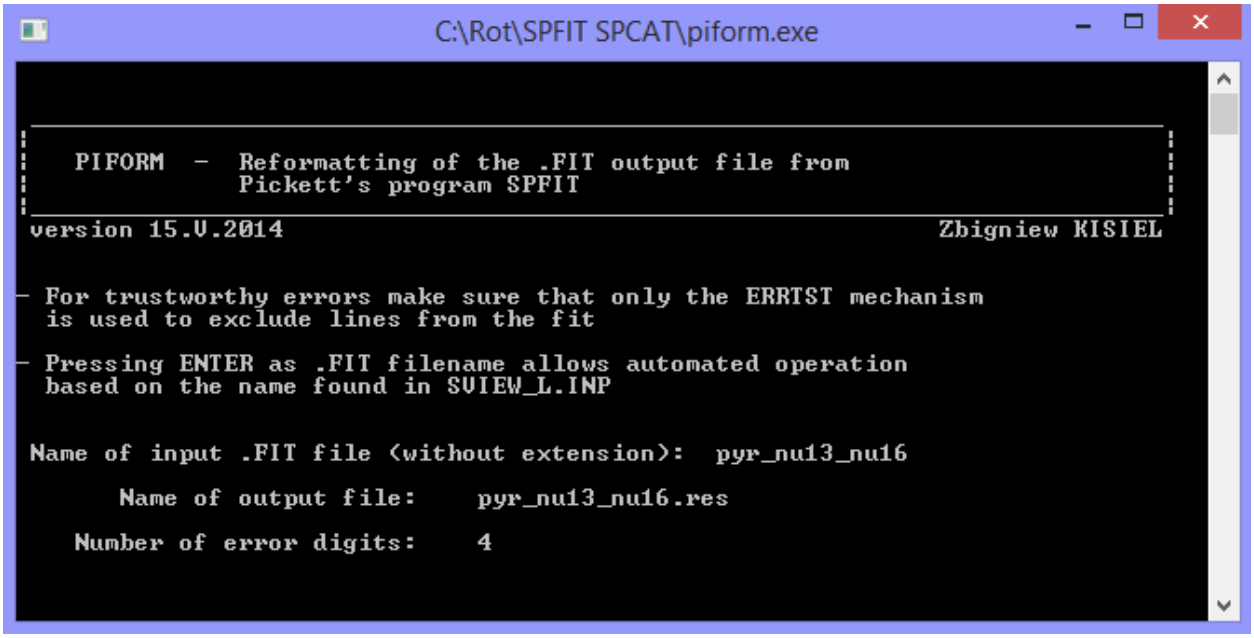
The original and converted constants will be listed on the screen, as well as output into CONVC.OUT. If the file existed previously, the most recent conversion will be appended to the end of the file.



PIFORM.exe

PIFORM translates the SPFIT least-squares fitting output file (.FIT) into a more reader-friendly form akin to the ASFIT output.

To run PIFORM, open the program and enter the name of the .FIT file without the extension. Press ENTER. Next, type the desired name of the output file *including* the extension (.txt or.res suggested). The user may use the up arrow key to automatically enter the file name entered above; then add the desired extension. Press ENTER. If the line file has over 10,000 lines, the program will ask whether to provide the full line number (1) or to leave off the ten-thousands digit for compatibility with AC.EXE or ACIR.EXE (0); if the user is not planning to run the mentioned programs, option 1 (full kine number) is suggested. Finally, provide the number of error digits desired (typically 2) and press ENTER.



The program will run through displays of the constants and worst-fitting lines list, then close.

PISLIN.exe

PISLIN.exe is a line-sorting and checking program. It uses a .LIN input file and outputs either a .LIN or .RES file. In order to use PISLIN.exe, the user must first separate all vibrational states into separate .LIN files. It is suggested that the user create a .LIN file for use with PISLIN.exe (separate from that used for fitting) even if only sorting single state.

Run PISLIN.exe. It will prompt the user for the input file name (with .LIN extension), then ask whether the user wants checks on the file. Checks will inform the user of duplicate quantum numbers and split blends (lines with the same frequency that are not adjacent and should be). It is suggested that the user run checks if they have not already been run and addressed, otherwise, they may be skipped.

To run checks, select 1, First, the program checks for duplicate quantum numbers; if duplicate quantum numbers are found, the program will inform the user of the number of duplicates. It is suggested the user use CTRL+C to exit the program, find, and resolve duplicate quantum numbers. An efficient means of finding these lines is provided below, on page XX.

Example PISLIN readout if duplicate pairs found

```
Working ..............................................................................

     3 duplicate pairs of lines with identical quantum
       numbers have been found
```

If there are not duplicate lines, the program prompts the user for the resolution criterion in the units of the .LIN file's line frequencies (MHz for our group; the resolution criteria is typically 0.05 for our group). Once entered, the program reports the number of split blends and offers to do nothing (0), remove them (1), or bring them together (2). It is suggested to sort lines first, then bring split blends together (sorting them will split blend lines, anyway). To sort lines first, select zero (do nothing, 0) and press ENTER.

To sort lines, provide a name for the output file, including extension (PISLIN.RES suggested, although over-writing the input file is also an option) as prompted. The program offers several line-sorting option:

- Sort by error (-2)
- Sort by frequency (-1)
- Sort by one of the quantum numbers (use values 1 – 12 to select which)
- Sort by selection rule (20)
- Sort by transition sequence (30) (*i.e.* Sort by K_a, then J,)

Option 30 (transition sequence) is used for multi-state fits; single-state distorted rotor fits are typically sorted by frequency (-1). If transition sequence is selected, the user must enter the indices of the six transition quantum numbers (J, K_a, and K_c upper and lower); these are typically 1, 2, 3, 5, 6, and 7 (separated only by spaces when entered into program: 1 2 3 5 6 7). The sorted lines will be output file named previously.

To combine split blends, run PISLIN.exe and request checks. When split blends are reported, select two (2, bring lines together). An example screen capture of running the program to this

point is provided on the next page. If bringing split blends together is selected, PISLIN.exe will output the sorted lines into PISLIN.RES.

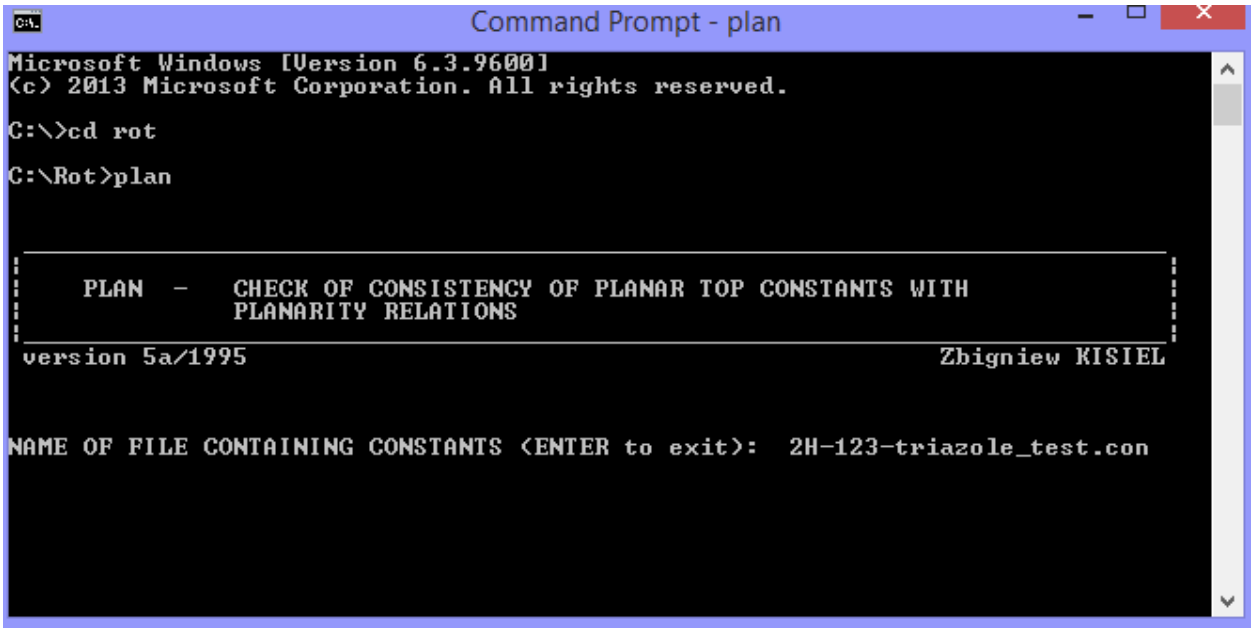
Copy-paste the entire PISLIN.RES output into the input file and run the program again. *Because PISLIN.exe is not always successful in combining the split blends, it is important to run checks again and repeat the bring together, copy-paste, rerun PISLIN.exe sequence as many times as necessary to combine split blends.*

Example run of PISLIN.exe

```
 ____________________________________________________________________________
|                                                                            |
|  PISLIN - Checking and Sorting of .LIN files for Herb Pickett's            |
|           SPFIT program                                                    |
|____________________________________________________________________________|
 version 14.I.2014                                              Zbigniew KISIEL



 Name of the lines file:  bniso00.lin

 8618 lines read in
    0 extra lines after the main .LIN block


Do you want checks on this .LIN file (0/1) ? .... 1

**** Results will be listed to file PISLIN.OUT


Working ..................................................................

     2 duplicate pairs of lines with identical quantum
       numbers have been found


Resolution criterion for blend identification = 0.05

Working ..................................................................

     1 pairs of split blends with frequencies closer than
       the resolution criterion of    0.05

     2 lines are implicated in split blends:

  7346  7348

       In order of line number the split blend lines are:

  7346  7348

Treatment of split blends:

              0 = do nothing
              1 = remove
              2 = bring together       .... 2

  8618 lines in original dataset
  8618 lines written to file PISLIN.RES

****** RERUN the program to check, and repeat blending
       operation if necessary
```

PLAN.exe
(http://www.ifpan.edu.pl/~kisiel/asym/asym.htm#plan)

PLAN.exe calculates the correct error value in inertial defects, among other values.
Type "PLAN" and press ENTER.
Type in the name of the appropriate .CON file and press ENTER.



The output will be displayed on the screen and appended to the end of the PLAN.OUT file.
To run another calculation, press ENTER and proceed to type another .CON file name. To exit, press ENTER twice.

PLANM.exe

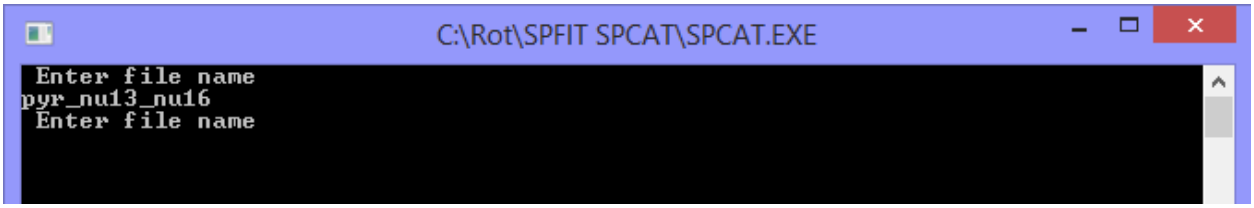
PLANM.exe is a manual version of PLAN.exe; the user inputs rotational constants and errors and the program outputs the inertial defect, error, and *kappa*.

PMIXC.exe

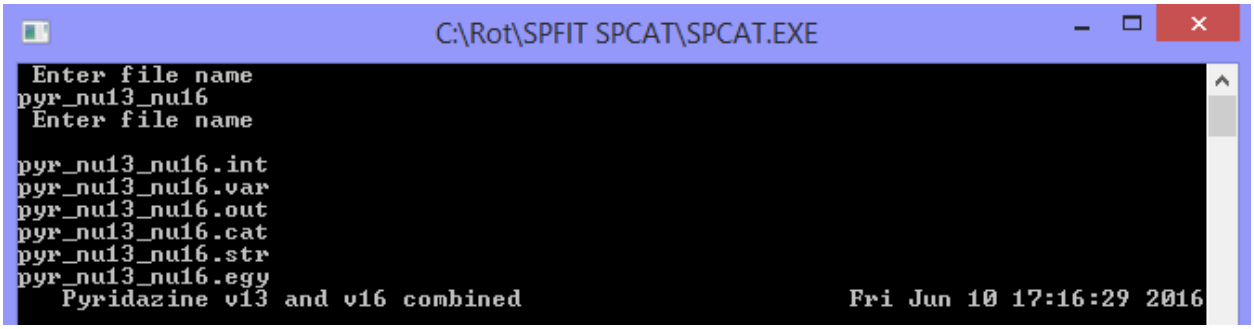
SPCAT.exe

SPCAT is the analog to ASROT. It requires a .INT and .VAR files with the same name preceding the extension. SPCAT write .CAT files, which can contain multiple states and be read by ASCP_L.

To generate a .CAT file, place the .INT and VAR files (with same file name) in the same directory as SPCAT and run the program. . Type in the file name (without the extension) and press ENTER twice.



The program will run and close.

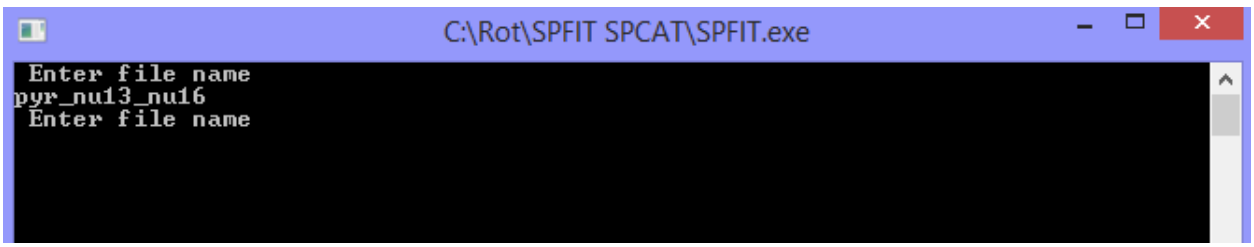


Four new files (.OUT, .STR, .EGY, and .CAT) will appear in the directory with the same name as the staring files.

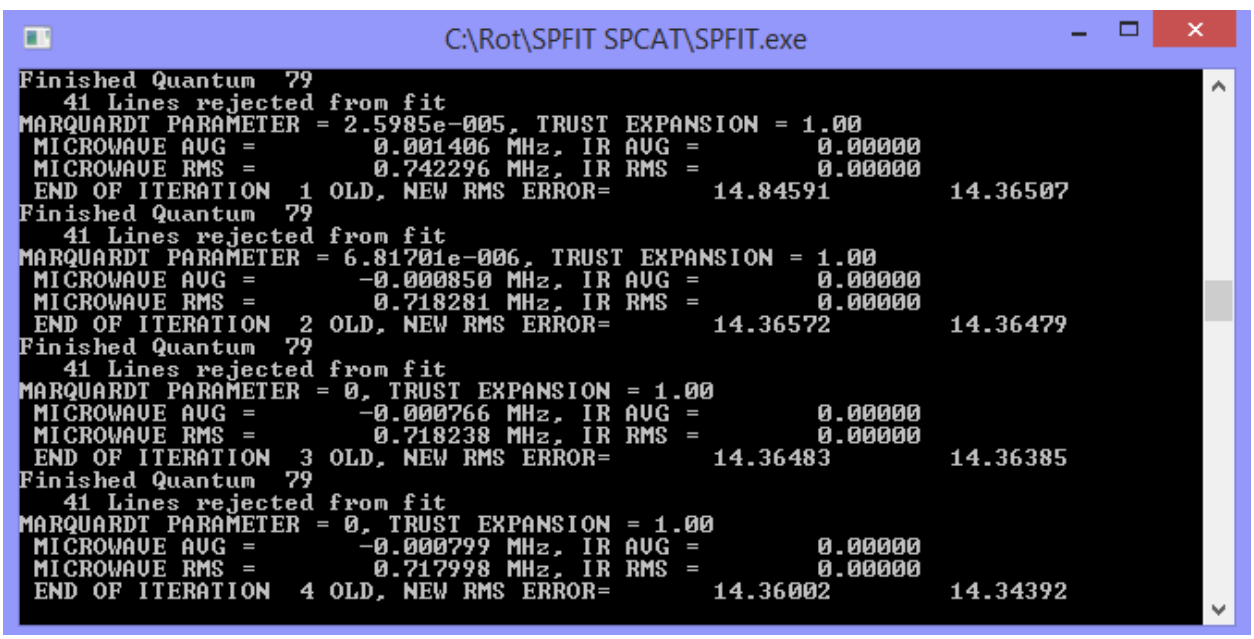
SPFIT.exe

SPFIT is a program analagous to ASFIT, but is capable of multi-state least-squares fitting. The program requires a .LIN file and corresponding .PAR file. If the user intends to complete a multi-state fit, the states should be combined into a single .LIN file and single .PAR file with appropriate vibrational quantum numbers. *The .LIN and .PAR file must have the same file name (preceding the file extension).* These files should be located in the same directory as the program. It is suggested that the user keep a separate backup folder and copy the .PAR and .LIN files into it before running SPFIT, as the program will overwrite the files in the directory from which it is run.

The program may be run via the Command Prompt, as ASFIT, or by opening the program (by double-clicking or select-and-entering) directly from the directory. When prompted, enter the file name *without the extension.* Press ENTER. The program will again display "Enter file name". Without entering anything more, press ENTER.



SPFIT will run least-squares fitting. The root mean square error value of each iteration of the least-squares fitting will be displayed on the screen.



The user should note that the program will stop when it has reached the maximum number of iterations stated in the .PAR file. The end of iterations does not mean a converged least-squares fit. It is recommended to continue running the least-squares fitting process until it only takes a

single iteration to converge. *Prof. Zbigniew Kisiel has suggested that if nothing useful happens in 10 – 15 iterations, the fit is unlikely to converge or yield anything useful after one hundred of more iterations, either*. The program will create .BAK, .FIT, and .VAR files with the same file name as the .PAR and .LIN files. The .FIT contains the output; it is suggested to immediately run PIFORM on the output.

ASCP_L.exe and SVIEW_L.exe
(http://www.ifpan.edu.pl/~kisiel/aabs/aabs.htm)

Having properly linked files in the SVIEW_L.INP file, open the desired .ASR (code 1) or .INP (code 2) file followed by the .SPE file. Align the windows such that the horizontal borders line up and the spectra are clearly visible while the stick spectrum is focal. Maneuvering in ASCP (the stick spectrum) will equivalently manipulate the experimental spectrum window.

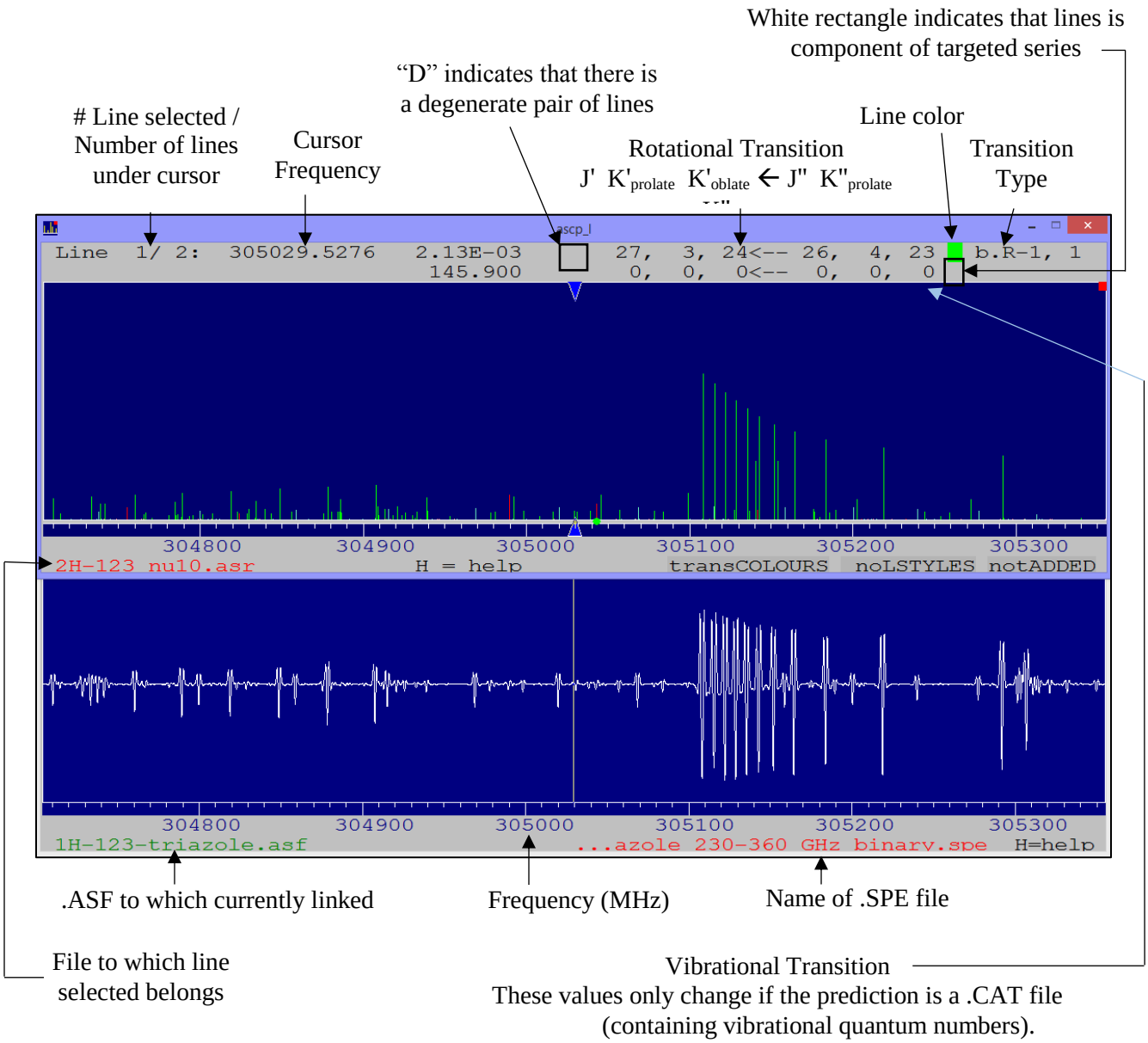
The user may adjust the size of the windows using aabs.cfg. To do so, open the file using a plain text editor. Lists of "Identified working configurations" and "Sample configurations" are provided at the end of the file.

Enter the desired width and height into the corresponding lines immediately beneath the "Current configuration is to follow immediately below" heading or copy-paste an entire sample configuration in place of the given configuration. The fonts may also be changed per instructions in the file, if desired.

Width must be multiples of 80 pixels, height must be multiples of 30 pixels. Some combinations work better than others.

```
!  Current configuration is to follow immediately below
!-----------------------------------------------------------------------
!
!  For general use on screens from 17 to 22" at >=1280x1024 resolution
!
!----------------------------------_____-----------------------------------
 Width of ASCP  window in pixels:  1280
Height of ASCP  window in pixels:   660              <--- for double ht
              Name of font:  Courier New
      Extended font attribute:    1
      Extended font attribute:    5
      Extended font attribute:    9
      Extended font attribute:   12
!----------------------------------_____-----------------------------------
```

Layout


White rectangle indicates that lines is component of targeted series
"D" indicates that there is a degenerate pair of lines
Line selected / Number of lines under cursor
Cursor Frequency
Rotational Transition
J' K'prolate K'oblate ← J" K"prolate
Line color
Transition Type
Line 1/ 2: 305029.5276 2.13E-03 145.900
27, 3, 24<-- 26, 4, 23
0, 0, 0<-- 0, 0, 0
b.R-1, 1
304800 304900 305000 305100 305200 305300
2H-123 nu10.asr
H = help
transCOLOURS noLSTYLES notADDED
1H-123-triazole.asf
...azole 230-360 GHz binary.spe
H=help
.ASF to which currently linked
Frequency (MHz)
Name of .SPE file
File to which line selected belongs
Vibrational Transition
These values only change if the prediction is a .CAT file (containing vibrational quantum numbers).

Common Functions

General

Refresh/Update Stick spectrum – ESC

Note that if the linked .ASF has been changed, *both* ASCP and SVIEW must be updated using the ESC key.

Help (Extensive) – H

Help (quick keys) – Backspace

Save Experimental Spectrum – m while active in SVIEW

Create an image of the current spectral view – g (on either SVIEW or ASCP individually, or on ASCP while linked to an experimental spectrum)

Viewing

Holding down SHIFT will *accelerate* any of these functions; CAPS LOCK has an equivalent effect

Increase/decrease intensity – w / z

Zoom horizontally in/out – e / q

Move cursor left/right in stick spectrum– k / l

Move cursor left/right in experimental spectrum while stick spectrum active – 2 / 3

Jump left/right – a / s

Shift experimental spectrum vertically up/down – 3 / 2 while focused on SVIEW

Jump to frequency range – f followed by lower frequency limit (MHz), ENTER, upper frequency limit (MHz), ENTER

Append spectrum (.SPE) -- & (SHIFT+7). Appended spectrum must be in binary.

Toggle background color – c (in both ASCP and SVIEW)

Line Color Toggle – p

Stack degenerate lines toggle -- y

Flip experimental spectrum – dash/underscore key (-) while active in SVIEW

Target all lines belonging to the same molecule/state – r while cursor is on line belonging to molecule/state

Target a subset of lines belonging to a single molecule/state – t (lowercase) followed by values of quantum numbers to be held constant when cursor is on line to be included in subset (value is typically -2 or -3; see page XX for detail)

Target a subset of line sfollowing a particular selection rule – T (uppercase) followed by quantum number, the value of that quantum number, the changing quantum number, the value by which the quantum number changes (see page XX for detail)

Remove lines from experimental spectrum – CTRL+b while active in SVIEW, with corresponding ECHO.ASC file

Loomis-Wood Mode

Loomis-Wood Plot – Having targeted a series of lines using T [value], F5

View different markers in Loomis-Wood Plot – p
View experimental spectrum in Loomis-Wood Plot – Left bracket, [
Scroll up/down in Loomis-Wood Plot – PGUP / PGDN
View more data/vertically condense data in Loomis-Wood Plot – equivalence/plus key, =
View less data/vertically expand data in Loomis-Wood Plot – dash/underscore key, -
Make image (.GLE file) of current Loomis-Wood view – g
Make plot for targeted series – t (a more detailed description is provided in *Making* $\nu - \nu_0$ *Plots*, page XX)

+1 – Frequency difference between vibrational satellite and corresponding ground state transitions divided by (J″ + 1) (P- and R- branches), K_a (Q-branch)

$$\frac{\nu-\nu_0}{J''+1}\ \text{(MHz)}\ \text{ vs. }\ (J''+1)$$

-1 – Raw frequency difference between ground and vibrational state for corresponding transitions

$$(\nu-\nu_0)\ \text{ vs. }\ (J''+1)$$

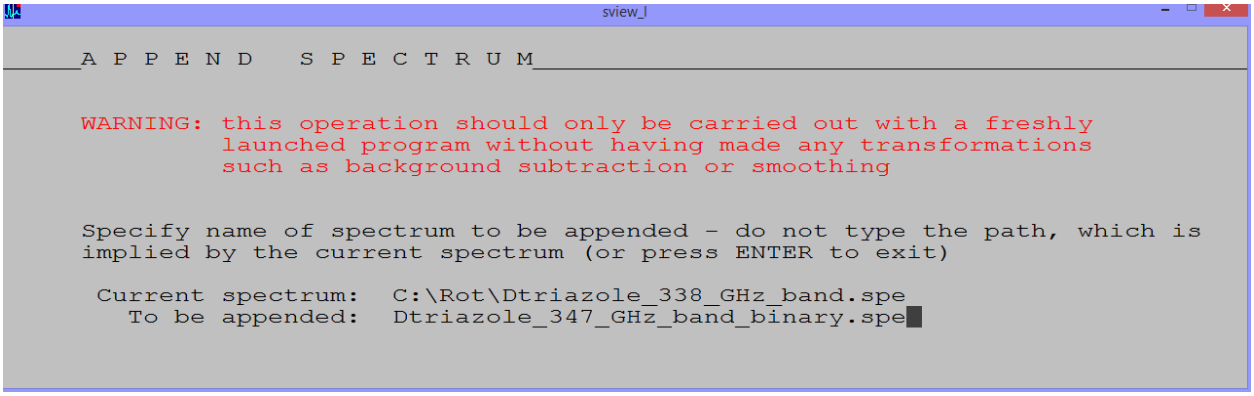
Measuring Lines
Enter line in .ASF file as excluded from least-squares fitting – F3
Enter line in .ASF file as unique line for least-squares fitting – F1
Enter line in .ASF file as blended component or degenerate line for least-squares fitting – F2
Measure a line frequency without entering into .ASF file – O while active in SVIEW

Appending Experimental Spectra

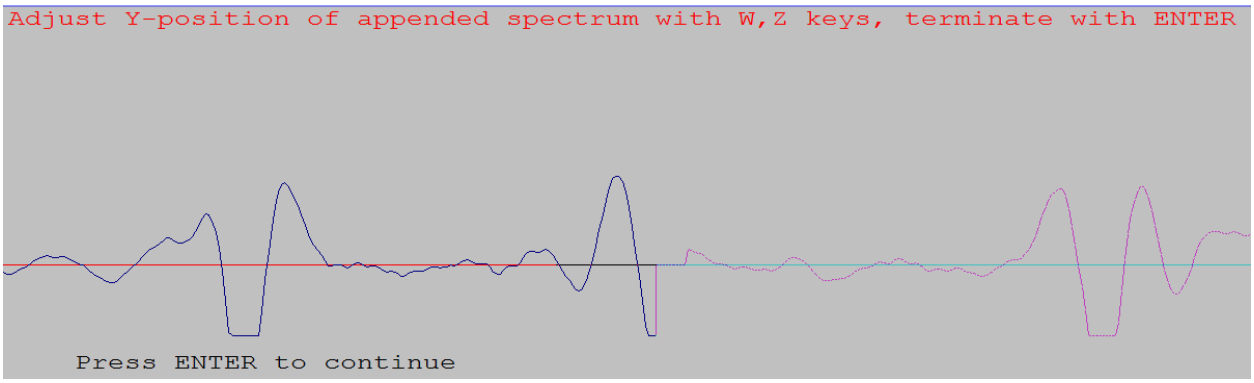
All experimental spectra to be appended must in be in binary form.

Open the lowest frequency .SPE file in SVIEW. Press the *& key* (SHIFT+7). Enter the maximum number of points contained in files to be appended. Entering a negative number matches matches the intensities of the spectra to be appended. *The maximum allowable absolute value is 76,595,200.* Press ENTER.

Enter the name of the file name to be appended and press ENTER.

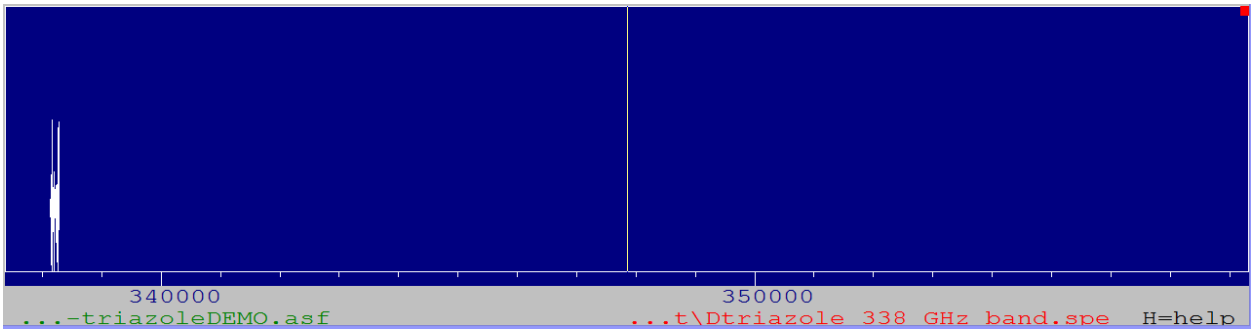


Press ENTER again and the program will prompt the user to set the baseline. Use the Z (down) and W (up) keys to match the baseline of the appended spectrum to that of the initially opened spectrum. Press ENTER to set.

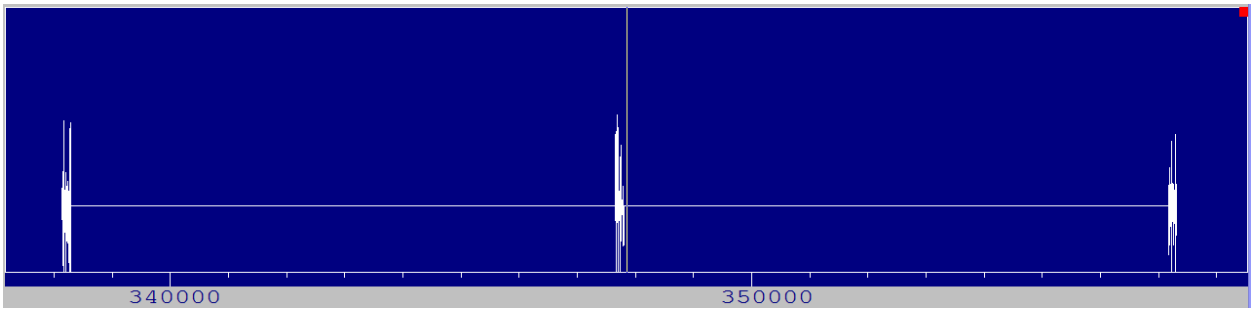


In this way, multiple spectra may be appended, as depicted on the following page.

Initially opened spectrum



Spectrum with two additional spectra appended

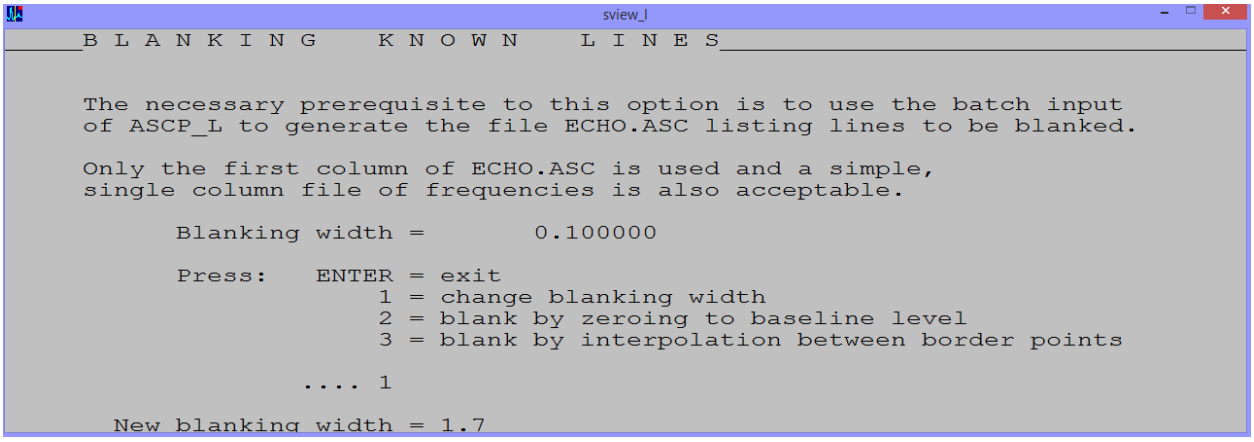


This function allows the user to jump between bands, easily viewing the associated individual scans. Unfortunately, Loomis-Wood plots may not be viewed in this way.

Removing Lines from Experimental Spectrum using ECHO.ASC

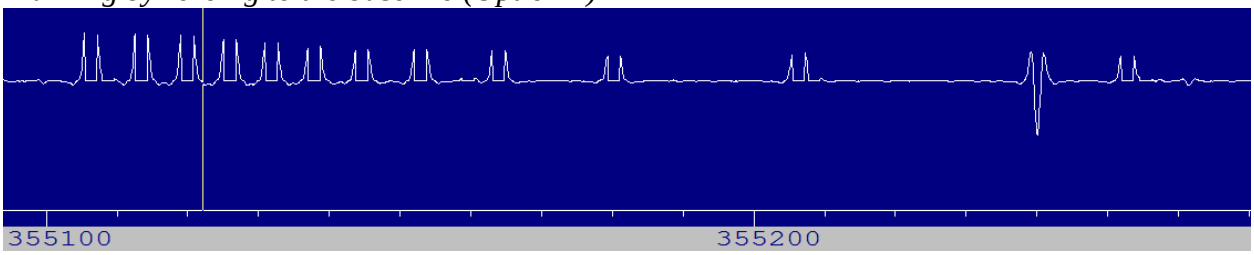
While active in SVIEW, press CTRL+B

Select option "1" and ENTER to change the blanking width. The program's default value is 0.1 MHz and is reset in each instance of SVIEW. Enter double the frequency modulation value (typically 1.7 for our spectrometer) and press ENTER.

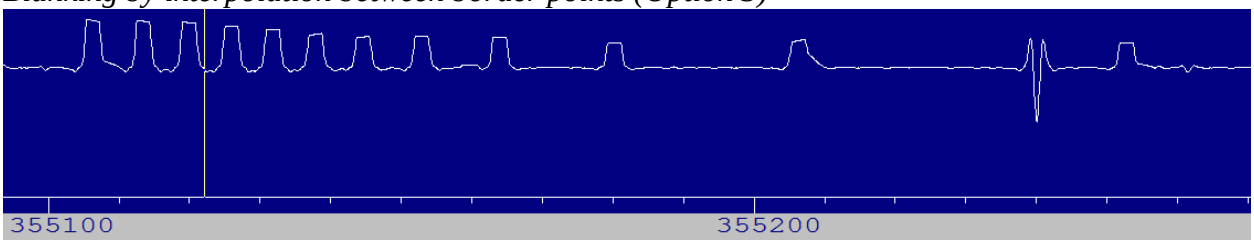


Select option "2" for blanking by zeroing to the baseline or option "3" for blanking by interpolation between border points. Option 3 is recommended, but the choice is up to the user. Press ENTER. In the sample below, the lines belonging to 2H-1,2,3-triazole have been removed.

Blanking by zeroing to the baseline (Option 2)



Blanking by interpolation between border points (Option 3)



The spectrum may be returned to its original form by pressing the *R key*.

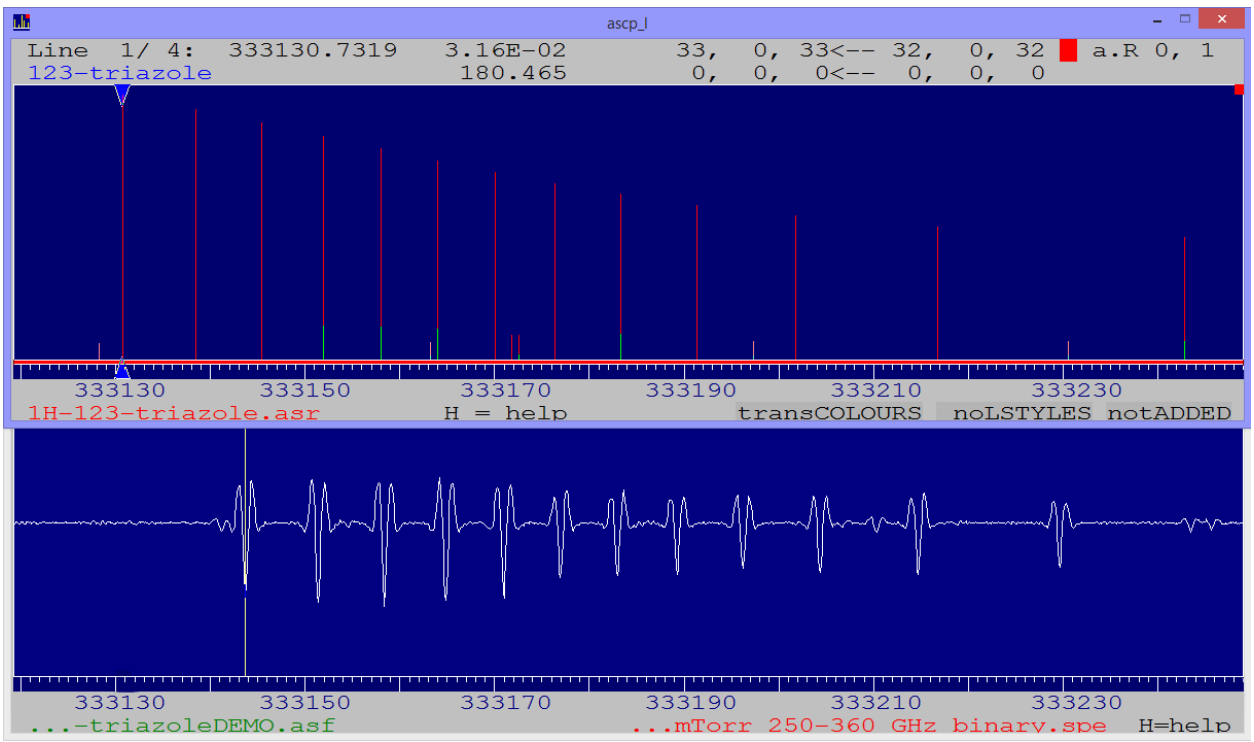
Taking Experimental Measurements viewing spectra

F1 enters measurement to be fit (1)

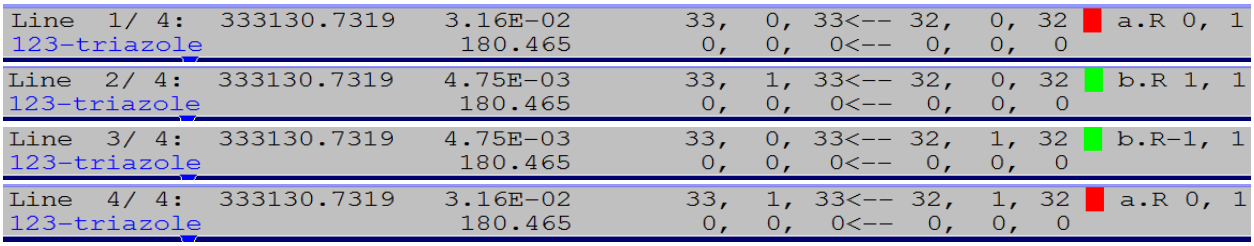
F2 enters measurement to be fit as a blended or degenerate line (2)

F3 enters the measurement, but excludes it from the fit (0)

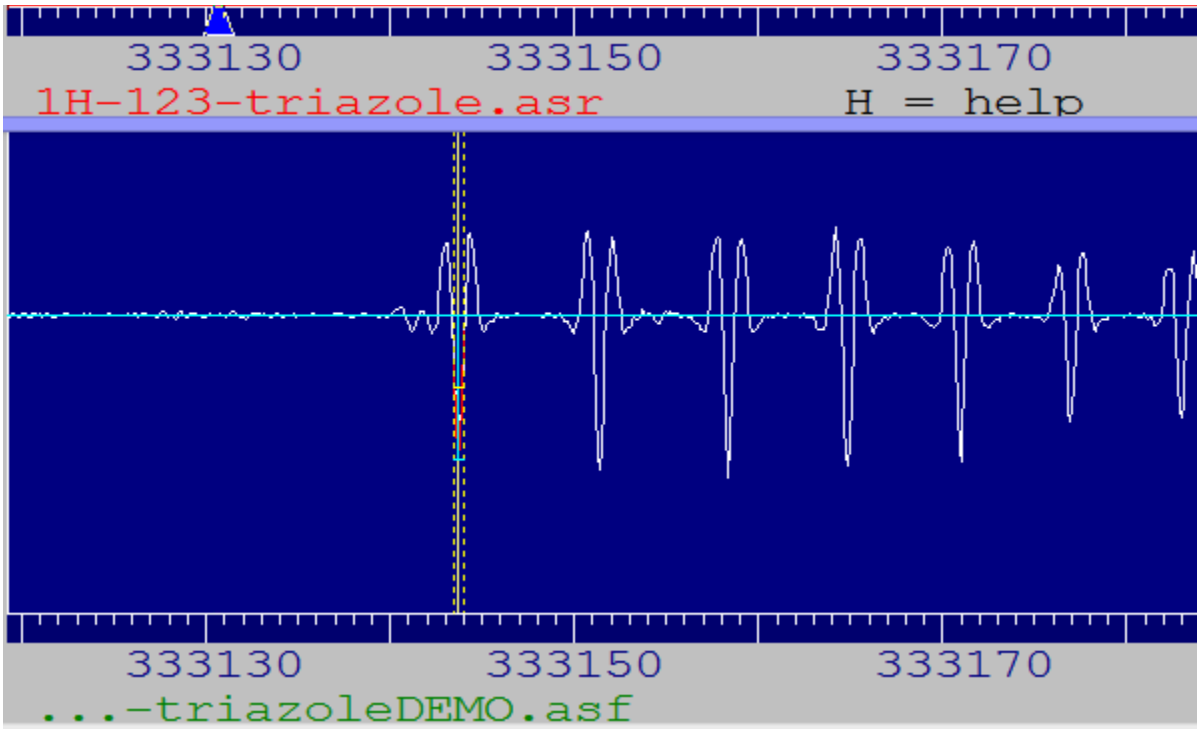
Having properly linked the programs and .ASF file, move the cursor in ASCP until it resides on the desired line (using K and L keys). The line's information will appear at the top of the ASCP window. If necessary, scroll the cursor in SVIEW to the approximate minimum of the desired peak to be measured using the 2 and 3 keys. It is possible the user will need to do this for multiple measurements until a predictive fit is obtained of if a perturbed state is being fit.



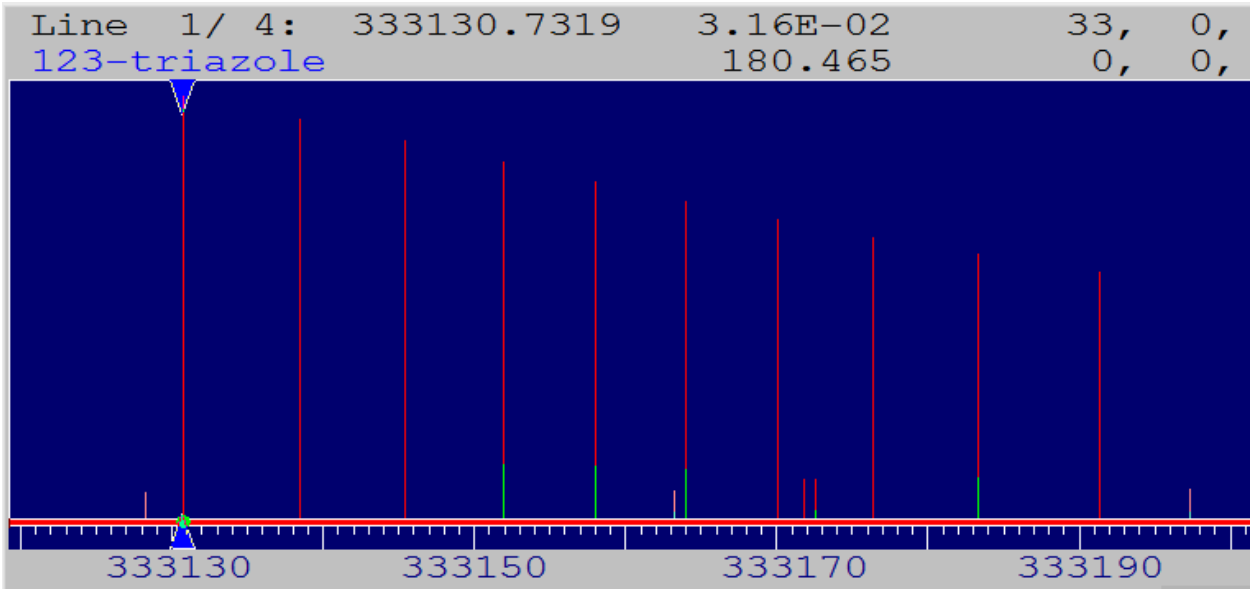
Note the number of lines under the cursor, listed along with the line information. By pressing the UP and DOWN arrow keys, the user is able to cycle through the lines being selected. The red .ASR file name at the bottom of the ASCP window tells the user to which molecule/state (file) the line belongs.



When the desired line is selected, press the appropriate function key (F1, F2, or F3, described at the beginning of this section). The selected peak will be marked in the SVIEW window.



In this example, all four lines belong to the same molecule and are degenerate pairs. Thus, each of the four lines was measured using F2. Refreshing ASCP using the ESC key, the first line is now marked as having been measured and included in the least-squares fitting.



The four lines are now included in the .ASF file (added as the last rows in the file).

```
-------------------------------------------------------------------------------
   J' K-1' K+1'    J  K-1  K+1        frequency          error    fit?   note/RI
+++++-----+++++-----+++++-----+++++++++++++++++++---------------+++++----------


-------------------------------------------------------------------------------
-------------------------------------------------------------------------------
   33    0   33   32    0   32       333130.749429       0.050000    2 3.16E-02
   33    1   33   32    0   32       333130.749429       0.050000    2 4.75E-03
   33    0   33   32    1   32       333130.749429       0.050000    2 4.75E-03
   33    1   33   32    1   32       333130.749429       0.050000    2 3.16E-02
```

In some cases, degenerate lines may not show up individually in the stick spectrum, but are marked with a "D" preceding the transition.

Only one line beneath cursor

"D" indicates degenerate line

```
ascp_l
Line   1/ 1:   285112.6424    3.48E-03    D    34,   6, 28<-- 34,   5, 29   b.Q 1,-1
2015-08-20-2                   256.803          0,   0,   0<--  0,   0,  0
```

Alternately, multiple lines may be beneath the cursor, but one accounts for a degenerate pair.

```
Line   2/ 3:   343071.0301    3.22E-02    D    34,   0, 34<-- 33,   0, 33   a.R 0, 1
123-triazole                   191.577          0,   0,   0<--  0,   0,  0
```

Then, the user must measure that line twice using F2 and change the transition in the .ASF file manually. Recall that if transitions are marked to be included in the least-squares fitting multiple times in the .ASF file, ASFIT will display a detailed warning before beginning least-squares fitting.

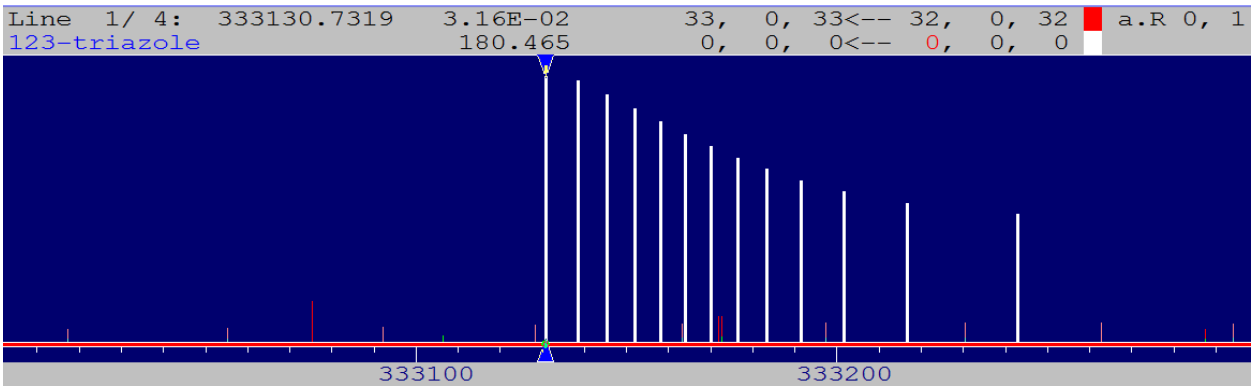
It is often convenient to cycle between bands and measure all of the corresponding transitions in those bands, both because lines with the same $K_{prolate}$ value tend to be similarly well predicted and because this action may be done with no right/left cursor scrolling.

To target a series of lines, press t (lowercase) followed by a value (±1-6) and ENTER. The choice of value is described in the following table. The most commonly used values are -2 (mainly used for R-branch transitions), which allows the user to cycle through n^{th} line of every band (n is the $K_{prolate}$ value), and -3 (mainly used for Q-branch transitions), which allows the user to cycle through lines with the same K_{oblate} value. The -4 value will allow a user to cycle through the lines within a vibrational state. If viewing a files with multiple vibrational states, the program will prompt whether to fix an additional quantum number, in which case, the user may select 4 to select a series within a vibrational state.

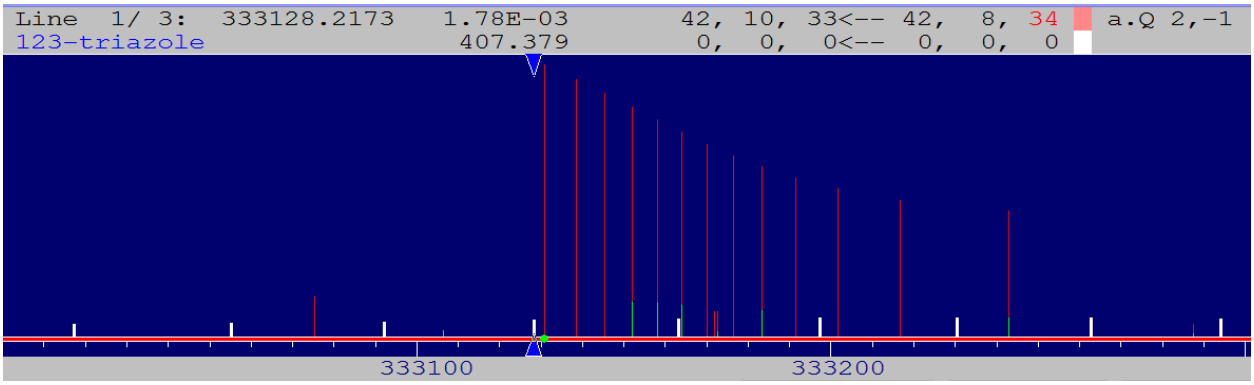
Enter Value	Series with Fixed Value(s)	Example of two lines in series
1	J'	33, 0, 33 ← 32, 0, 32 and 33, 2, 32 ← 32, 2, 31
2	$K'_{prolate}$	33, 0, 33 ← 32, 0, 32 and 34, 0, 34 ← 33, 0, 33
3	K'_{oblate}	33, 0, 33 ← 32, 0, 32 and 34, 1, 33 ← 33, 1, 32
4	Vibrational quantum$'_1$	
5	Vibrational quantum$'_2$	
6	Vibrational quantum$'_3$	
-1	J' and $(J' - K'_{prolate} - K'_{oblate})$	33, 0, 33 ← 32, 0, 32 and 33, 1, 32 ← 32, 1, 31
-2	$K'_{prolate}$ and $(J' - K'_{prolate} - K'_{oblate})$	33, 0, 33 ← 32, 0, 32 and 34, 0, 34 ← 33, 0, 33
-3	K'_{oblate} and $(J' - K'_{prolate} - K'_{oblate})$	33, 0, 33 ← 32, 0, 32 and 34, 1,33 ← 33, 1, 32
-4	VQ'_1 and $(J' - K'_{prolate} - K'_{oblate})$	
-5	VQ'_2 and $(J' - K'_{prolate} - K'_{oblate})$	
-6	VQ'_3 and $(J' - K'_{prolate} - K'_{oblate})$	
Selecting a negative number also fixes the value of $(J' - K'_{prolate} + K'_{oblate})$.		

The targeted series will be highlighted in the stick spectrum.

Stick spectrum with T-4 selected on R-branch line.



Stick spectrum with T-3 selected on Q-branch line.



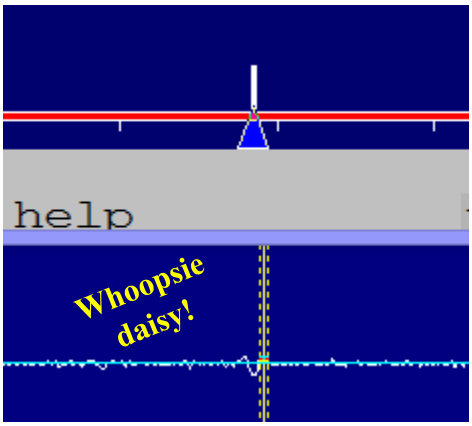
To target lines following a particular selection rule, press T (capital – shift+t) while the cursor is on a line in the same .ASR or .CAT file. When prompted, enter the value associated with a quantum number (1 – 6), a comma, then the value to set for the quantum number in the lower state (*e.g.* "2,0" to set $K_a = 0$ for the lower rotational state), then ENTER, then another value associated with a quantum number (1 – 6), a comma, and the change in that quantum number (*e.g.* "2,0" to set $\Delta K_a = 0$), then ENTER. For example, to jump to an $^{a}R_{0,1}$ branch with $K_a = 0$, move the cursor to a line in the same .ASR or .CAT file, type capital T, "2,0 ENTER 2,0 ENTER" and press a left or right arrow key to move to nearest such lines. Note that this will initially select other line types with $K_a = 0$ and $\Delta K_a = 0$, but by using the arrow keys until the desired line type is found, one may proceed to use the lowercase t command to select the desired series. This command is particularly useful for finding nominal interstate transitions.

[Insert examples here]

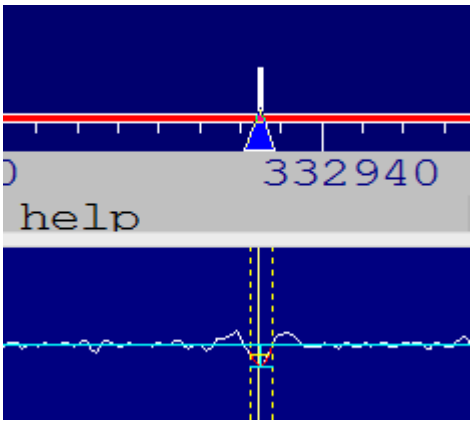
Alternately, the *r key* may be used to target all lines belonging to the molecule or state whose line is selected by the cursor when the R is pressed.

The user may then use the RIGHT and LEFT arrows keys to skip between lines in the series. Run least-squares fitting and update the .IN and corresponding .ASR files frequently to expedite line selection and avoid mis-assignments.
Sometimes, the minimum of a peak is not clear and the program measures an adjacent maximum or other incorrect frequency. It is sometimes possible to correct this by zooming in horizontally and increasing the intensity of the experimental spectrum.

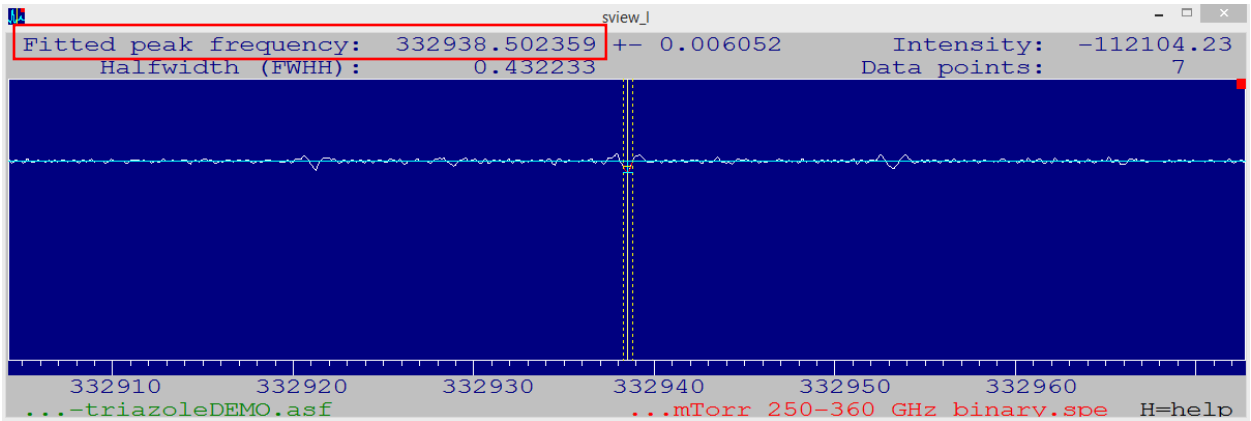
Adjacent manximum measured



Zoomed, minimum measured



To see the frequency of a minimum without measuring, make SVIEW the active window and scroll to the desired peak using K and L keys. Press the O key to measure the frequency.

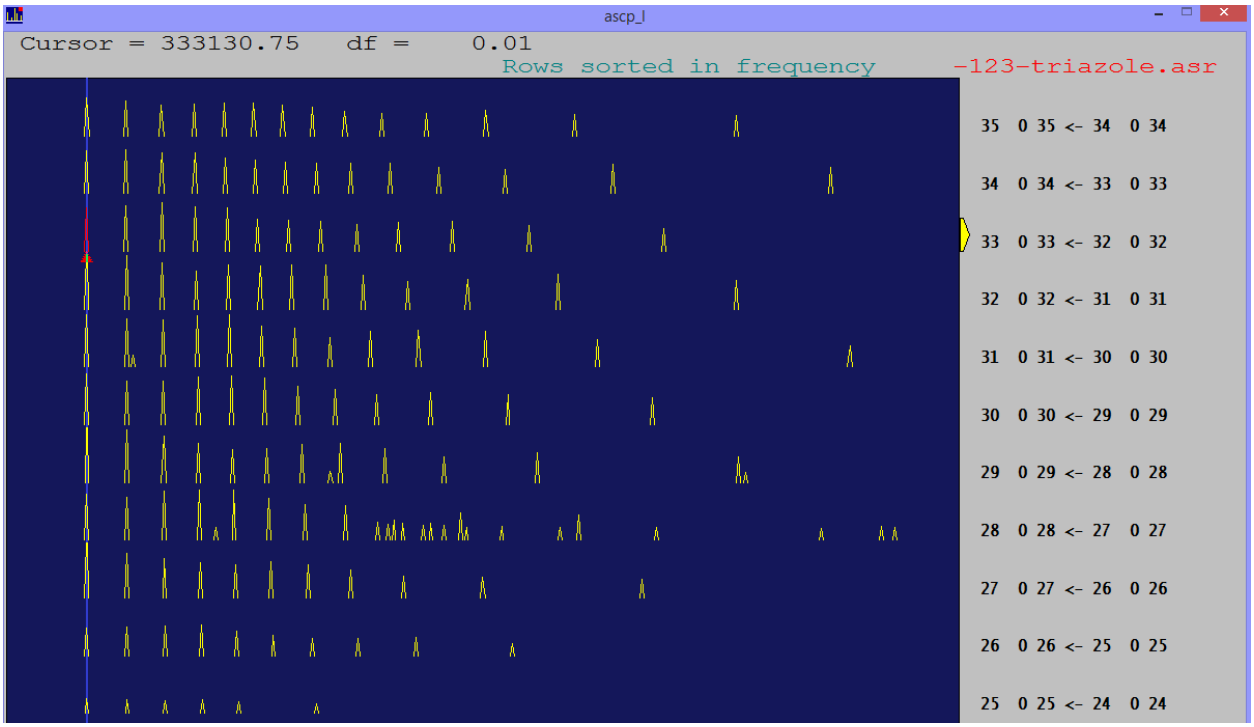

sview_l
Fitted peak frequency: 332938.502359 +- 0.006052
Intensity: -112104.23
Halfwidth (FWHH): 0.432233
Data points: 7
332910
332920
332930
332940
332950
332960
...-triazoleDEMO.asf
...mTorr 250-360 GHz binary.spe
H=help

Loomis-Wood Plots

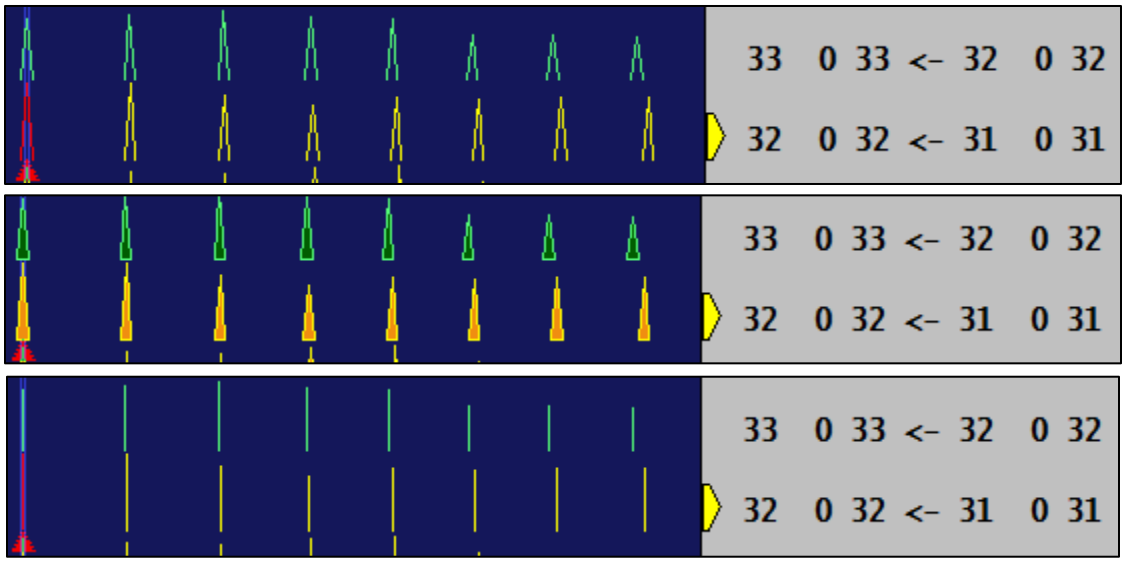
To view Loomis-Wood plots, the user must have a properly linked .FRE file in SVIEW_L.INP for the symbolic plot and must be using a binary .SPE in SVIEW to view the experimental spectrum plot.

Select a series of lines using the T-value command (described on pages 27-28).

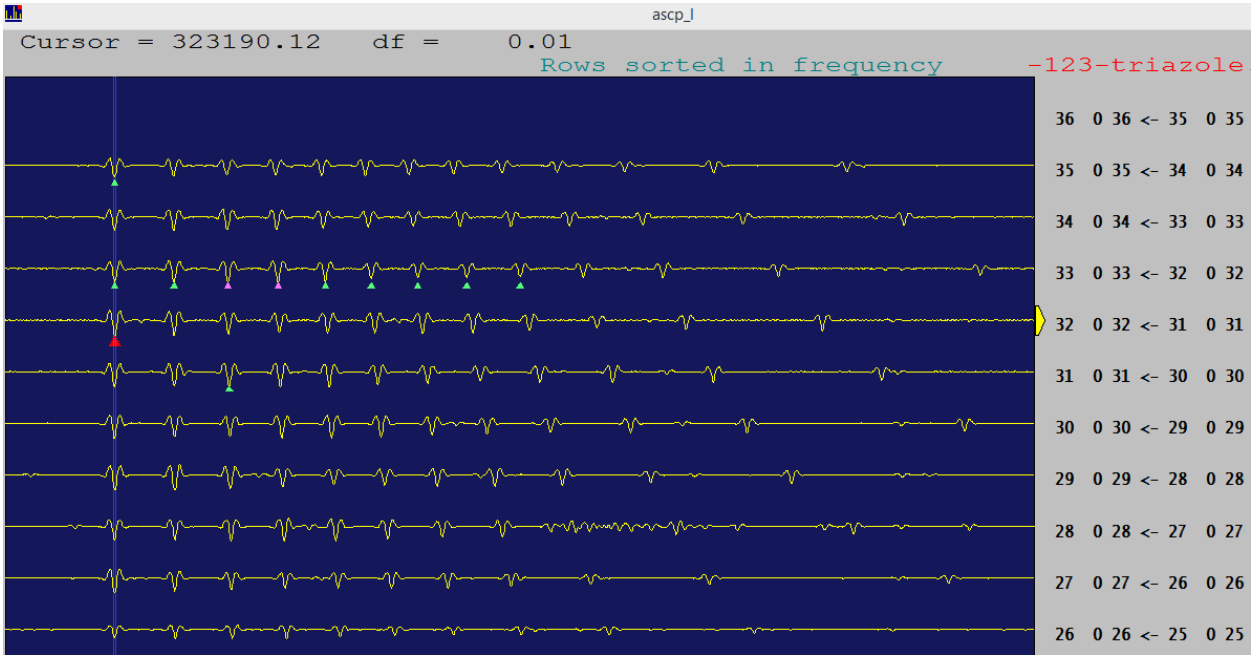
F5 (in ASCP) will bring up the Loomis-Wood Plot.



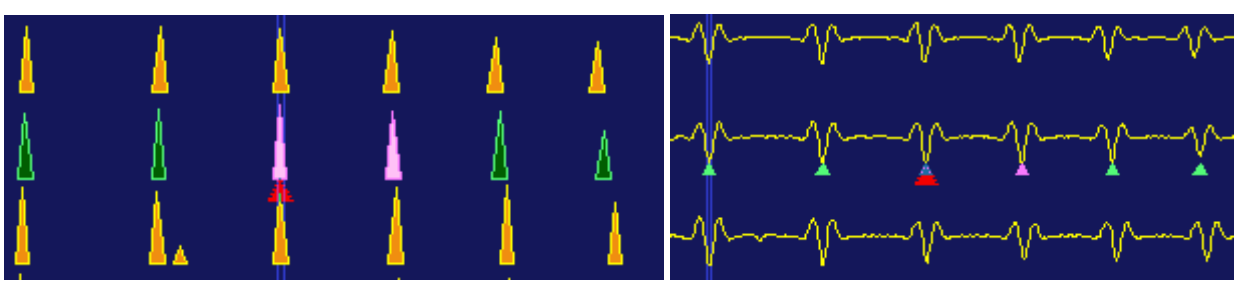
Hitting **P** will toggle different markers.



If using a binary .SPE file, the Loomis-Wood plot may also be viewed in spectral view by pressing the left bracket key, [. This view can be especially useful in seeking low-intensity molecules or states.



The red triangle is the cursor and may be moved using the arrow keys. Lines that have been measured are marked by triangles below the peak in spectral view and either by triangles below the marker or by non-yellow coloring of markers in symbolic view. The marker colors change depending upon the background color. With the default blue background, green triangles indicate lines included in least-squares fitting (*i.e.* marked with 1 or 2 in the .ASF file) and lilac indicates lines excluded from least-squares fitting (*i.e.* marked with 0).

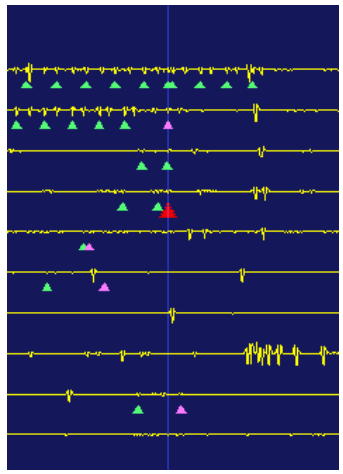


To measure a line from the Loomis-Wood plot, move the red cursor to the desired line and use the function keys (F1, F2, or F3) as usual. Only one set of transition quantum numbers can be entered into the .ASF file this way. However, doing so allows the user to mark the measured line and easily find it in the stick spectrum/experimental spectrum view. This can greatly simplify measuring lines in a dense region of lines with similar intensities.

Note that ASCP cannot be updated/refreshed using the ESC key in Loomis-Wood plot mode.

Making $\nu - \nu_0$ Plots

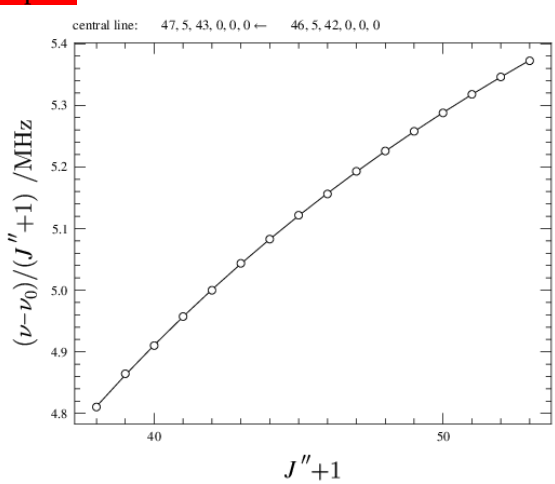
Using the *T* key, target the desired series in ASCP_L.exe. Enter the Loomis-Wood plot using *F5* and proceed to the spectral view using the left bracket, *[*. With at least four transitions in the targeted series, use the arrow keys to move the cursor to a line that is neither the highest nor lowest frequency, as shown.



Then press the *T* key and make a selection: *-1* for a plot of $\nu - \nu_0$ vs. $J''+1$, *1* for a plot of $(\nu - \nu_0)/(J''+1)$. Press ENTER and type a file name that is up to 8 characters long; press ENTER twice.

```
Select:        -1 = straight F-F0
                0 = exit
                1 = (F-F0)/(J+1) for p,R or (F-F0)/Ka for Q

                     ....  1

Up to eight character generic file name for GLE output for
this line sequence (ENTER = exit):

                     ....  demo

    Data file written to  C:\Rot\demo.dat
 Control file written to  C:\Rot\demo.gle

                     f i n i s h e d,   Press ENTER to continue
```

Open the .GLE file to view the plot.

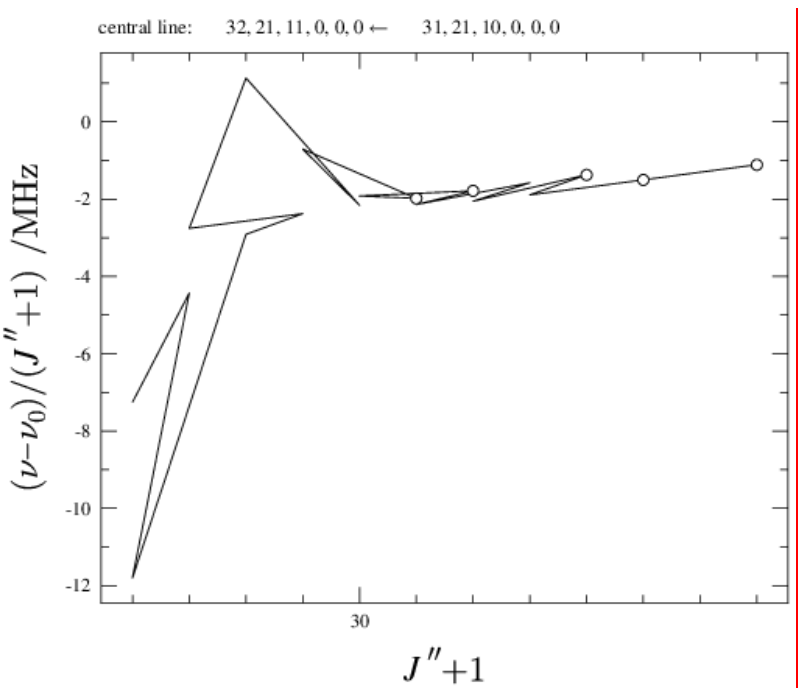


Two common errors that come up in making these plots have been noted.

1) If there are four or fewer transitions in the targeted series or if the cursor is on the highest or lowest lines in the Loomis-Wood plot, ASCP_L.exe often crashes when attempting to make the plot and must be restarted.
2) In some cases when a series contains degenerate lines and the user targets either the *J* or the *J + 1* series separately, it will appear as though the plot has been successfully written. Upon opening the file, however, the user finds the following error displayed at the bottom of the .GLE window.

```
>> demo.gle (27) |xaxis min ************** max ************** dticks 10.00...|
>> Error: illegal variable name '**'
>> In: '**************'
>>       ^
```

The user should then target the series including both *J* and *J + 1* lines (*i.e.* use positive values when targeting the series using the *T* key from the spectrum). This will result in two series being plotted as one – a disconcertingly ugly plot.



In order to correct it, the user should copy both the .GLE file and the corresponding .DAT file. Then, rename each descriptively (*e.g.* demo_J and demo_J+1). Open both the .GLE and .DAT files in a text editor. In the .DAT file, delete a lines to leave only the *J* transitions or the *J + 1* transitions.

							CALCULATED			MEASURED	
						J+1	F	F0	(F-F0)/(J+1)	F	(F-F0)/(J+1)
26	21	6	25	21	5	26	282278.2095	282466.8759	-7.256401	*	*
27	21	7	26	21	6	27	290249.2616	290368.9034	-4.431178	*	*
26	21	5	25	21	4	26	293480.3536	293787.1155	-11.798534	*	*
28	21	8	27	21	7	28	296713.8102	296795.3622	-2.912573	*	*
29	21	9	28	21	8	29	302641.8635	302710.8245	-2.377966	*	*
27	21	6	26	21	5	27	302854.7600	302929.2224	-2.757868	*	*
28	21	7	27	21	6	28	306700.2059	306668.5435	1.130799	*	*
30	21	10	29	21	9	30	308515.6815	308580.5332	-2.161725	*	*
29	21	8	28	21	7	29	310212.8184	310233.3598	-0.708324	*	*
31	21	11	30	21	10	31	314443.3133	314504.7254	-1.981035	314443.2448	-1.983244
30	21	9	29	21	8	30	315207.0253	315264.6125	-1.919573	*	*
32	21	12	31	21	11	32	320420.9964	320478.1703	-1.786685	320420.9680	-1.787572
31	21	10	30	21	9	31	320862.6835	320929.0518	-2.140913	*	*
33	21	13	32	21	12	33	326432.6618	326484.7804	-1.579351	*	*
32	21	11	31	21	10	32	326729.2785	326795.0861	-2.056488	*	*
34	21	14	33	21	13	34	332466.5645	332513.2422	-1.372874	332466.4751	-1.375504
33	21	12	32	21	11	33	332679.0680	332741.4947	-1.891720	*	*
34	21	13	33	21	12	34	338672.9464	338730.7312	-1.699552	*	*
35	21	14	34	21	13	35	344694.0297	344746.6554	-1.503591	344693.9824	-1.504941
36	21	15	35	21	14	36	350733.4161	350780.4249	-1.305799	*	*
37	21	16	36	21	15	37	356785.4986	356826.6177	-1.111326	356785.3649	-1.114940

Lines removed leaving only *J + 1* transitions.

							CALCULATED			MEASURED	
						J+1	F	F0	(F-F0)/(J+1)	F	(F-F0)/(J+1)
27	21	7	26	21	6	27	290249.2616	290368.9034	-4.431178	*	*
28	21	8	27	21	7	28	296713.8102	296795.3622	-2.912573	*	*
29	21	9	28	21	8	29	302641.8635	302710.8245	-2.377966	*	*
30	21	10	29	21	9	30	308515.6815	308580.5332	-2.161725	*	*
31	21	11	30	21	10	31	314443.3133	314504.7254	-1.981035	314443.2448	-1.983244
32	21	12	31	21	11	32	320420.9964	320478.1703	-1.786685	320420.9680	-1.787572
33	21	13	32	21	12	33	326432.6618	326484.7804	-1.579351	*	*
34	21	14	33	21	13	34	332466.5645	332513.2422	-1.372874	332466.4751	-1.375504

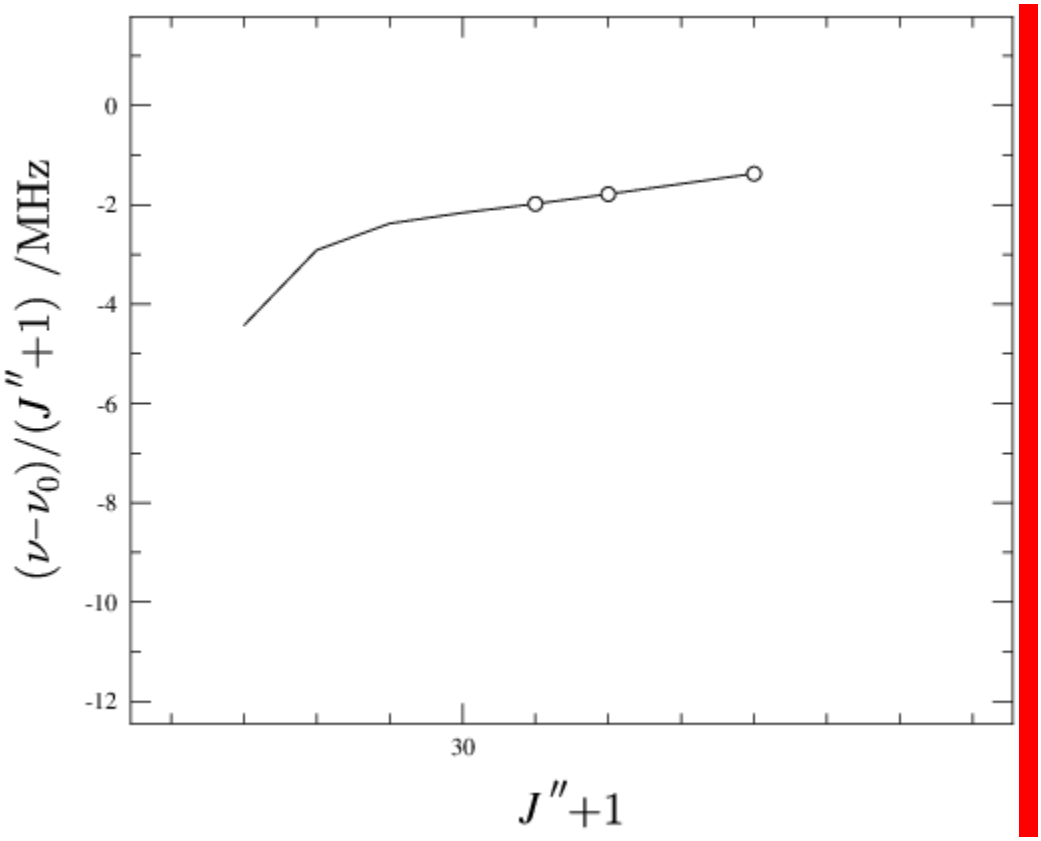
Save the file under the appropriate filename. Go to the .GLE file in the text editor and find the "data" commands in the lower half of the file.

```
    data demo.dat d1=c7,c10
    data demo.dat d2=c7,c12
!
    d1 lstyle 1
    d2 marker wcircle msize sizem color black
!
end graph
```

Change the name of the .DAT files to the appropriate new filename.

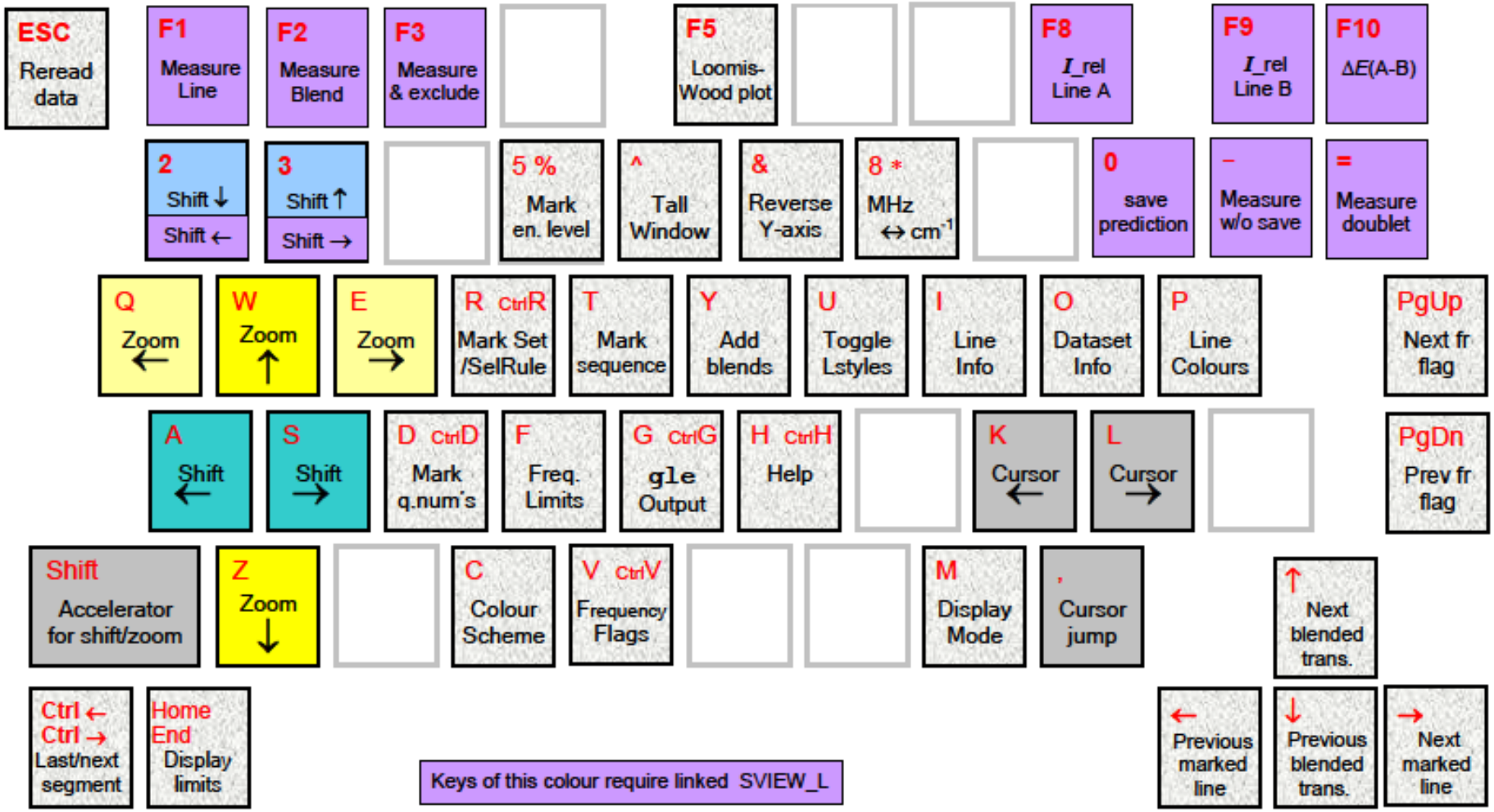
```
    data demo_J+1.dat d1=c7,c10
    data demo_J+1.dat d2=c7,c12
!
    d1 lstyle 1
    d2 marker wcircle msize sizem color black
!
end graph
```

It is recommended, but not absolutely necessary, to change or delete the "central line" text that appears at the top of the plot to match the displayed plot. The .GLE file (viewed in QGLE.exe) will now display a single series.

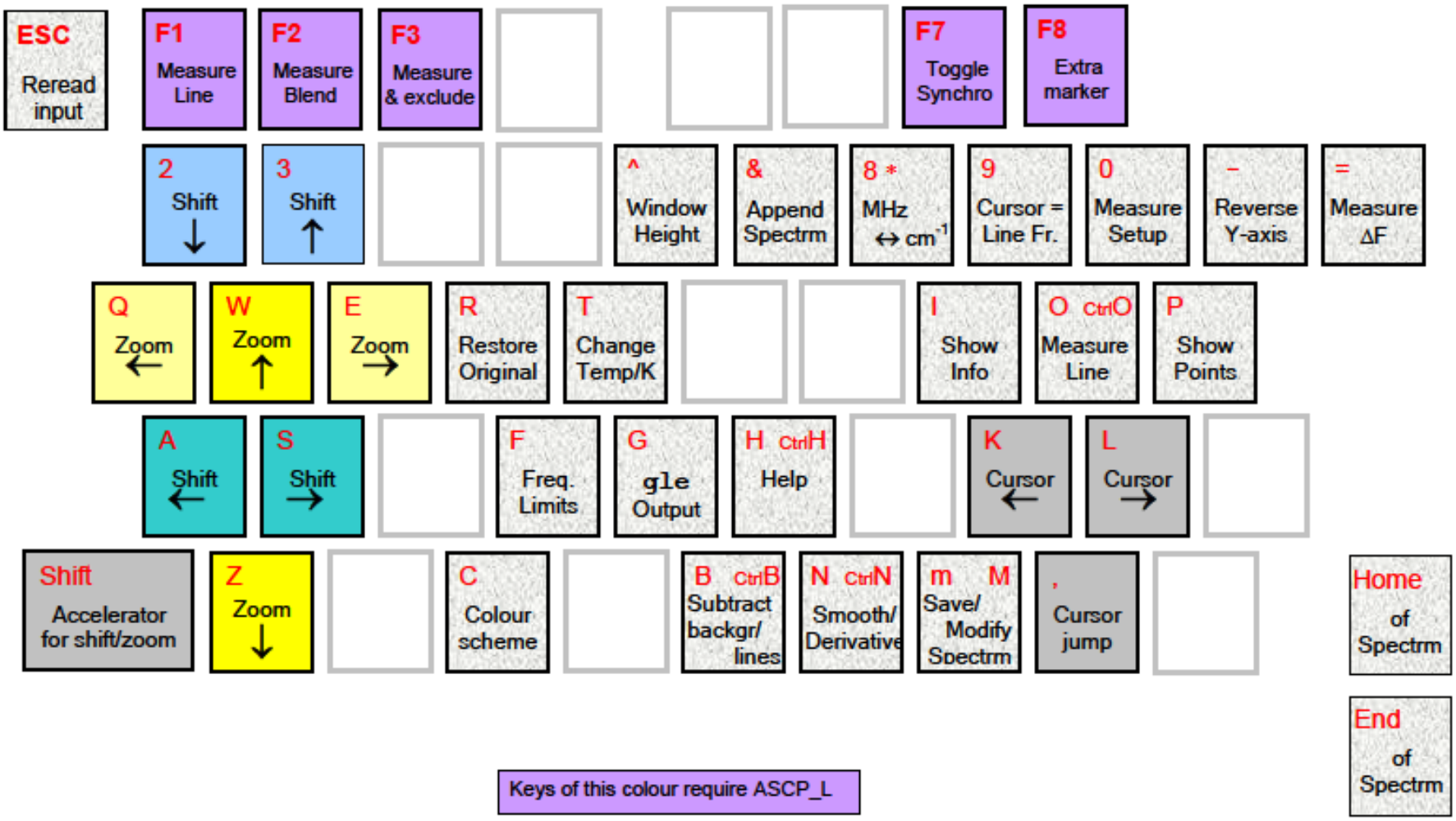


ASCP_L keys

Main screen



SVIEW_L keys


ESC Reread input
F1 Measure Line
F2 Measure Blend
F3 Measure & exclude
F7 Toggle Synchro
F8 Extra marker
2 Shift ↓
3 Shift ↑
^ Window Height
& Append Spectrm
8 * MHz ↔ cm-1
9 Cursor = Line Fr.
0 Measure Setup
– Reverse Y-axis
= Measure ΔF
Q Zoom ←
W Zoom ↑
E Zoom →
R Restore Original
T Change Temp/K
I Show Info
O CtrlO Measure Line
P Show Points
A Shift ←
S Shift →
F Freq. Limits
G gle Output
H CtrlH Help
K Cursor ←
L Cursor →
Shift Accelerator for shift/zoom
Z Zoom ↓
C Colour scheme
B CtrlB Subtract backgr/ lines
N CtrlN Smooth/ Derivative
m M Save/ Modify Spectrm
, Cursor jump
Home of Spectrm
End of Spectrm
Keys of this colour require ASCP_L

GLE Previewer (QGLE.exe)
(http://www.ifpan.edu.pl/~kisiel/prospe.htm#gle)

The GLE previewer allows viewing of various plots and figures coded in LaTeX. It can export the plot or figure in the window to other file formats, such as .PNG. To edit the plot or figure, the user need to edit the .GLE file in a text editor and potentially the .DAT file(s) from which the .GLE draws its data. The program automatically updates, so a user with two monitors may edit the .GLE or .DAT file(s) on one monitor and conveniently watch the plot or figure update on the other monitor each time the user saves.

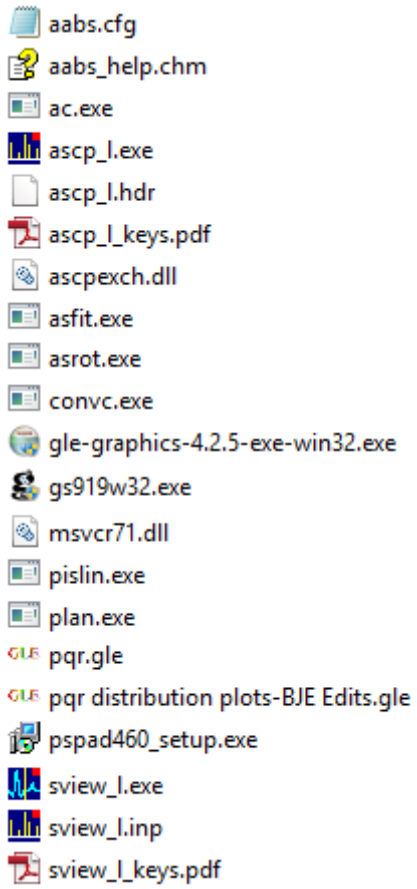
Where to Obtain the Required Files

The necessary AABS software may be obtained directly from http://www.ifpan.edu.pl/~kisiel/prospe.htm

PSPad, the recommended text editing software that allows column highlighting is available at http://www.pspad.com/ or the installer is available in the folder noted below. A user may simply paste the PSPAD.exe file into one's Program Files (x86) files on the main hard drive, but it will open in Slovak.

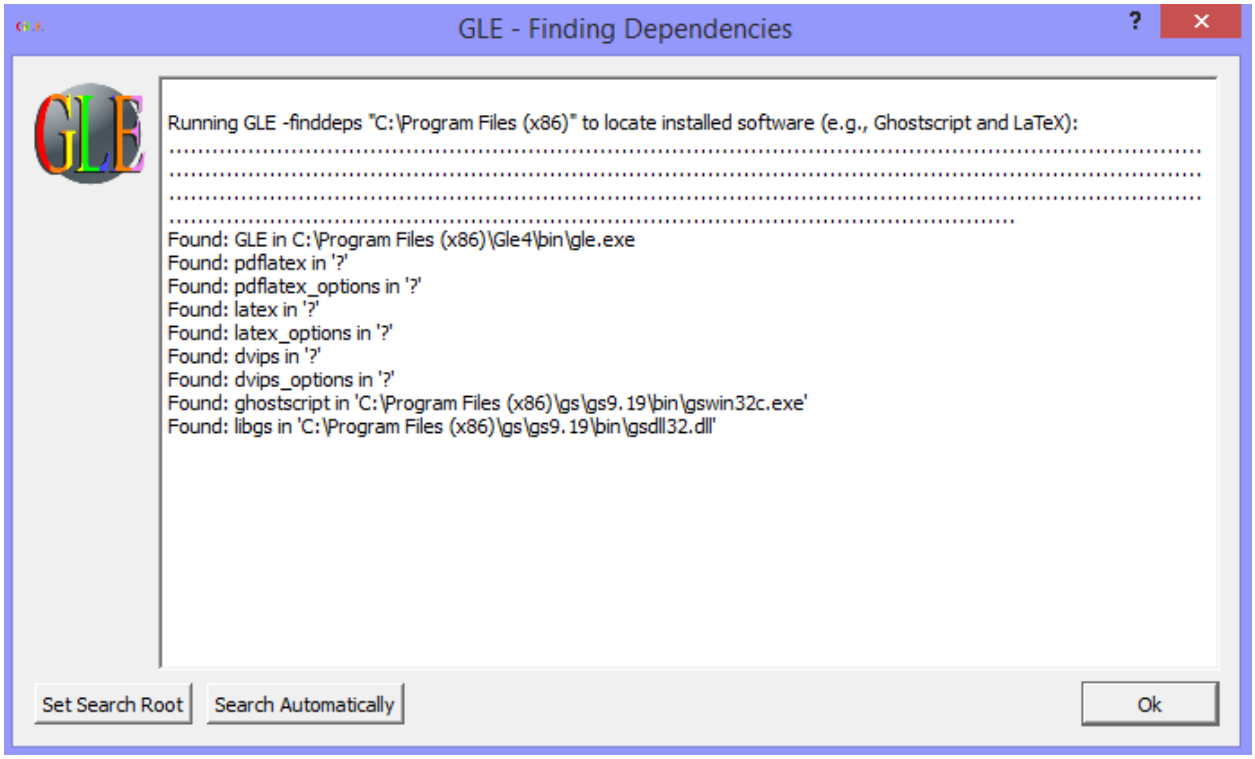
All necessary files are also located in *\\chemistryfs3.chem.wisc.edu\Groups\McMahon Group\Rotational Spectrometer\AABS Suite*

Required Files in ROT folder	
aabs.cfg	
aabs_help.chm	
ac.exe	
ascp_l.exe	
ascp_l.hdr	
ascp_l_keys.pdf	
ascpexch.dll	
asfit.exe	
asrot.exe	
convc.exe	
msvcr71.dll	
pislin.exe	
plan.exe	
pqr.gle	
Brian's edited .GLE file	
sview_l.exe	
sview_l.inp	
sview_l_keys.pdf	
Other Programs	
QGLE.exe gle-graphics-4.2.5-exe-win32.exe	
GhostScript gs919w32.exe	
PSPad.exe	



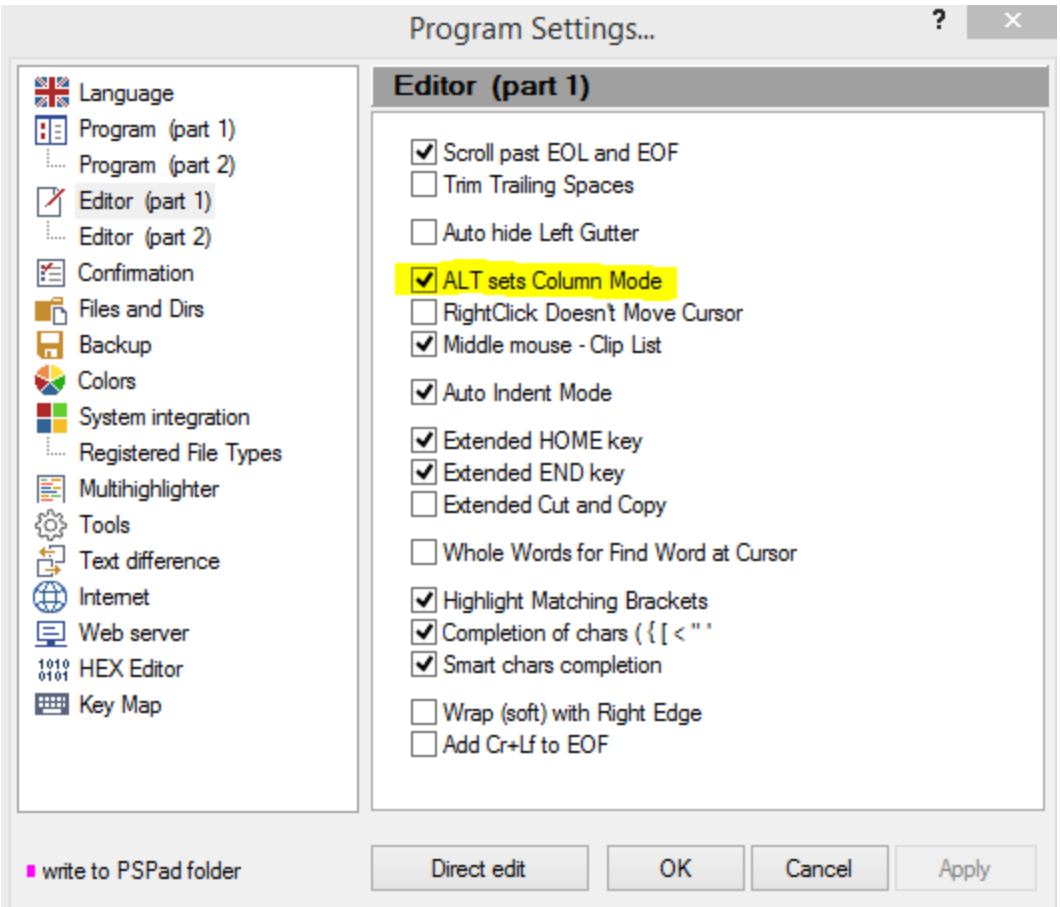
Proper Installation of Ghost Script and QGLE.exe

Copy the QGLE and GhostScript files to a directory on the computer, such as the Desktop. (Many computers have security settings that prevent copying directly to the necessary program files folder from an external source.) Transfer the files to the main drive (typically the C: drive) folder "Program Files (x86)". Run and install the Ghostscript file first. After is has installed, install QGLE. Near the end of the installation process, the program will search for GhostScript and other files. Check that it locates the follows as in the image below.



Activating Column-Highlighting in PSPad

To activate column highlighting in PSPad, open the "Program Settings" window under the "Settings" menu. Select "Editor (part 1)" on the left. Check the "ALT sets Column Mode" box and click OK.



Citing PROSPE software

Z.Kisiel, PROSPE - Programs for ROtational SPEctroscopy, http://info.ifpan.edu.pl/~kisiel/prospe.htm.

Z.Kisiel, in: J.Demaison et al. (Eds.), *Spectroscopy from Space*, Kluwer Academic Publishers, Dordrecht, 2001, pp.91-106.

ASFIT_Line_Analysis Excel Macro

The Excel Macro ASFIT_Line_Analysis.xlsm parses and analyzes ASFIT output and PIFORM output text files.

Features:

- Lists all of the lines and states the line type (*e.g.* $^{a}R_{0,1}$) in the "Lines" worksheet, including an "OOPS!" for mis-entered transitions (*e.g.* J′ and J″ written in opposite order, unknown line types, mistaken quantum numbers...)

- Removes lines excluded from the fit from the "Lines" worksheet, but leaves the line numbers for easy tracking.

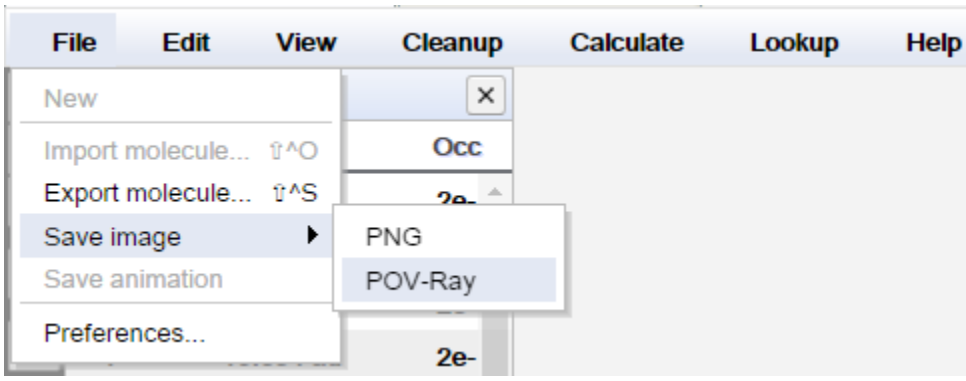
- Gives a line count by line type, including "OOPS!" count in the "Tables" worksheet.

- Gives the minimum and maximum J″, K″prolate, and K″oblate values in the "Tables" worksheet.

- Makes a table of constants and corresponding errors with correct notation for A- and S-reductions in the "Tables" worksheet.
    - There are two tables of constants -- one with typical publication units and another all in MHz
    - Values that were fixed in the least-squares fit are marked "FIXED"
    - B', Δ_t, K, σ, weighted σ, the number of lines, and the degrees of freedom are calculated and/or tabulated

POV-Ray Image Editing

In WebMO, run a geometry optimization followed by a Natural Bond Orbitals calculation at the desired level of theory. Select and view any molecular orbital (or the orbital of interest, should one be desired). Select File → Save image → POV-Ray. Give the file a meaningful name and save it to the POV-Ray folder located in the Documents directory.



Open the newly-saved .POV file. Replace the first five lines of the file (down to, but excluding, the first *cylinder* command

```
#default { texture { finish { specular 0.5 } } }
light_source {<-6.412,-6.315,-12.488> color rgb <1,1,1>}
camera { location <-6.837,-6.695,-13.310> sky <-0.905,0.173,0.388> look_at <-6.412,-6.315,-12.488> right -4/3*<-0.905,0.173,0.388> up <0.005,-0.909,0.417>}
background { color rgb <1.0,1.0,1.0> }
#macro C(r,g,b,a) texture{pigment{rgbt<r,g,b,a>}} #end
```

…With the text provided below

```
camera {
      location <0,9,0>
      look_at <0.0,0.0,0>
      right <-1.33,0,0>
      rotate <160,22,35>
}

light_source {
      <5,-10,5>
      color rgb <1,1,1>
}
background { color rgb <1.0,1.0,1.0> }
#macro C(r,g,b,a) texture{pigment{rgbt<r,g,b,a>}finish {
      ambient .2
      diffuse .6
      phong .75
      phong_size 25}} #end
```

Insert space between the last *sphere* command and the *mesh2* lines.

```
sphere { <-2.212,-0.780,-1.321>, 0.096 C(1.00,1.00,1.00,0.00)}
sphere { <0.128,-1.700,-2.152>, 0.096 C(1.00,1.00,1.00,0.00)}
mesh2 {vertex_vectors { 5376,<-2.29,-0.83,1.02>,<-2.29,-0.91,1.06>
normal_vectors { 5376,<-0.78,0.01,-0.62>,<-0.82,-0.20,-0.54>,<-0.8
texture_list { 5376,C(0.00,0.00,1.00,0.30),C(0.00,0.00,1.00,0.30),
face_indices { 1792,<0,1,2>,0,1,2,<3,4,5>,3,4,5,<6,7,8>,6,7,8,<9,1
}
mesh2 {vertex_vectors { 5376,<-2.29,-0.83,-1.50>,<-2.38,-0.83,-1.4
normal_vectors { 5376,<-0.58,-0.10,-0.81>,<-0.84,-0.10,-0.53>,<-0.
texture_list { 5376,C(1.00,0.00,0.00,0.30),C(1.00,0.00,0.00,0.30),
face_indices { 1792,<0,1,2>,0,1,2,<3,4,5>,3,4,5,<6,7,8>,6,7,8,<9,1
}
```

In the space between, enter the following macro.

```
#macro C(r,g,b,a) texture{pigment{rgbt<r,g,b,a> transmit 0.5} finish {
      ambient .2
      diffuse .6
      phong .75
      phong_size 25}}#end
```

It should appear as shown below.

```
sphere { <-2.212,-0.780,-1.321>, 0.096 C(1.00,1.00,1.00,0.00)}
sphere { <0.128,-1.700,-2.152>, 0.096 C(1.00,1.00,1.00,0.00)}

#macro C(r,g,b,a) texture{pigment{rgbt<r,g,b,a> transmit 0.5} finish {
      ambient .2
      diffuse .6
      phong .75
      phong_size 25}}#end

mesh2 {vertex_vectors { 5376,<-2.29,-0.83,1.02>,<-2.29,-0.91,1.06>,<-2.
normal_vectors { 5376,<-0.78,0.01,-0.62>,<-0.82,-0.20,-0.54>,<-0.81,-0.
texture_list { 5376,C(0.00,0.00,1.00,0.30),C(0.00,0.00,1.00,0.30),C(0.(
face_indices { 1792,<0,1,2>,0,1,2,<3,4,5>,3,4,5,<6,7,8>,6,7,8,<9,10,11:
}
mesh2 {vertex_vectors { 5376,<-2.29,-0.83,-1.50>,<-2.38,-0.83,-1.42>,<-
normal_vectors { 5376,<-0.58,-0.10,-0.81>,<-0.84,-0.10,-0.53>,<-0.62,-(
texture_list { 5376,C(1.00,0.00,0.00,0.30),C(1.00,0.00,0.00,0.30),C(1.(
face_indices { 1792,<0,1,2>,0,1,2,<3,4,5>,3,4,5,<6,7,8>,6,7,8,<9,10,11:
}
```

Save the file and click Run to render the image. To view the image as you make adjustments, a small image is suggested (*e.g.* [640x480, No AA]).

Further Image Adjustments

Angle, lighting, and size within frame

These can be adjusted by changing the *camera* and *light source* functions, located slightly above the bond- and atom-drawing commands.

```
camera {
    location <7.309945,9.57232,-8.093034>
    look_at <0.09236193,0.13256334,0.0>
    right <-1.33,0,0>
}

light_source {
    <7.217583052510278,9.439756848820501,-8.093033436841925>
    color rgb <1,1,1>
}
```

For simplification, the *location* and *look_at* may be changed to simple values, as below, and the *rotate* command added to rotate the molecule.

```
camera {
      location <0,10,0>
      look_at <0.0,0.0,0>
      right <-1.33,0,0>
      rotate <135,0,145>
}

light_source {
      <10,13,10>
      color rgb <1,1,1>
}
```

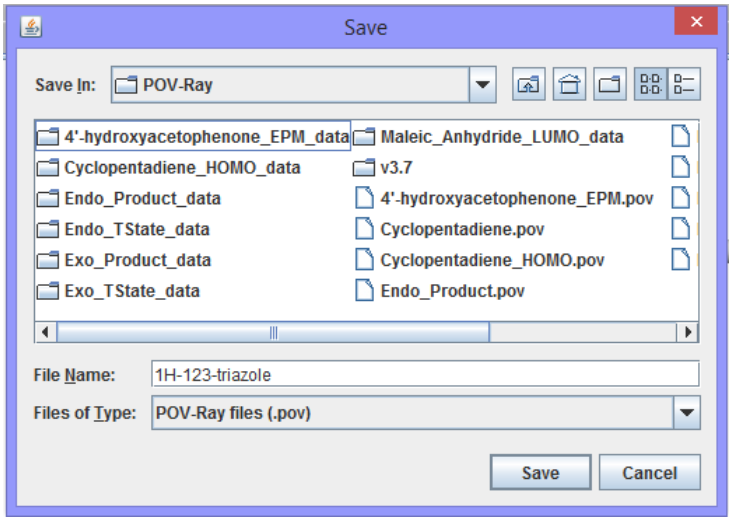
Colors

A table of codes for commonly used colors is provided below.

Color	Code
Red	<1.0,0.0,0.0,0.0>
Yellow	<1.0,1.0,0.0,0.0>
Green	<0.25,1.0,0.0,0.0>
Blue	<0.0,0.0,1.0,0.0>

POV-Ray Image Editing (JAVA WebMO Version)

In WebMO, run a geometry optimization followed by a Natural Bond Orbitals calculation at the desired level of theory. Select and view any molecular orbital (or the orbital of interest, should one be desired). Select File → Save As to the POV-Ray folder located in the Documents directory. Give the file a name and select "POV-Ray Files (.pov)" from the file type dropdown menu, then save.



Open the newly-saved file using POV-Ray. Replace the highlighted text (everything above the camera, depicted in the image on the following page) with the text provided in the upper part of the file *POV-Ray_Code.txt* (in the AABS Suite folder).

```
#declare ATOM_SIZE = 0.3;
#declare BOND_SIZE = 0.1;
#declare POS_ISO_COLOR = <0.0,0.0,1.0,0.33>;
#declare NEG_ISO_COLOR = <1.0,0.0,0.0,0.33>;
#declare COLOR_H = <1.0,1.0,1.0>;
#declare COLOR_C = <0.3,0.3,0.3>;
#declare COLOR_N = <0.0,0.0,1.0>;

#macro DECLARE_ATOM(X1,Y1,Z1,RADIUS,COLOR1)
sphere {
    <X1,Y1,Z1>, RADIUS*ATOM_SIZE
    pigment {color rgb COLOR1 }
}
#end

#macro DECLARE_BOND(X1,Y1,Z1,X2,Y2,Z2,BO,COLOR1,COLOR2)
#local MID_X = (X1+X2)/2.0;
#local MID_Y = (Y1+Y2)/2.0;
#local MID_Z = (Z1+Z2)/2.0;
cylinder {
    <X1,Y1,Z1>, <MID_X,MID_Y,MID_Z>, BO*BOND_SIZE
    pigment {color rgb COLOR1 }
}
cylinder {
    <X2,Y2,Z2>, <MID_X,MID_Y,MID_Z>, BO*BOND_SIZE
    pigment {color rgb COLOR2 }
}
#end

camera {
    location <7.309945,9.57232,-8.093034>
    look_at <0.09236193,0.13256334,0.0>
    right <-1.33,0,0>
}

light_source {
    <7.217583052510278,9.439756848820501,-8.093033436841925>
    color rgb <1,1,1>
}
background { color rgb <1.0,1.0,1.0> }

DECLARE_ATOM(1.1174802,-1.8703885,0.0,1.4550897,COLOR_C)
DECLARE_ATOM(2.0793638,0.53488886,0.0,1.4550897,COLOR_C)
```

If the image is to have molecular orbitals, replace the *pigment* and *finish* portions of the Isosurface code with that provided in the aforementioned file. Additionally, the *interpolate* value (1, just below the density filename) should be changed to 4. If the molecule is not to have orbitals, the isosurface code should be deleted.

Highlighted code to be replaced in Isosurface code for better orbital images (yellow – interpolation value; black – pigment and finish functions)

```
#declare ISO_FUNC=function
{
    pattern
    {
        density_file df3 "C:\Users\Ballooneytick\Documents\POV-Ray\1H-123-triazole_data
        interpolate 1
        scale <14.377055,14.1928005,10.185628>
        translate <-6.9364204,-6.806139,-4.9336634>
    }
}

isosurface {
    function { ISO_FUNC(x,y,z) }
    threshold 0.29271623
    contained_by{box{<-6.776675,-6.6484413,-4.7745132>,<6.961399,6.913568,4.7745132>}}
    max_gradient 1.0
    open
    pigment {rgbf POS_ISO_COLOR }
    finish {}
}
```

Adjust the code to draw double bonds properly by adding the number "2" after "DECLARE_BOND". If the orientation of the molecule does not allow the double bond to be visible, "2alt" should be used in place of the "2". The double bonds are most easily located by searching for "2" immediately preceding the color.

In the image below, both the edited code to draw double bonds and the "2" used to locate the double bonds are highlighted in yellow.

```
DECLARE_ATOM(1.1174802,-1.8703885,0.0,1.4550897,COLOR_C)
DECLARE_ATOM(2.0793638,0.53488886,0.0,1.4550897,COLOR_C)
DECLARE_ATOM(0.0,2.0205386,0.0,1.4172952,COLOR_N)
DECLARE_ATOM(-2.1356325,0.6323666,0.0,1.4172952,COLOR_N)
DECLARE_ATOM(-1.4537683,-1.7280053,0.0,1.4172952,COLOR_N)
DECLARE_ATOM(-0.120231934,3.9219413,0.0,0.94486344,COLOR_H)
DECLARE_ATOM(3.9738672,1.2735639,0.0,0.94486344,COLOR_H)
DECLARE_ATOM(2.0911086,-3.6568148,0.0,0.94486344,COLOR_H)
DECLARE_BOND2alt(-2.1356325,0.6323666,0.0,-1.4537683,-1.7280053,0.0,2,COLOR_N,COLOR_N)
DECLARE_BOND(-1.4537683,-1.7280053,0.0,1.1174802,-1.8703885,0.0,1,COLOR_N,COLOR_C)
DECLARE_BOND2(1.1174802,-1.8703885,0.0,2.0793638,0.53488886,0.0,2,COLOR_C,COLOR_C)
DECLARE_BOND(2.0793638,0.53488886,0.0,0.0,2.0205386,0.0,1,COLOR_C,COLOR_N)
DECLARE_BOND(0.0,2.0205386,0.0,-2.1356325,0.6323666,0.0,1,COLOR_N,COLOR_N)
DECLARE_BOND(0.0,2.0205386,0.0,-0.120231934,3.9219413,0.0,1,COLOR_N,COLOR_H)
DECLARE_BOND(2.0793638,0.53488886,0.0,3.9738672,1.2735639,0.0,1,COLOR_C,COLOR_H)
DECLARE_BOND(1.1174802,-1.8703885,0.0,2.0911086,-3.6568148,0.0,1,COLOR_C,COLOR_H)
```

Colors are defined at the very top of the code file. They may be adjusted by changing the values within the chevrons corresponding to the color to be changed. *POS_ISO* and *NEG_ISO* colors correspond to lobes of the isosurface.

```
#declare ATOM_SIZE = 0.3;
#declare BOND_SIZE = 0.1;
#declare BOND_SPLIT = 0.25;
#declare POS_ISO_COLOR = <1.0,1.0,0.0,0.0>;
#declare NEG_ISO_COLOR = <0.25,1.0,0.0,0.0>;
#declare COLOR_H = <1.0,1.0,1.0>;
#declare COLOR_C = <0.3,0.3,0.3>;
#declare COLOR_N = <0.0,0.0,1.0>;
#declare COLOR_O = <1.0,0.0,0.0>;
```

To view the image, select a small type (*e.g.* [640x480, No AA]), save the file, and click “Run”.